ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2024)6560347 - 17/09/2024

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101178059
Project name:	A First of a kind Hub for circularity demonstrator for Attica and peripheral regions
Project acronym:	Theseus
Call:	HORIZON-CL4-2024-TWIN-TRANSITION-01
Topic:	HORIZON-CL4-2024-TWIN-TRANSITION-01-38
Type of action:	HORIZON-IA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 December 2024
Project duration:	60 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

Industrial-Urban symbiosis (I-US) involves municipalities and regions cooperating with industries on the management of resources, waste, energy, water, infrastructures, services, transportation and networks. This inter-disciplinary and cross-sectoral character is implemented by Hubs4Circularity (H4C) in a FOAK approach, as key instruments for delivering territorial I-US solutions in Europe by Processes4Planed partnership. Theseus will establish first-of-a-kind H4C in Greece originating from the capital region of Athens/Attica (incl. Athens metropolitan area -Greek capital, 68 municipalities, 2.928 km, 3,7 million inhabitants 35% of national- several industrial and semi-urban areas) towards the whole country, with overarching aim of closing loops through I-US to become climate neutral by 2050, in line with P4P partnership objectives. Theseus will validate new or existing technologies and develop solutions for water, energy and materials based on regional needs (Innovation Pillar (IP)
2 , 3) and digital/robotic technologies (IP4). The hub will operate under innovative governance and cooperation models of involved stakeholders, to enable pilots, upscale solutions, exploit outcomes, attract further companies and expand existing ecosystem (IP1,5). Water, energy and material flows were targeted selected, being widely spread, having high potential all over Europe and included in EU/P4P strategies. The overall systemic, based on regional science approach and IP1,IP5 activities will ensure strong insights to smoothly replicate Theseus solutions to other EU regions. Several existing innovations from H4C, I-US and Circular Cities and Regions Initiative (CCRI) projects will be used, retrofitted and expanded.. The systemic solution of Theseus is expected to have significant impact on the region, while metaphorically recalling the mythical figure of Theseus, king and great reformer of Athens.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	ICCS	EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON	EL	999654356
2	BEN	CIBOS	CIBOS INNOVATION IDIOTIKI KEFALAIOUCHIKI ETAIREIA	EL	891970776
3	BEN	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142
4	BEN	ATTICA	PERIFEREIA ATTIKIS	EL	954443723
5	BEN	ADDMA	ETAIREIA ANAPTYXIS KAI TOURISTIKIS PROVOLIS ATHINON - ANAPTYXIAKI ANONYMOS ETAIREIA ORGANISMOU TOPIKIS AFTODIOIKISIS	EL	996713801
6	BEN	SusCITY	DIKTYO POLEON GIA TI VIOSIMI ANAPTYXI KAI KYKLIKI OIKONOMIA	EL	907509788
7	BEN	DHCP	DIMOTIKI EPICHEIRISI TILETHERMANSIS PTOLEMAIDAS-DIMOU EORDAIAS (DETIP)	EL	928584493
8	BEN	CLEAN	CLEAN ENERGY EPE DEVELOPMENT	EL	972886430

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
			APPLICATION AND CONTROL OF CLEAN ENERGY TECHNOLOGIES		

9	BEN	CERTH	ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS	EL	998802502
10	BEN	NCSR	NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS"	EL	999978239
11	BEN	HMU	ELLINIKO MESOGEIAKO PANEPISTIMIO	EL	899132771
12	BEN	DUTH	DIMOKRITIO PANEPISTIMIO THRAKIS	EL	999659109
13	BEN	UAEGEAN	PANEPISTIMIO AIGAIOU	EL	999840693
14	BEN	TUC	POLYTECHNEIO KRITIS	EL	924773848
15	BEN	TUD	TECHNISCHE UNIVERSITAET DRESDEN	DE	999897729
16	BEN	LiU	LINKOPINGS UNIVERSITET	SE	999852236
17	BEN	A.SPIRE	A.SPIRE	BE	939785180
18	BEN	CORE	CORE KENTRO KAINOTOMIAS AMKE	EL	891712756
19	BEN	EMVIS	EMVIS SYMVOULOI MICHANIKOI ANONYMI ETAIREIA	EL	952799670
20	BEN	RISA	RISA SICHERHEITSANALYSEN GMBH	DE	999512930
21	BEN	TLNV	TELENAVIS ANONYMI EMPORIKI ETAIREIAPAROCHIS TILEMATIKON YPIRESION KAIEIDON PLIROFORIKIS	EL	965534218
22	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
23	BEN	CIRCE	FUNDACION CIRCE CENTRO DE INVESTIGACION DE RECURSOS Y CONSUMOS ENERGETICOS	ES	999516907
24	BEN	HSRT	HYPERTECH KENTRO EPISTIMONIKON KAI TECHNOLOGIKON EREVNON AEIFORIAS ASTIKI MI KERDOSKOPIKI ETAIREIA	EL	889954146
25	BEN	WATT	MONOPROSOPI ANONYMI ETAIRIA EPEXERGASIAS APORRIMMATON PROS ANAKYKLOSI & PARAGOGIS ENALLAKTIKON KAFSIMON AE	EL	878932715
26	BEN	RECYCOM	VANGELIS ARAPIS KAI SIA - OMORRYTHMI ETAIRIA EMPORIAS KAI ANAKYKLOSIS	EL	893479611
27	BEN	EYDAP	ETAIREIA YDREYSEOS KAI APOCHETEFSEOS PROTEYOYSIS ANONIMI ETAIREIA	EL	957970449
28	BEN	Halcor	ELVALHALCOR ELLINIKI VIOMIHANIA HALKOU KAI ALOUMINIOU ANONYMOS ETAIREIA	EL	912255319
29	BEN	ThraceMarble	MARMARA THRAKIS A.V.E.E.	EL	879325274
30	BEN	ThracePoly	THRACE POLYFILMS MONOPROSOPI	EL	902089719

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
			ANONYMI VIOMICHANIKI KAI EMPORIKI ETAIREIA		
31	BEN	KEBE	KERAMOURGIA VOREIOU ELLADOS ANONYMI ETAIRIA	EL	893482036
32	BEN	TSIMIS	TSIMI ANONYMOS ETAIREIA VIOMICHANIA PARAGOGIS EFKAMPTIS SYSKEVASIAS	EL	878833581
33	BEN	PANCHART	AFOI PANOPOULOI ANONYMI VIOMICHANIKI KAI EMPORIKI ETAIREIA	EL	879742956
34	BEN	Fibran	FIBRAN INSULATING MATERIALS INDUSTRY DIMITRIOS ANASTASIADIS SA	EL	962894460
35	BEN	COMO	COMO MONOPROSOPI IKE	EL	882827168
36	BEN	AB	ATHINAIIKI ZYTHOPIIA ANONYMOS ETAIRIA	EL	904942974
37	BEN	RISE	RISE RESEARCH INSTITUTES OF SWEDEN AB	SE	999613422
38	BEN	MALMO	MALMO STAD	SE	993964045
39	BEN	SHAS	SINTEF HELGELAND AS	NO	916377528
40	BEN	SINTEF	SINTEF AS	NO	910945140
41	BEN	KPH	KUNNSKAPSPARKEN HELGELAND AS	NO	890668745
42	BEN	HFB	HELGELAND FERDIGBETONG AS	NO	879541584
43	BEN	DS	FONDEN DANSK STANDARD	DK	951792131
44	BEN	SN	Standard Norge	NO	907260886
45	BEN	ICL	IMPERIAL COLLEGE OF SCIENCE TECHNOLOGY AND MEDICINE	UK	999993468
46	BEN	HUD	UNIVERSITY OF HUDDERSFIELD	UK	999452208
47	BEN	TWI	TWI ELLAS ASTIKI MI KERDOSKOPIKI ETAIREIA	EL	906183701
48	AP	TVP	TVP SOLAR SA	CH	916449211
49	AP	MIP	Mo Industripark AS	NO	916026970
50	AP	NFK	NORDLAND FYLKESKOMMUNE	NO	947735688
51	AP	RU	RANA UTVIKLING AS	NO	892528041
52	AP	RANA	Rana kommune	NO	913537853

LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Project Management & Ethics	1 - ICCS		1	60	D1.1 – Theseus PM, QA & Ethics v1 D1.2 – Theseus PM, QA & Ethics v2 D1.3 – Theseus PM, QA & Ethics v3 D1.4 – Theseus PM, QA & Ethics v4 D1.5 – Theseus PM, QA & Ethics final D1.6 – Data Management Plan v1 D1.7 – Data Management Plan v2 D1.8 – Data Management Plan v3 D1.9 – Data Management Plan v4 D1.10 – Data Management Plan final D1.11 – Short Interim Management Report v1 D1.12 – Short Interim Management Report v2 D1.13 – Short Interim Management Report v3
WP2	Development of the Theseus framework	16 - LiU		1	36	D2.1 – Governance Model v1 D2.2 – Governance Model v2 D2.3 – Governance Model final D2.4 – Participatory approaches v1 D2.5 – Participatory approaches v2 D2.6 – Participatory approaches final D2.7 – Material flow analysis v1 D2.8 – Material flow analysis v2 D2.9 – Material flow analysis final D2.10 – Regional/interregional pathways v1

						D2.11 – Regional/interregional pathways v2

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D2.12 – Regional/interregional pathways final D2.13 – Characterization of value chains v1 D2.14 – Characterization of value chains v2 D2.15 – Characterization of value chains final D2.16 – Digital interactions and planning v1 D2.17 – Digital interactions and planning v2 D2.18 – Digital interactions and planning final

WP3	Digital Tools as enablers of Greek Attica H4C and common initiatives	1 - ICCS		1	48	D3.1 – GIS tool v1 D3.2 – GIS tool v2 D3.3 – GIS tool final D3.4 – Symbiotic models v1 D3.5 – Symbiotic models v2 D3.6 – Symbiotic models final D3.7 – Dynamic DPP and MR app v1 D3.8 – Dynamic DPP and MR app v2 D3.9 – Dynamic DPP and MR app final D3.10 – Prediction logistics module v1 D3.11 – Prediction logistics module v2 D3.12 – Prediction logistics module final D3.13 – Data Exchange Channels v1 D3.14 – Data Exchange Channels v2 D3.15 – Data Exchange Channels final D3.16 – Theseus H4C Markeplace v1 D3.17 – Theseus H4C Markeplace v2 D3.18 – Theseus H4C Markeplace final

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables

WP4	Separation and Sorting technologies	1 - ICCS		1	48	D4.1 – Technologies requirements D4.2 – Sorting lines (TRL 5-ready) D4.3 – Sorting lines (TRL 6-ready) D4.4 – Sorting lines (TRL 7/8-ready) D4.5 – Autonomous Platform v1 D4.6 – Autonomous Platform v2 D4.7 – Autonomous Platform final D4.8 – Smart bins for Attica v1 D4.9 – Smart bins for Attica v2 D4.10 – Smart bins for Attica final D4.11 – XR for human-robot interaction alpha version D4.12 – XR for human-robot interaction beta version
WP5	Material Processing Technologies	23 - CIRCE		6	48	D5.1 – Biological recycling for textiles v1 D5.2 – Biological recycling for textiles v2 D5.3 – Biological recycling for textiles final D5.4 – Biodegradable masterbatches v1 D5.5 – Biodegradable masterbatches v2 D5.6 – Biodegradable masterbatches final D5.7 – Construction materials v1 D5.8 – Construction materials v2 D5.9 – Construction materials final D5.10 – Biobased thin films v1 D5.11 – Biobased thin films v2 D5.12 – Biobased thin films final D5.13 – Thin-film materials v1 D5.14 – Thin-film materials v2 D5.15 – Thin-film materials final
WP6	Energy Symbiotic Flows	9 - CERTH		1	48	D6.1 – RES technologies

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D6.2 – Solar thermal technology v1 D6.3 – Solar thermal technology v2 D6.4 – Solar thermal technology final D6.5 – CHP fuel cell system v1 D6.6 – CHP fuel cell system v2 D6.7 – CHP fuel cell system final D6.8 – DH smart technologies v1 D6.9 – DH smart technologies v2 D6.10 – DH smart technologies final
WP7	Water Symbiotic Flows	3 - NTUA		1	60	D7.1 – Analysis for Attica water needs D7.2 – Decentralized Water Reuse v1 D7.3 – Decentralized Water Reuse v2 D7.4 – Decentralized Water Reuse final D7.5 – NBS for Water Reuse v1 D7.6 – NBS for Water Reuse v2 D7.7 – NBS for Water Reuse final D7.8 – Fertigation of reclaimed water v1 D7.9 – Fertigation of reclaimed water v2 D7.10 – Fertigation of reclaimed water final D7.11 – Valorization of sewage sludge v1 D7.12 – Valorization of sewage sludge v2 D7.13 – Valorization of sewage sludge final D7.14 – Master Plan for CE in Water v1 D7.15 – Master Plan for CE in Water final

WP8	Piloting and Demonstration	3 - NTUA		25	60	D8.1 – Biobased materials and Packaging flow initial report D8.2 – Textiles flow initial report D8.3 – CDW & Glass flow initial report D8.4 – Energy flow initial report D8.5 – Water flow initial report

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D8.6 – Biobased materials and Packaging flow report v2 D8.7 – Textiles flow initial report v2 D8.8 – CDW & Glass flow report v2 D8.9 – Energy flow report v2 D8.10 – Water flow report v2 D8.11 – Biobased materials and Packaging flow final report D8.12 – Textiles flow final report D8.13 – CDW & Glass flow final report D8.14 – Energy flow final report D8.15 – Water flow final report D8.16 – Validation of Frameworks V1 D8.17 – Validation of Frameworks final report D8.18 – Impact assessment V1 D8.19 – Impact assessment final report

WP9	Synergies, replication and Cooperation	37 - RISE		9	60	D9.1 – Synergies v1 D9.2 – Synergies v2 D9.3 – Synergies final D9.4 – Replication areas v1 D9.5 – Replication areas v2 D9.6 – Replication areas final D9.7 – Replicability Roadmap v1 D9.8 – Replicability Roadmap v2 D9.9 – Replicability Roadmap final
WP10	Nexus Modelling Social Awareness and Training	45 - ICL		7	60	D10.1 – H4C models initial D10.2 – H4C models final D10.3 – TM-WEMW nexus tool initial D10.4 – TM-WEMW nexus tool final

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D10.5 – Cities climate modelling v1 D10.6 – Cities climate modelling final D10.7 – AR app for awareness raising alpha version D10.8 – AR app for awareness raising beta version D10.9 – Training toolkit v1 D10.10 – Training toolkit v2 D10.11 – Training toolkit final D10.12 – Impact on health v1 D10.13 – Impact on health final D10.14 – De-risking assessment v1 D10.15 – De-risking assessment final

WP11	Exploitation and Dissemination activities	24 - HSRT		1	60	D11.1 – Communication Dissemination v1 D11.2 – Communication Dissemination v2 D11.3 – Communication Dissemination v3 D11.4 – Communication Dissemination final D11.5 – Policy Recommendations v1 D11.6 – Policy Recommendations final D11.7 – Exploitation Strategy v1 D11.8 – Exploitation Strategy v2 D11.9 – Exploitation Strategy v3 D11.10 – Exploitation Strategy final D11.11 – Market analysis v1 D11.12 – Market analysis final D11.13 – Standardization activities v1 D11.14 – Standardization activities final D11.15 – Stakeholder Engagement v1 D11.16 – Stakeholder Engagement v2

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D11.17 – Stakeholder Engagement final

Work package WP1 – Project Management & Ethics

Work Package Number	WP1	Lead Beneficiary	1 - ICCS
Work Package Name	Project Management & Ethics		
Start Month	1	End Month	60

Objectives

Project Management, Consortium coordination; Monitoring the overall technical progress; Managing legal, contractual, financial, and administrative aspects; Quality Assurance (QA); Establish Executive Board, assign managers (Tech, Risk, QA); Develop data management plan, open science, GDPR and gender balance requirements, manage legal or ethics matters.

Description

Task 1.1 Project Coordination & Management (incl. quality assurance, risk management) M1-M60 (ICCS, all): It will provide the overall project coordination, management, and administration by developing and implementing processes for an efficient and successful project execution, including QA and risk management. It includes: i)preparation of technical and financial interim reports every 6-month to ensure the achievement of deliverables and milestones and that technical activities follow resource consumption; ii)preparation of external EC reports, technical/financial monitoring and audit reports; iii)distribution of the EC financial contribution and costs controlling; iv)communication and submission of deliverables to EC; v) ensuring the communication between partners and the proper information exchange, organisation of consortium meetings, maintenance of appropriate communication and collaborative environment for the consortium (e.g., mailing lists, repository/wikis, reporting tools); vi) support work plan evolution; vii)implementation of QA procedures with particular focus on monitoring review activities and reports vi) ensuring proactive measures for risks, monitoring potential critical implementation risks; viii) management of the external advisory board to provide consulting and validation on vision, impact, and outcomes; ix)coordinate WP and pilot leaders and interact with T2.1,T2.2,T11.6,WP9 activities to effectively coordinate Consortium. Outputs: D1.1

Task 1.2 Technical Management and Scientific Connection (incl. data management) M1-M60 (NTUA, all): It will monitor the technical progress and quality to support Coordinator, guarantee the smooth technical cooperation between partners and WP/tasks, participate in the contributions of the relevant working groups, and suggest alternative routes to possible obstacles. It will also track milestones and coordinate the technical meetings, will ensure that scientific and technological outcomes comply with the project's work plan, and the results fulfil the technical requirements for effective progress towards project's goals. In accordance with EC Guidelines task will set up a Data Management Plan (DMP) defining how research data collected and/or generated throughout the project will be handled during (and after) the project duration. Following FAIR principles, task will elaborate on data types, accessibility to datasets, use of standards and publication methodology. Special emphasis will be put on making the research data accessible through open repositories, to maximise the visibility of the project's data results, aiming also, to assist other research activities to find and utilise the knowledge produced by Theseus. The task will deliver open science guidelines capturing the envisioned practices early with regular updates. Outputs: D1.1, D1.2

Task 1.3 GDPR, Gender Balance, Legal and Ethical Issues Management M1-M60 (ICCS, all): It will monitor the procedures for handling ethical issues during the project. A framework of ethical requirements will be constructed to ensure that project is carrying out of responsible research practice with specific reference to the European Code of Conduct for Research Integrity (ALLEA) and the Data Protection legislations. It will set out ethics requirements (e.g. criteria to identify/recruit participants, templates for consent, etc.) and manage the ethical issues concerning the design of information and consent forms to be used in case of involvement of humans through demonstrations, while ensuring GDPR and Privacy Compliance Policy by properly disseminate informed consent forms for all project participatory activities (demonstration activities, online surveys, evaluation forms, workshops, etc.) according to the EU GDPR regulations and national data protection legislation. It will also ensure that regional and social aspects are embedded in Theseus. Outputs: D1.1

Work package WP2 – Development of the Theseus framework

Work Package Number	WP2	Lead Beneficiary	16 - LiU
Work Package Name	Development of the Theseus framework		

Start Month	1	**End Month**	36

Objectives

Project Management, Consortium coordination; Monitoring the overall technical progress; Managing legal, contractual, financial, and administrative aspects; Quality Assurance (QA); Establish Executive Board, assign managers (Tech, Risk, QA); Develop data management plan, open science, GDPR and gender balance requirements, manage legal or ethics matters.

Description

Task 2.1 Governance for Regional and Intra Regional I-US M1-M36 (LiU, ICCS, UAEGEAN, WP partners): Governance experiences from different contexts will be reviewed and assessed for impacts, strengths and shortcomings taking notice of the contextual characteristics of relevance. Learnings gained will be used to propose a governance approach to effectively assist the emergence and development of a sustainable I-US system and community, continually deliver environmental, economic and social value in the long term. More specifically: Propose a core coordination body with overarching governance responsibilities; Identify key stakeholders and respective roles for different development stages and needs and plan their active engagement; Develop shared objectives; Formulate thematic workgroups; Develop plans for relevant activities. ICCS will apply model principles throughout project, piloting activities and results. Outputs:
D2.1
Task 2.2 Participatory approaches and co-creation M1-M36 (LiU, UAEGEAN, ADDMA, SusCITY): This task will research and propose approaches enabling active participation and collaboration of diverse stakeholders for effective co-creation of necessary building blocks for I-US developments. These approaches will be applicable to the Attica H4C but also to similar hubs in other European regions. The task will: Review literature to collect information on the stakeholders of I-US developments as well as documented participatory and co-creation approaches; Research 2-3 IUS developments, in different stages of developments, to better understand the role of different stakeholders as well as the details and potentials of different participatory and co-creation approaches; Identify, analyse and characterise stakeholders and structures that are relevant for Attica HUB for circularity; Building on the above and on the framework of Regional institutional capacity (RIC), task will provide suggestions for the Attica HUB for circularity regarding participatory approaches and cocreation, which will be used a direct input into T11.6 Stakeholder engagement; Deliver a guideline for future hubs to make this replicable; Outputs: D2.2
Task 2.3 Material flow analysis and 2050 projections M1-M36 (ICCS, TUC, CIBOS, ATTICA, CERTH, HUD): In this Task a detailed mapping of materials, water and energy flows, based on Material flow analysis (MFA), will be performed in Attica, considering also imports and exports with other regions. To facilitate comprehensive MFA, Process Simulation and Modelling (PSM) and its Material Flow Networks (MFN) framework will be utilized, to simulate the complex interplay of material and energy flows of Theseus H4C, creating an interconnected and circular ecosystem, where waste is minimized, and resources are optimally utilized. Offering capabilities for real-time process industry model simulation and monitoring, PSM integration to the digital platform will be explored. Based on this mapping and with task 2.5 input, future projections in short, middle and long term will be developed for the region of Attica as well as for Greece, aligned with the general vision of EU and Local authorities and proposing actions and strategies to be adopted. Outputs: D2.3
Task 2.4 Regional and interregional pathways for symbiotic flows linked to EU and local regional plans M1-M24 (UAEGEAN, ICCS, WP partners): Based on Regional Science and Circular economy roadmaps (CERMs) approaches, it will develop pathways and action plans to integrate the activities of Theseus H4C in Attica and other regions to EU, National, Regional or Municipal Plans (e.g. PEP Attica, RIS3, SUMPs etc). The path towards regional sustainable development will be ensured by measuring the impact of project activities to regions in environmental, social and economic level, how this impact affects their resilience and their position related to nearby regions, national and EU average level, using classic but also innovative ML enabled methods. Strategic blueprints including actions, initiatives, timelines with a focus on resource optimization and business value creation will be created. KPIs will be identified measuring sustainability and resilience (including economic & social aspects)linked and calculated by circularity aspects and KPIs defined in T2.3, T2.5, T2.6 ,WP10. Outputs: D2.4
Task 2.5 Characterization of value chains and KPIs definition M1-M36 (RISE, CORE, TUC, CERTH, HUD CIRCE): Theseus value chains will be analysed, characterized, detailed described and validated in a collaborative workshop. All the involved stakeholders will serve as additional input. Firstly, they will focus on added values and risks of future changes made to transaction points (existing/new) in value chains. Then a facilitated workshop will be performed to elicit diverse perspectives regarding benefits and risks of alterations to transaction points from stage one. Stakeholders will be guided through collaborative discussions regarding how their different values can be balanced, and responsibility

distributed between the different actors in the value chain. This includes the identification of additional stakeholders interested in or affected by future transaction point changes. Task will identify key KPIs that can be utilized both in the

description of the value chain, but also enable impact monitoring in all the different dimensions required. The outcome will be used directly in T2.3, T2.6 to provide insights. Outputs: D2.5

Task 2.6 Digital interactions and planning M1-M36 (CORE, TLNV, RISA, CIRCE,): It will coordinate digital plans for each pilot and comprehensive mapping of data requirements enabling the finalisation of data model. CORE will coordinate the participants involved in pilots, in line with the specifications developed for secure, safe and efficient data exchange in T3.5. It will develop, with WP3 partners, the data-model and integration of different solutions on platform level (GIS, models). Partners from WPs 4-7 will provide inputs for data of their processes to implement appropriate data acquisition measures prioritising cost-efficiency and robustness. The integration component of heterogenous data sources such as PEF, DPP, field or remote sensor data, in-situ observations and enrichment with data collected from open-access repositories and databases (such as pricing, climate and weather data) will be prepared for deployment in T3.5. By M12, the mapping will be completed, addressing needs for each use case and a deployment plan (extra sensors,

connectivity, restrictions, etc.). By M18, operational data availability will be ensured for critical parameters, to enable training of models and deployment of digital tools of WP3. Outputs: D2.6

Work package WP3 – Digital Tools as enablers of Greek Attica H4C and common initiatives

Work Package Number	WP3	**Lead Beneficiary**	1 - ICCS
Work Package Name	Digital Tools as enablers of Greek Attica H4C and common initiatives		
Start Month	1	**End Month**	48

Objectives

Create a digital framework for the holistic management, optimization and modelling of waste resources and logistics; Set up dynamic DDPs; Develop sophisticated modules and services; Ensure access to information and secure data exchange.

Description

Task 3.1 GIS tool for dynamic symbiotic maps and reverse logistics optimization M1-M40 (TLNV, ICCS, HUD, NTUA): Based in expertise from REDOL, ICCS will develop a digital I-US map, an GIS tool to be integrated in T3.6 platform and used as the territorial visualization of H4C activities. Data on mapping waste resources, flows, symbiotic chains, analysis and assessment of barriers, needs on reverse logistics etc. will be integrated and gathered from different sources, analysed and visualized. The GIS tool will be fed with existing I-US examples and related activities documented in Attica. Data collected will also include waste types and quantities from national databases (such as HMA), waste collection vehicle fleets routes and bins locations from municipal databases (available by TLNV), transportation data from regional such as traffic management center of Attica, as well as data from the IMPETUS tool mentioned below. Different optimization algorithms (heuristics/genetics) will be evaluated and implemented to optimize logistics (and reverse) operations considering static and dynamic data: positions of symbiotic stakeholders, providers and services, road network and traffic, scheduling, capacities, characteristics of flows and processes. Models will simulate different approaches in a dynamic context in terms of sustainability. The tool will be linked to IMPETUS Tool (NTUA for Attica) benefiting from analytics and functionality for Climate Adaptation for Region and inform the symbiotic map with relevant data and services. It will allow regional stakeholders to visualise symbiotic loops acting as a spatially aware, 'live inventory' of interventions throughout Attica, informing stakeholders for initiatives and provide extra input in T3.2 module. Outputs: D3.1

Task 3.2 Matching frameworks and development of symbiotic models M1-M40 (ICCS, IRIS, HUD, RISE): Based on matchmaking frameworks from previous projects and related research from ICCS and IRIS, dedicated modules will be developed and will be specialized for the different value chains tackled in Theseus. Symbiotic flows will be identified and evaluated by new matchmaking and recommender algorithms to be developed and deployed. Module will identify and pre-assess potential symbiosis between organisations (e.g. similarities in NACE and Classification), resource needs and compatibility, capacities, supply and demand; aspects such as aspects such as processes, services required, distance etc. by interacting with GIS (T3.1) and PRLM (T3.3) to enable sophisticated features. The evaluation and impact assessment of the proposed symbiotic models will be based on the further development of ISRL (industrial symbiosis readiness level)-Matrix developed in CORALIS project for I-US developments and value chains. ISRL-Matrix will be tested and validated in 3-5 different cases, and it will be developed to a VCRL (value chain readiness level) in 3 Theseus value chains. Outputs: D3.2

Task 3.3 DPPs and end-users interactions M1-M40 (ICCS): It will develop adjustable schemes for dynamically updated DPP modified accordingly per circular product and related material flow as per: i) DPP design at data and architecture level. Define static and dynamic information to be included based on existing regulations (EU /national), state of the art analysis and consultations with stakeholders. Special attention will be given to energy related data; ii) technical development of a DPP alpha version with live update mechanism and user-friendly interface; iii) demonstrate DPP and necessary modifications per case to be performed; iv) testing by demonstrators and collecting feedback from the users; v) present final version of the DPP tailored to the needs of users. Finally, an MR application will be developed to facilitate value chain interactions as the events in product lifecycle take place through immersive visualizations, assisting enduser to obtain information about available resources and enabling informed decisions for resource efficiency (e.g. for repurposing, recycling, reduce, etc.). MR tool will be fed via T3.5 platform with information on user interactions, ML object detection and positional awareness. Outputs: D3.3

Task 3.4 Prediction of resource logistics M1-M40 (CORE, HALCOR): CORE will design and operate a predictive resource logistics module (PRLM), to predict the material and water flows between the different actors of the H4C, using (and producing) quantitative and qualitative indicators and assigning weighed values to produce socioeconomic and environmental indicators. PRLM will comprise the operationalization of the material flow analysis, calculating the associated emissions, costs, and other variables based on dynamic values (provided by models e.g. LCIA, PEF, DPP or public data). The results will provide useful insights to the users, such as early identification of potential bottlenecks, business opportunities, risks and scalability / replication options, that on the pre-commercial, project-level implementation will be analysed by the project's experts. Additionally, a plan for the automation of this process postproject (TRL8) will be delivered, using tools such as NLP. Beyond material flow, cost and emission data, resiliencefocused indicators (supply / demand indicators, origin, risk-related indicators) will also be considered. PRLM will be hosted in T3.6 and will provide outputs through a dedicated analytics interface, with dynamic elements to provide clear and pertinent information to each stakeholder by type and sector. Outputs: D3.4.
Task 3.5 Data Harmonization and Secure data exchange M9-M48 (ICCS, CORE, IRIS, RISA, AB): It aims to harmonization, homogenisation and secure exchange of data between diverse sources, modules T3.1-3.4 and the T3.6 platform. It begins with data ingestion, followed by crucial data cleansing, involving tackling challenges (missing values, format inconsistencies). Activities will be based on defined common data models of T2.6. To enhance security, services, APIs, and platform are designed with security as a priority and secure communication via HTTPS protocol, encrypting data transmitted between clients and servers. Also, the implementation of the Security Enabler (MATRYCS tool, ICCS) will drive identity access management (IAM) strategy. This tool, an enhanced layer added to KeyCloak, empowers Service Owners to define role-based access policies, determining user access rights to specific datasets and functionalities based on roles. By utilising OAuth 2.0 protocol, the tool's capabilities are significantly expanded, offering comprehensive IAM functionalities. Outputs: D3.5
Task 3.6 Marketplace and modules integration across digital H4C M9-M48 (IRIS, ICCS, CORE, HSRT): A digital platform (web/mobile) will be built based on a microservices architecture ensuring scalability and modularity. UI/UX will be defined and agreed via a co-creation process with end-users, back-end and front-end will be developed. The work will follow iterative sprints following scrum methodology, each culminating in functionality releases through MVP (Minimum Viable Product) for end-users testing and their feedback will be used for optimization and refining the platform. The core functionalities of the marketplace will be built, including sophisticated modules and services including prediction, interaction, knowledge exchange, collaboration and intelligent matchmaking. Modules and algorithms from existing platforms developed by IRIS (ShareBox, ISH4C) and ICCS (Accelwater, CORALIS) and new matchmaking and predictive tools and models developed in the project -within T3.1-3.4 (GIS, IS models, DPP, logistics) will be incorporated in the platform though the secure data channels of T3.5. Combining all the available tools, data will be inputted in a searchable database of all organizations (in or out of consortium) and their resources, including waste streams, by-products, available capacities and services. Analytics tools for tracking the use of resources and measuring the environmental and economic benefits of the industrial symbiosis will also be deployed in an analytics suite, which will enable the visualisation and tracking of I-US synergies, also include information of products through DPPs. Outputs: D3.6

Work package WP4 – Separation and Sorting technologies

Work Package Number	WP4	**Lead Beneficiary**	1 - ICCS
Work Package Name	Separation and Sorting technologies		
Start Month	1	**End Month**	48

Objectives

Develop novel sensing, separation and sorting mechanisms and solutions to improve secondary streams valorisation in different material or sources; Develop methodologies and algorithms for the analysis and optimization of the secondary materials.

Description

Task 4.1 Separation and sorting at multiple levels analysis and Materials Upcycling, optimization and design M1M9 (NTUA, ICCS TUC, HALCOR, AB, HUD): It involves in-depth requirements analysis, conceptual design and optimization of sorting processes for each of the 5 distinct sorting lines as well as the analysis of the upcycling of materials (textiles, packaging and cement materials) of WP5. NTUA based on T2.3 outcomes and feedback of the involved partners, will analyse the proposed upcycling processes of WP5 and set the goals and acceptance for each one. ICCS will select and streamline sorting techniques for each case in collaboration with hosts, by analyzing data collected within T4.2-4.6 to develop tailored innovative strategies for efficient waste separation for each stream. Output: D4.1

Task 4.2 Mixed urban waste separation M6-M48 (ICCS, WATT, TUC, HUD): ICCS will initially deploy optical monitoring systems at designated locations. These modules will employ a combination of RGB and HSI over the conveyor lines to collect both qualitative and quantitative data from the passing waste stream, towards the development of a comprehensive dataset. The collected data will be utilized to analyze the flow and train ML classification algorithms. DL models, based on convolutional kernels, will be utilized to extract the spatio-spectral content of both HSI and RGB images and encode it in a sparse vector representation. Subsequently, the vectors will be passed through multiple fully connected layers to categorize the materials in their respective classes. In conjunction with T4.1, optimization interventions will also be devised and positioned strategically along the line, introducing the trained classifiers to accurately categorize metallic, paper, plastic waste, and disposed textiles using the installed optical systems and direct a combination of integrated delta robots, with effective payload up to 5kg, equipped with electrical or pneumatic gripping to automatically extract and redirect them into designated containers and a pneumatic actuation system using controlled pressurized air discharge nozzles to handle small sized waste, up to 3mm. Sorting accuracy is expected to exceed 95% and integrated system will be TRL8 validated. Output: D4.2

Task 4.3 Textiles sorting M6-M48 (ICCS, RECYCOM, HUD, WATT, CERTH, TUC, CIRCE): ICCS will integrate a machine vision module based on RGB and HSI exploring a spectrum range between 700-2500nm into RECYCOM's current textile sorting line. A rotary joint, 6 Degrees of Freedom (DOF) industrial robot, equipped with rigid grasping end-of-arm-tooling (EoAT) suitable to handle textile up to 5kg will be introduced. Guided by signals from instance segmentation model's (e.g. YOLOv8, SAM), the robot will lift individual garments from the line to increase exposure. The HSI classifier will employ deep neural architectures e.g. CNN with skip connections, Vision Transformers, to extract and analyze the spectral features of images and ultimately categorize the item by fibre type, (thermoplastic, natural, mixed fibres) with a precision rate over 90%. Subsequently, the robots will deposit the textiles into their respective containers. The data acquired by the vision system will be used in further analysis and optimization of the line in T4.1. Output: D4.3 Task 4.4 Plastics recovery from mixed waste M6-M48 (ICCS, WATT, TWI): It involves the enhanced distinguishing of specific types of plastic from generic waste streams stemming from WATT's deployed blue bins. ICCS will integrate a multi-sensor vision classifier that employs combined RGB and HSI into WATT's current sorting line along with delta robots capable of rapid pick-n-place operations using pneumatic suction grippers. For plastic polymers' localization, a SoTA neural network, i.e. YOLOv8, will be retrained on RGB images feeding the robotic system with information regarding the position, shape, size, etc. Low-cost delta robot alternatives, tailored to the payload and speed requirements of plastic sorting will be explored by TWI. A DL model based on CNN architecture trained by ICCS will jointly process the image data from both HSI and RGB cameras to identify PET, PE and bio-based plastics among the mixed waste will digitally label the passing items. Then robots will extract designated items from the stream, into dedicated bins
(negative sorting). Output: D4.4

Task 4.5 Metals Separation/ Glass separation M6-M48 (ICCS, HALCOR, AB, TWI): It will focus on the extraction of metals from mixed CDW utilizing an expanded version of the W2W sorter. A custom XRT cell will also be employed, designed by TWI, alongside pre-processing algorithms, and middleware interfaces required for interoperability with ICCS' software modules. ICCS will conduct comprehensive system upgrades, including the installation of pre and posttreatment modules for shredding and balling respectively, implementing a vibrating conveyor to ensure even distribution of waste, integrating a multi-sensor classification system for the precise categorization and grading of specific metals and robotic material handling. The extended sorting line will be directly integrated into HALCOR's facilities, undergoing testing to validate its functionality up to TRL8. The task also includes the implementation and testing of the glass bottle sorting line. ICCS will design and integrate all necessary upgrades in AB facilities. The line will include an HSI optical system, equipped with suitable lighting over a conveyor. Bottles guided along the conveyor will be scanned to identify shape and type. Intact bottles will be extracted and grouped using an industrial robot manipulator fitted with tailor-made gripping tooling. Damaged instances will continue until the end of the line and result in a container for recycling. Upon installation, ICCS will ensure TRL8 compliance. Output: D4.5, D4.6

Task 4.6 Resource optimization for symbiotic flows management at plant level M6-M48 (DUTH, ICCS): This task aims to explore and orchestrate autonomous platforms, e.g., AGVs, UAVs thus optimizing the material flows within plants to achieve the designed sorting and symbiotic processes. The main goal is to transfer the dedicated bins from the sorting stations to the targeted production line based on the outcomes of the tasks belonging to WP4. On the one hand, a) a comprehensive analysis of the existing material flow within the plants will be conducted to identify areas where

such platforms can be effectively applied. On the other hand, b) different solutions available in the market will be investigated and assessed in terms of suitability and efficiency, considering the corresponding pilot scenarios. Toward this goal, this task c) will also develop the corresponding navigation algorithms, e.g., path planning and obstacle avoidance, and perception techniques, e.g., object detection, human estimation, etc., required to enable the platform's integration in the plant. The overall focus is on enhancing efficiency, reducing manual intervention, and improving operational performance. Output: D4.7

Task 4.7 Smart bins for Attika M6-M48 (ICCS, ADDMA, ATTICA, SusCity, WATT, HUD): It will conduct in-depth data collection and mass flow analysis of waste within Athens municipality. Data will be utilized to develop an AIdriven digital tool identifying and quantifying the distinct valorisation streams. ICCS will employ this tool for waste flow calculations and predictive estimation, providing valuable insight for optimal placement of smart bins developed by ICCS, IoT and automatic waste segregation enabled , focusing to space restrictions of Attica. Output: D4.8

Task 4.8 XR for human-robot interaction/collaboration M6-M48 (ICCS, ADDMA): ICCS will develop an application for XR Head Mounted Device to interact with the robot and other sensors during i) textile, and ii) urban waste separation. A user-friendly interface will deliver fast and clear guidelines to a human operator about the actions to be performed to optimize and facilitate a more efficient sorting process. Object detection algorithms will be embodied to the XR system, to facilitate the positional awareness of the requested action. The XR application will optimize the training of workers during sorting to significantly optimize human intervention and improve efficiency. Assessment and evaluation reporting will ensure a structured quality controlling. Task will include the configuration layer and communication layer among the XR application and the sorting components, as well as the XR interface for intuitive actions. Output: D4.9

Work package WP5 – Material Processing Technologies

Work Package Number	WP5	**Lead Beneficiary**	23 - CIRCE
Work Package Name	Material Processing Technologies		
Start Month	6	**End Month**	48
Objectives			
Design, test and validate innovative solutions for waste flows recycling and valorisation into circular symbiotic synergies.			
Description			
Task 5.1 Textiles upcycling M6-M48 (CIRCE, HUD, TUC, Fibran, COMO, HUD, CERTH): Post-consumption textile based on PET will be biologically recycled by CIRCE through a depolymerization process, incorporating the use of various metallic catalysts under MW-assisted heating. PET textile will be depolymerized by glycolysis in 75ml reactors for initial trials and optimized in a 1L scaled up batch reactor. This process will use ethylene glycol together with metallic catalysts to improve the yield and quality of the monomer obtained from PET waste. The task will explore the modification of several variables (e.g. microwave intensity, reaction time, catalyst concentration) to optimize a) reaction yield and speed and b) recovery and reuse of solvents and catalysts. The potential impact of scale-up to yield, purity, and overall process efficiency will be assessed. The yield and purity of BHET will be established using characterization techniques such as NMR, DSC, GC-MS . Post-consumer textile waste will be respun by HUD with a spinning process To achieve high value-added circular products, the new fibre will be enhanced with active compounds, either at spinning, where the delivery of the actives could be modelled and designed as a factor of the chemical and morphological structure, or post-spinning, by developing novel linker chemistry to attach the actives with the fibrous materials and functionalise them using technologies such as plasma treatment. The active compounds that can be used include polydimethylsiloxane (water and oil-repellent finishing), polyhexamethylene biguanide (antimicrobial finishing) or different nanoparticles. TUC will develop AI algorithms to critically relate recycling process parameters with material properties to be recycled (as indicated by sorting) and achieve their maximum performance. The bio-chemical recycling by HUD will be optimized for different components of textile waste streams. An AI-assisted framework will be created to critically understand the structure-property relationship of the recycled materials as a factor of recycling process parameters, towards appropriate high-value technical products. Output: D5.1 Task 5.2 Packaging Upcycling M6-M48 (HMU, Panchart): HMU will receive recycled bio-based or synthetic biodegradable polymers, e.g. bio-based ones: PLA, PHA, PBS, PBAF; synthetic ones: Polycaprolactone (PCL),			

Polybutylene adipate terephthalate ester copolymer (PBAT). These plastics will dry at 60oC in a vacuum over overnight and will be fed then to a mechanical shredder machine to be chopped resulting into a powder-like material, being kept in dry environment. The following biodegradable recycled materials/ formulations will be developed by HMU: (1) neat polymers, (2) mixed polymer powders, towards biodegradable polymer blends (with co-continuous and bicontinuous morphologies with/ without compatibilizers), and (3) polymer powders mixed in solid state or via a twinscrew melt compounding process with different low-cost (nano-)additives e.g. CaCO3, ZnO, SiO2 nanoparticles, etc., in different wt.% filler loadings..Cost of additives on the above formulations will be considered as a function of the potential increase of the properties e.g. mechanical, thermomechanical, etc. The recycled based thermoplastic polymeric materials / compounds that will be developed will be tested by melt-rheology, DMA, DSC, FT-IR, GPC and tensile testing on dog-bone as well as on melt-blown film samples [KPIs: (1) at least 50% reduced cost of the recycled compounds, and (2) >10% increased tensile modulus and 25% strength, due to the additives that can have a bi-functional role, e.g. as reinforcement due to the rule-of-mixtures (RoM) principle or acting as nucleating agents]. Optimum melt-mixing/ compounding processes by a twin screw-extrusion process (screw speed>200 rpm and at optimum compounding times: t and temperatures: T) will be applied for the proper additive dispersion [means of verification: TEM, SEM, melt-rheology, Raman, DMA, DSC]. The nanocomposites will be prepared in one-step mixing process, while some representative mixing parameters could be the following: rotation speed>100 rpm, mixing time of 5 min and temperature, T> the Tm of the thermoplastic used each time. The materials will be fed via a hopper into the microcompounder at the processing speed. After completing the mixing process, the materials will be taken out from the compounder as an extruded strand.
Output: D5.2
Task 5.3 Innovative construction/cement materials M6-M48 (TUD, KEBE, SINTEF, MarbleThrace, Fibran): CDW and waste glass will be refined with both mechanical separation and wet separation technologies, to develop different combinations of alternative raw materials and binders for brick production and concrete production. Theseus approach will apply the whole fraction of CDW. Fine fractions of CDW and pulverised waste glass will be systematically designed for brick and concrete production by mineralogy (XRD, TGA), microstructure (SEM/EPMA) and pozzolanic reactivity by determining the chemical bound water . Binder combinations will be optimised based on ceramic content, waste glass content, mineralogy and chemical leaching for both application routes. Task 5.3.1 will develop a brick where the quality is optimised regarding mechanical strength, water absorption, linear shrinkage, bulk density, colour changes. Task 5.3.2 will develop three grades of concrete mix designs, targeting the 28-day compressive strengths of 30 MPa, 40 MPa, 50 MPa. Fresh and hardened concrete properties will be documented according to EN 1235052, EN 1239053. The durability of two final recipes (Recipe 1,2) will be tested according to EN standard testing methods for key durability parameters including resistance to chlorides, carbonation, sulphates, reactive alkalis and frost. Task 5.3.3 will assess the environmental acceptability (i.e. leaching of dangerous substances) will be determined for both product applications. Laboratory leaching tests will be conducted for both single waste materials and as applied in final building products (i.e. concrete and brick). The leaching properties will be measured on the materials in granular (EN 14429) and monolithic form (CEN/TS 16637-2:2014) to disclose solubility and diffusion controlled leaching behaviour, respectively. Both are important for products in primary service life and in the recycling stage to cover the life cycle leaching. Geochemical speciation modelling (e.g. Orchestra) will be used for the leaching data to identify certain crucial leaching mechanisms (e.g. sorption to Fe/Al oxide surfaces, solid solution formation etc.). Output: D5.3
Task 5.4 Biobased thin films for packaging M6-M48 (NCSR, TSIMIS): It involves the development of a biobased thin film using bioplastics extracted. i)Manual development & testing: Food waste rich in cellulose and polysaccharides e.g. fruit peels will be extracted and converted in usable form for materials. The extracted polysaccharides will be blended with other biopolymers (chitosan, gelatin) and additives (solvents, plasticizers, crosslinking agents) in various ratios to achieve desired mechanical and barrier properties. ii)Automated dev/testing: Technical designs and prototyping of the automated synthesis and testing will be realized. Synthesis (pumps, liquid handling, powder dispenser, magnetic and mechanical mixing) and metrology modules (pH, rheology, characterization) will be prototyped and calibrated. The automated robotic synthesis will be validated by comparing to manual synthesis with respect to accuracy, repeatability for various biomaterials. iii)AI Planner: Distributed IoT modules will be used for flexible automation and planning during every cycle. A Bayesian Optimizer will be integrated for the decision making of recipes to be tested. iv)Autonomous Biobased thin film optimization for packaging: Reduction of development cycles for the creation of various biobased thin films with targeted properties. Output: D5.4
Task 5.5 Electrospinning for thin films M6-M48 (NTUA, ThracePoly): NTUA pilot line is capable to prepare pellets and fibres from recycled polymers with different functionalities for a wide range of applications using electrospinning process and extruder line. Through these process lines, various compounds will be incorporated into the polymeric structures to adapt or enhance specific properties, such as antimicrobial or barrier properties, and, thus, to extend the shelf life of the thin polymeric films. The incorporation of the compounds will be performed in three different ways, each one described in a separate Subtask. Subtask 5.5.1 involves the direct encapsulation of compounds in the polymeric matrix through extrusion to produce batches of enhanced polymeric pellets. In Subtask 5.5.2, the encapsulation of compounds in the polymeric matrix will be done through electrospinning and a micro/nano-composite powder will be

developed that will be co- extruded with conventional polymeric pellets. Thus, batches of enhanced polymeric pellets will be produced. In

Subtask 5.5.3, the encapsulation of compounds in the polymeric matrix through electrospinning will be performed for the development of a nanofiber film that will be coated onto conventional polymeric films. Thus, functional multilayer films will be fabricated. The developed products can have potential applications in packaging (garbage bags, medical packaging, food packaging), agricultural films and textiles sectors. Finally, in Subtask 5.5.4, the involved materials and produced structures will be characterized in terms of: (a) their mechanical properties (shrinkage, tensile strength, etc.), (b) their thermal stability through Differential Scanning Calorimetry (DSC), (c) the chemical characteristics and the interaction of the used matrix with the included agents through Infrared Spectroscopy using the technique of Attenuated Total Reflectance (ATR) and Infrared Spectroscopy and mainly (d) the encapsulation efficiency of the antimicrobial compounds using High Performance Liquid Chromatography (HPLC) and Gas Chromatography coupled with Mass Spectrometry, (e) the viscoelastic behaviour using Dynamic Mechanical Analyzer (DMA). Output: D5.5

Work package WP6 – Energy Symbiotic Flows

Work Package Number	WP6	**Lead Beneficiary**	9 - CERTH
Work Package Name	Energy Symbiotic Flows		
Start Month	1	**End Month**	48
Objectives			
Design, test and validate a set of innovative solutions for the reuse and valorisation of energy flows.			
Description			

Task 6.1 Study for Introduction of innovative RES technologies to District heating (DH) systems M1 -M9 (CERTH, TVP, RISE, CLEAN): It aims to provide technical, economic, and environmental insights to a sustainable transition towards cleaner and resilient energy. A comprehensive study on the penetration of RES in DH systems will be performed aiming to replace lignite and fossil fuels with transition fuels into a strategic and gradual substitution plan. Solar thermal systems, biomass boilers, geothermal heat pumps, wind power can be integrated into the energy grid. Biogas injection into the natural gas grid will be studied including aspects like origin of the biogas and end users/consumers perception. The synergy of RES offers a diversified and resilient approach. The systematic recovery of zero urban waste heat is included to decrease GHG emissions and create a healthier environment for citizens. This involves the strategic reintegration of industrial processes heat and synergies across the value chain. Also, emphasis on low-temperature heat signifies a commitment to energy efficiency and reducing the carbon footprint of DH. This task aims to enhance DH systems by not only substituting fossil fuels with RES, but also by innovative strategies such as recovering, IS and valorisation.
Output: D6.1
Task 6.2 Innovative solar thermal technology for DH covering heating and cooling purposes M6-M48 (CERTH, TVP, DCHP, CLEAN): It will develop the technical specifications for the adaptation of a Solar District Heating and Cooling (SDHC) solution based on 2nd gen HVFPs. This solution will have dual mode operation: provision of heating during the winter (October-April) and cooling during the summer for carbon-free tertiary, industrial and/or residential applications in line with local decarbonisation strategies. TVP with CLEAN and CERTH will improve and define all the crucial technical characteristics to be adapted in the pilot site (preparation, final studies for integration of the technology in 2 buildings, permissions and regulation preparation for T8.4). In alignment with EU goals (KSOs of Cluster 4), the adaptation of this SDHC aims to put the EU on track to achieve climate neutrality in energy sector. This task encompasses technical advancements in SDHC, emphasizing its potential to decrease GHG emissions, promote CE and create a healthier urban environment. Output: D6.2
Task 6.3 Utilization of green H2 for production of heat and electricity through a CHP fuel cell system M6-M48 (CERTH): A comprehensive study will be conducted initially to identify possible realistic solutions for utilization of green hydrogen (produced from RES) for electricity and heat production through technology of fuel cells. Different types of fuel cells will be examined such as Proton Exchange Membrane Fuel Cells (PEMFC), Solid Oxide Fuel Cells (SOFC), Alkaline Fuel Cells (AFC)- non exhausted list. Each type of fuel cell has its own set of advantages and challenges, and the choice of technology depends on factors such as the intended application, operating conditions, and specific requirements of the system. The application of fuel cell systems in DH will be mostly examined towards decreasing its carbon footprint in the area. Considering the fuel cell system of 5 kW (upon installation via EVELIXIA project in CERTHs building) an adaptation study will be performed to integrate the system properly to DH of the building. The integration of green hydrogen production with fuel cell systems offers a promising pathway toward sustainable and low-carbon heat and electricity generation. As technology advances and economies of scale are realized, these systems are likely to play an increasingly important role in transition to cleaner energy. This installation introduce hydrogen as an alternative solution to the existing carbon emission fuels and making the community familiar. The above aligns with the promotion of I-US through circular and clean energy, contributing to "EU Green Deal" and "Fit for 55" and Cluster 4 KSO. Output: D6.3
Task 6.4 Use of digital smart technologies for DH towards improved energy management and control M6-M48 (CERTH, DHCP): It aims the digitalization of DH systems, enhancing energy management, control, and optimal utilization of energy sources, including renewables. A software tool will be developed towards the optimised utilisation of thermal sources comprised by a load prediction assessment module, utilisation of storage capacity based on renewables and conventional energy sources availability and hourly storage capacity and considerations of the thermodynamic characteristics of the DH network and exit points requirement. Determination of locations and types of additional sensors and telemetry will be considered to support input information and control. The software will analyse key aspects of thermal production economics, based on different hourly based cost of heat enabling the selection of the most viable solution concerning total production, heat delivery cost and CO2 emissions. The integration of digital technologies into the existing network aims to improve processes control, create synergies among the nodes for a more interconnected and efficient DH infrastructure, recycle energy streams, align with LCA principles, maintain low-temperature process heat through smart control and energy management. Software will implement cost-optimal pathways and propose new value chains for the engineering of a net-zero while using encryption protocols, secure communication channels, and regular cybersecurity audits to ensure system integrity. Output: D6.4

Work package WP7 – Water Symbiotic Flows

Work Package Number	WP7	**Lead Beneficiary**	3 - NTUA

Work Package Name	Water Symbiotic Flows		
Start Month	1	**End Month**	60

Objectives

Design, test and validate a set of innovative solutions for recovery and valorisation of water flows aiming at its inclusion in different circular symbiotic value chains for both centralized and decentralized cases.

Description

Task 7.1 Analysis for water needs per use for the whole Attica region M1-M9 (NTUA, EYDAP, ADDMA): In this task, a comprehensive water flow analysis of non-domestic water flows will be performed for the entire region of Attica that will explore the different needs in respect to the demand of the various users and will estimate the water needs per sector/ use. This analysis will also be integrated with the qualitative dimension, meaning that the calculated quantities of water demand in time will be associated with "fit for purpose water use" and to the relevant EU and National Legislation (e.g. EU Water Reuse Regulation 2020/741). The quality of water used for industrial purposes depends on the application with some applications ranging from ultrapure water (e.g. pharmaceutical production) to lower quality water. This Task will build upon the existing work implemented by EYDAP in Eastern Attika with the detailed mapping of the relevant stakeholders and their needs. In this aspect, alliance and coalitions with relevant quadruple helix stakeholders will be boosted through a survey that will include a questionnaire tool and targeted interviews to co-create value from circular utilization of water, but also energy, waste, and by-products at regional scale. The main aim is to identify the water needs per use in terms of quantity and quality and enable efficient future I-U symbiotic links and collaborations with reduced water flows. Output: D7.1

Task 7.2 Decentralized reclaimed water valorisation for different uses M6-M48 (NTUA, EYDAP): Integration of decentralized solutions within an urban environment for the valorisation of both treated wastewater and harvested rainwater will be investigated. The penetration of decentralized water reuse will be supported by implementing the sewer mining technology as well as by capitalizing on the existing sewer mining systems that are currently operating at the Goudi Park and will be operational in the Municipality of Markopoulo in Attika by abstracting wastewater from the sewerage system treating it at the point of demand and reuse it to irrigate green spaces and parks (NextGen & IMPETUS project). Furthermore, based on the findings of the CIRC4FOOD project, the integration of rainwater harvesting solutions in residential buildings and in schools will be investigated. The system will be installed in 3 residences and one school. The harvested rainwater will be used to irrigate an urban farm that will be developed at household level promoting the concept of urban farming. Suitable composters will be installed and used to treat the produced biowaste and convert it into compost to be applied to the urban farm as soil additive. In this decentralized system, the concept of water and nutrient recycling and urban food production is shown. Decentralized wastewater treatment and reuse has the bottleneck that it is more energy intensive than centralized water reuse. Details on the Risk Management Plan are given in Task 7.4 as this approach is applied for both for centralized and decentralized water reuse as it is a requirement for both cases (Water Reuse Regulation 2020/741) as a condition to set up a water reuse system. Output: D7.2

Task 7.3 NBS for water recovery and reuse M6-M48 (NTUA, EYDAP): Decentralised NBS will be deployed in the urban and peri-urban environment to collect and store rainwater in the subsurface, from rooftops and/or from surface runoff, both as a flood mitigation measure and for recovery and reuse for urban and industrial use. The recovered water may either be used to decrease the pressures from water availability or for artificial recharge to protect the coastal aquifers from sea water intrusion and improve groundwater quality in general. Furthermore, NBS will be leveraged in the urban environment of Athens towards a redesign plan for of the Athens' parks (including new pocket parks) promoting tested solutions for water management, harvesting rainwater and modifying runoff patterns to alleviate urban floods. These activities are of direct benefit to EYDAP in symbiotic sense, as they reduce stress both on potable water resources and combined sewers (still in operation at the centre of the city of Athens). Output: D7.3

Task 7.4 Centralized water treatment for reuse in Eastern Attica M6-M48 (EYDAP, NTUA): This Task will consist of a series of activities that will support the centralized water reuse in Eastern Attica. It will build on the existing activities which EYDAP has implemented concerning reclaimed water reuse in the Eastern Attika and will be serviced in the next years by 3 medium scale WWTPs (~3.6 million p.e.). EYDAP has implemented a detailed mapping of the potential users (hotels, farmers, nurseries etc) of the reclaimed water and has already launched a communication process with them to increase the acceptance of water reuse. This Task will (i) implement a detailed RMP on water reuse as required by the EU Water Regulation, (ii) will evaluate the fate and removal of emerging contaminants (pharmaceuticals, personal care products, etc) in water, soil, soil and crops) showing the potential accumulation of these compounds in crops, (iii) will evaluate the impact of reclaimed water to the crops, (iv) will investigate the fate and removal of Antibiotic

Resistance Genes and Antibiotic Resistant Bacteria, (v) will investigate the impact of reclaimed water on soil health (i.e. biodiversity). The RMP will follow a systematic approach that includes a structured analysis of the water reuse system, the identification of potential hazards and hazardous events along with the populations and environments at risk and the related exposure routes, and the management of the assessed risks with the use of existing and/or possible preventive measures and barriers, when appropriate, to mitigate them. It includes communication and cooperation among the parties involved to ensure that corrective actions are taken and communicated opportunely. To implement activities (ii)-(v) a demonstration site will be set up in Eastern Attika consisting of an agricultural field irrigated with Class A reclaimed water (i.e. suitable for unrestricted irrigation). Output: D7.4

Task 7.5 Novel Valorisation Pathways for Sewage Sludge in Attica Region M6-M48 (NTUA, EYDAP): It will test and demonstrate novel valorisation pathways for materials and energy (if possible) recovery of sludge that is produced in wastewater treatment plants. Fast pyrolysis and torrefaction will be tested as processes to produce high added value products from dried sludge originating from Psyttalia WWTP and dewatered sludge from Metamorfosis WWTP in Attica. Fast pyrolysis mainly produces biooil which can be Integrated into existing petroleum refineries to produce fuels and chemicals. Slow pyrolysis produces significant quantities of all three products (bioil, gas and biochar). Torrefaction mainly produces char that can have a high added value. The above processes will be tested in a three-stage pyrogasification furnace which has recently been installed and operated at the premises of the Sanitary Engineering Laboratory. Output: D7.5

Task 7.6 Strategic Master Plan on Circularity M6-M60 (NTUA): Task partners will collaborate to formulate a detailed Strategic Master Plan on Circularity enabling investment proposals, including pricing, systems design, along with the Regulatory Framework for all sectors included i.e. water and wastewater services, energy and materials. Output: D7.6

Work package WP8 – Piloting and Demonstration

Work Package Number	WP8	**Lead Beneficiary**	3 - NTUA
Work Package Name	Piloting and Demonstration		
Start Month	25	**End Month**	60
Objectives			
Establish material/water/energy of the 5 pilots in T8.1-T8.5, create action plans as described in sections above, leaders to coordinate all the activities and engage the proper stakeholders; Install and operate the new technologies and mechanisms, demonstrate them for the proper duration and producing the required results; Evaluate the performance and impact of processes; Develop upscaling plan.			
Description			

Task 8.1 Biobased materials and Packaging flow implementation M25-M60 (HMU, NTUA, WATT, TSIMIS, Panchart)
Implementation of pilot as described in section 1.2.1. Outputs: D8.1
Task 8.2 Textiles flow implementation M25-M60 (TUC, RECYCOM, HUD, CERTH)
Implementation of pilot as described in section 1.2.1. Outputs: D8.2
Task 8.3 CDW & Glass flow implementation M25-M60 (ICCS, AB)
Implementation of pilot as described in section 1.2.1. Outputs: D8.3
Task 8.4 Energy flow implementation M25-M60 (CERTH, TVP, CLEAN, DHCP)
Implementation of pilot as described in section 1.2.1. Outputs: D8.4
Task 8.5 Water flow implementation M25-M60 (NTUA, EYDAP)
Implementation of pilot as described in section 1.2.1. Outputs: D8.5
Task 8.6 Validation of digital governance and collaboration schemes M25-M60 (ICCS): It will focus on the smooth integration and validation of digital, governance and collaboration schemes within the pilots. All the digital technologies developed in WP3-4 will be operated and optimized based on pilot feedback and aligned with T2.6, T3.6. Innovations on governance, collaboration, expertise and replication knowledge transfer, developed in WP2,WP10 will be applied, adjusted and optimized in the different cases. Output: D8.6
Task 8.7 Impact Assessment of Theseus H4C M25-M60 (CIRCE, WP partners): Based on insights from WP2 and demonstration, this task seeks to underscore the importance, adequacy, and efficacy in evaluating and improving the circularity of value chains. This evaluation will be extended to include an assessment of sustainability performance, with comparisons drawn to the findings in T10.3. The project will explore alternative scenarios to predict circularity performance (ICL will develop and apply the circularity assessment methodology based on multi-sectoral circularity assessment framework -developed in CSSBOOST, HYDROUSA, DEEP-PURPLE projects- and the new ISO 59020) and emissions reduction, offering a roadmap for future optimizations within each value chain. Moreover, a comprehensive dynamic economic feasibility analysis, centred around Life Cycle Cost (LCC), will be undertaken for each value chain, spanning from waste streams to final products. Leaders of the demonstrations will be responsible for calculating Key Performance Indicators (KPIs) and executing LCC calculations, guided by the task leader. This analytical framework will remain adaptable, considering various factors like logistics, process technology, and end-product utilization, to evaluate the economic feasibility for individual stakeholders and the global balance of the value chains. These findings will provide valuable insights to facilitate replication and exploitation efforts. Output: D8.6

Work package WP9 – Synergies, replication and Cooperation

Work Package Number	WP9	**Lead Beneficiary**	37 - RISE
Work Package Name	Synergies, replication and Cooperation		
Start Month	9	**End Month**	60

Objectives

Establish synergies and cooperation with related initiates; Establish the 2 replicators and guidelines for further replication.

Description

Task 9.1 Synergies with existing H4C and ECoP towards common tools and knowledge exchange M9-M60 (A. SPIRE, ATTICA, SHAS): Alignment of industrial and research stakeholders engaged in development of H4Cs and their supporting technologies, across Europe is essential. Several projects granted under HEU Processes4Planet are now working on Hubs in multiple locations. The 2 CSA projects behind “H4C Knowledge Community” are doing key work on connecting urban and industrial stakeholders and sharing know-how to support such efforts. However, not enough attention has been paid to the specific coordination of technological innovation needs of different Hubs. Each Hub requires a tailored solution, but often similar-purpose industrial technology innovations are being worked on in parallel by the different project consortia. This affects both digital tools and infrastructure, as well as process technologies. Theseus proposes to coordinate existing Processes4Planet funded Hubs4Circularity projects, specifically in the types of similar technologies being developed in parallel. The goal is for IP owners to understand what others are doing, how to specialize their own product, and potentially to find synergies that allow for mature technologies, or a combination thereof, to be deployed to other Hubs and, eventually, widespread market adoption. Output: D9.1

Task 9.2 Synergies with Cities mission and CCRI initiative towards tools and methodologies M9-M60 (CIRCE, RISE, TUD, SHAS): This task establishes synergies with the pilots and key partners involved in the CCRI and Cities mission

initiatives and projects, many of which already count with the participation of our partners. Cooperation and mutual support activities will be initiated by the Task leader with the critical support of all partners as needed (e.g. for establishing the initial contact or presenting the technical tasks in webinars and events). Examples of synergies include joint communication and dissemination activities (e.g., through CCRI channels), invitation of other projects to participate in our events, presentation of THESEUS in mutual learning workshops, organisation of joint webinars, presentation of Theseus in CCRI-CSO's Thematic Working Groups, contribution to CCRI-CSO's multi-stakeholder dialogues, etc. Output: D9.1

Task 9.3 Replication to Malmö and MO areas M9-M60 (RISE, SINTEF, MALMO, KPH, all WP partners): Malmö: A) resource hub and assessment: co-creation approach including stakeholders from Malmö.: A1) Stakeholder mapping and engagement via existing networks i.e Malmö resource hub, to expand and capitalise on a broader and more diverse knowledge base. A2) Further adaption of existing system dynamics model (WECOS project) focusing on future scenarios for waste & energy to identify pathways for achieving the EU Plastic and Waste Framework Directives: A3) Validation of system dynamics model transferability and inclusion of additional modules directed by the Malmö Resource Hub. A4) Convening a simulation testbed with user dashboards with KPIs of municipal authorities. A5) Scenario analysis of Theseus innovations to evaluate applicability A6) Based on scenarios, resource requirements will be mapped to reveal possibilities of (re)utilisation through facilitated workshops, involving further identification of collaboration actors. The ISRL/VCRL from T3.2 will be assessed. B) Adaptation of main hub innovations: Knowledge exchange between main hub and Malmö resource hub to identify adaptations and innovations for replication. MO: A) Replication simulation: A1) Workshops to detail the replication simulation and mapping data (energy, water, material). A2) Develop a simulation model for sustainability assessment and optimisation of I-US system, considering GIS data for identifying geographical challenges in a just transition. Optimisation will consider economic, environmental, and social parameters. A3) ISRL from T3.2 will be assessed based on methods from the H4C ECoP projects. B) Regional reference group and adaptation of main hub innovations: B1) Establish a I-US reference group with local, regional and state private and public sectors, to meet at least twice yearly. B2) Business case development, innovations evaluated by the reference group to construct business cases. B3) Matchmaking of resources replicating the marketplace of T3.6. B4) Feedback to main hub on needed adaptations for innovations to be replicated. B5) Produce policy recommendations for local I-US and just transition. Outputs: D9.2

Task 9.4 Roadmap for replicability to additional sites M9-M60 (ICCS, A. SPIRE, all WP partners): Based on T9.1 and on the experience gained from Theseus, task will propose a roadmap for replicability of H4C to synthesize the lessons learned, positive and negative, in Attica and to channel industrial practitioners R&I agendas into the overall goal of Processes4Planet (P4P) of developing H4C and FOAKs across Europe. The activity will be sought in partnership with other H4C projects and H4C Knowledge Community, as well as with the Theseus 2 replicators in Helsingborg (Sweden) and Nordland (Norway) – see also KPI 5.4 (Replication Roadmap Tool).The Common H4C guidelines will be a consultative document aimed at giving the first insights to what it means to develop a H4C from the industrial perspective, the synergies required, the technical and technological basis, the business case, an exploitation strategy, ways of assessing viability of a hub. It will also provide a checklist or roadmap type tool newcomers to the idea can use to start launching their H4C. The guidelines will be written based on the experience of Theseus and other H4C projects funded under P4P, connecting with the process industry sectors via the A.SPIRE governance and advisory structures, which gathers experts from 10 process industries sectors. The development of the guidelines will include the organization of at least 1 meeting with the relevant stakeholders. Also, TUC will develop a Replication Roadmap Tool (from ROBINSON project) a systematic guide designed to assist stakeholders in adopting Theseus model, a multi-stage collaborative framework offering a detailed approach to replication. At least three potential replicators are set to test this innovative tool, ensuring its effectiveness and adaptability across different environments. The tool assesses local resources and feasibility of Theseus solutions and formats stakeholder action groups. Then it assists in the selection of appropriate technological and organizational elements for the targeted area. It provides guidance through the crucial phases of contextual adaptation and business planning, modifying the Theseus solutions to align with local regulations, culture, and economic conditions. It is also embedded with decision support capabilities, providing access to expert guidance and proven workflows. The user-friendly and accessible interface promises navigation through the replication stages to effectively create and scale up sustainable and circular industrial-urban ecosystems. Outputs: D9.3

Work package WP10 – Nexus Modelling Social Awareness and Training

Work Package Number	WP10	**Lead Beneficiary**	45 - ICL
Work Package Name	Nexus Modelling Social Awareness and Training		

Start Month	7	**End Month**	60

Objectives
Develop models of H4C and I-US activities and their impacts; Develop Social Awareness and Training Activities.
Description

Task 10.1 Climate Change in relation to the future of H4C M7-M60 (ICCS, ADDMA): It upscales WP9 insights into European / global level and instigate a series of modelling exercises. Two global open-source integrated assessment models (IAMs) (the GCAM, CLEWs as well as their European variants GCAM-Europe , CLEWs-EU) currently developed in DIAMOND project, will be employed to understand implications of assumed I-US across possible H4C (a) throughout EU and (b) globally. Early, a database will be developed to document CE (socioeconomic, energysystem, technoeconomic, etc.) solutions and policy measures that can be modelled in the two IAMs, drawing from recent literature, advancements in research projects for representing CE in IAMs (e.g., CircoMod, CircEUlar) as well as in projects for enhancing the modelling capacity for the decarbonisation, circularity performance, and sustainability of European industry (e.g., TRANSIENCE, AMIGDALA). To enable modelling, task will establish open science guidelines for documenting and sharing parameters, and assumptions with research community, tenabling the evaluation, reproduction, and exploitation of the model outputs (D10.1.1). Building on the insights gained into Attica (T8.6) and additional sites (T9.3), it will attempt to upscale similar solutions and advancements in EU, using GCAM-Europe and CLEWs-EU. Next, similar assumptions will be developed for global sites in major economies, to assess global-level implications and warming potentials using global GCAM and CLEWs models (GLUCOSE). Outputs will be documented in the community's protocols for the IPCC's AR7 cycle processes (using protocol established in the IAM consortium).
(D10.1.2) Output: D10.1
Task 10.2 Water-Energy-Materials-Waste nexus decision support module M7-M60 (Imperial, RISE): Territorial metabolism (TM) including water-energy-material-waste (WEMW) nexus model will be integrated in T3.6 Platform to simulate I-US, providing valuable insights on resource utilisation, generation of waste/emissions and their interlinkages. TM will be leveraged for quantification and analysis (e.g. quality, source) of all resource flows (from extraction to disposal) within Theseus territory. Utilising WP3 data it will provide insights about current TM and enable the assessment of benefits arising from the implementation of Theseus solutions. TM model will empower decision-makers to better understand the efficiency of territorial resources use, areas for intervention to optimize system-level inputs, outputs and feedback loops. TM will be coupled with a WEMW nexus model to investigate interconnections between the different resources, enabling the identification of synergies and trade-offs. WEMW ensures an integrated and coordinated territorial approach, preventing interventions from shifting burdens across components or different sectors. TM-WEMW will incorporate growth scenarios projecting resource use and waste generation patterns, concurrently with the uptake of Theseus up to 2050, leveraging information generated in WP3 modules (reverse logistics, flows, stakeholders, predicted material). Scenarios will consider socio-demographic, economic, policy, population, housing, and behavioural factors, influencing the functioning of the investigated territory. With this tool, territorial and sectoral managers will make informed decisions by providing knowledge of the effectiveness of Theseus interventions in respective territory. . Output:
D10.2
Task 10.3 Cities climate modelling and evaluation of environmental and economic impacts M7-M60 (NTUA, RISA): It evaluates environmental and economic performance of Theseus solutions. LCA methodology will be applied on each material flow to evaluate environmental sustainability and impacts on humans' health of the processing and byproducts valorization solutions that will be proposed during Theseus. LCA will assess the environmental benefits of the developed approaches, considering all involved processes, from the generation of the by-products to the enduse of the final materials. All solutions implemented will be compared to the existing status of each material. A sensitivity analysis will identify bottlenecks & opportunities for the optimization of each material flow, feeding into the WPs responsible for materials optimization regarding future pathway modelling. Specifically, the methodology involves 4 distinct analytical steps: goal and scope definition, inventory analysis, impact assessment and interpretation. Existing database subscriptions and software tools will be used e.g. GaBi following ISO 14040, 14044. Hot spots will be identified and whenever possible new solutions will be proposed towards their optimization to develop the final strategies. Additionally, a precise city climate modelling approach will be developed. The application of LCC analysis will elucidate the economic feasibility and cost-effectiveness of proposed solutions accounting for relative cost distributions. Computation of pertinent financial metrics and establishment of suitable economic indicators will facilitate the estimation and mitigation of economic impacts and associated risks. This analysis aims to pinpoint specific material flow processes necessitating refinement to alleviate economic burdens and optimize the economic efficacy of the investigated system. Output: D10.3
Task 10.4 Social Awareness increase through AR M7-M60 (ICCS, RISA): The task includes the design, development and use of an Augmented Reality (AR) application to improve social awareness towards environmental sustainability and educate them through immersive visualizations of the impacts on humans' health of the processing and by-products valorization solutions. Flows and circular solutions produced during the Theseus implementations will be considered content inputs and will be curated upon the guidance of the technology partners to content customized for AR interactions and storylines (campaigns). The content will be supported by gamification and educational functionalities, aiming to support engagement and improve the self-efficacy of the immersed audience. Immersive campaigns will be:

i) assessing the knowledge and understanding of the importance of environmental sustainability and material flows, ii) showcasing 3D content, storylines and other data, iii) fostering awareness and engagement towards the Theseus technologies and solutions for water, energy and materials based on regional needs. To ensure social participation and engagement, the AR tool will be part of the training activities in T10.5 and will also assess the public understanding to increase awareness liaising with T10.6, T10.7 and T11.6. Output: D10.4

Task 10.5 Upskilling & Training activities in region cities and universities M7-M60 (Imperial, TVP, RISE, ICCS, LiU): It aims to enhance circular and digital upskilling for local population, workforce, high-level decision-makers by assessing the existing digital and circular competence of staff members and anticipating future market skillset requirements. Based on the status quo results, T10.5 will develop tailored enlightenment, guidance, training, and mentoring services for decision-makers, employees, and citizens using knowledge provided by internal and external experts to address specific sectoral, regional, or national needs. Regular validation of acquired skills and progress will occur through formal and informal channels, with workshops organized for co-creating and validating the educational materials in the identified areas for upskilling. AR app from T10.4 will be exploited in the workshops to support the training activities. This initiative aligns with EU digital compass and contributes to the development of a digitally skilled population and workforce, emphasizing the creation of upskilled pathways, including non-formal and informal learning. The project's results, lessons learned, and best practices will be summarized for knowledge documentation and sharing, while solutions demonstrated and business models presented will be utilized to create replication packages tailored to specific sectoral, regional, or national contexts. Output: D10.6

10.6 Impact on the health of citizens M7-M60 (RISA, ICCS): It is expected that citizens will benefit from Theseus implementation in many ways such as being a healthy environment. Deployment of I-US will contribute to lowering emissions and waste reduction thus to positive impact on the environmental health, ICCS will build on the World Health Organization (WHO)report, in which health effects of the circular economy transition are identified and analysed . Links between Theseus actions and their effect on the citizens health will be mapped and use of the WHO Driver, Pressure, State, Exposure, Effect, and Action framework will be investigated. In line with WHO report, ICCS will propose Theseus specific indicators relevant to measuring the health implications of the hub implementation, building on existing work from WHO and other countries , .Analysis performed will be supported by LCA related to health (related categories: human health is human toxicity (HT), particulate matter formation (PMF), photochemical oxidant formation (POF) ,ozone depletion (OD)) performed in T10.4. Output: D10.6

10.7 De-risking assessment M7-M60 (HSRT): In the 1st phase task involves a review of de-risking strategies for investments in circular economy models and technologies. The task will review the perceived high risks associated with investing in existing circular economy solutions in the fields addressed by Theseus. This includes identifying barriers to investment and understanding the reluctance of financiers. The assessment will then move to analyse existing policy and financial instruments, and on-going efforts to restructure them to better support circular economy investments over traditional linear models. An important component of this task is the analysis of fiscal policies and taxation to provide an understanding of how various tax policies, such as raw material taxes or incentives for reuse and repair activities, affect the viability and attractiveness of circular economy investments. In the 2nd phase, through stakeholder engagement, interaction with technology developers, investors, policymakers, and businesses will be sought to gather diverse perspectives on the risks and opportunities in circular economy investments particularly related to the models and technologies proposed. These insights will be integral in shaping the assessment's outcomes and recommendations. Outcome will be a comprehensive report, encapsulating all findings, stakeholder insights, and strategic recommendations including case studies and examples of successful de-risking strategies in circular initiatives, a guideline for future investments. Output: D10.7

Work package WP11 – Exploitation and Dissemination activities

Work Package Number	WP11	**Lead Beneficiary**	24 - HSRT
Work Package Name	Exploitation and Dissemination activities		
Start Month	1	**End Month**	60

Objectives
Map the market landscape, identifying opportunities and obstacles for commercialization and upscaling; Develop an efficient exploitation and IPR management plan towards sustainability; Shape a tailored dissemination and communication plan and activities.

Description
Task 11.1 Dissemination and communication activities M1-M60 (HSRT, CERTH, All WP partners): HSRT will collaborate early with all partners to draft a comprehensive communication, dissemination and exploitation strategy, involving identifying clear strategies for effectively reaching each target group, ensuring that achievements and lessons are widely available to communities of interest. To ensure all identified stakeholders and wider communities are regularly updated on the project's progress and achievements, it will implement various communication channels which be constantly enriched with content coming from all partners. Besides the project's dynamic website creation and activation of multiple social media platforms like Facebook, Twitter, and LinkedIn, regular (2 annually) e-Newsletters, will also be developed and the project will also feature videos and interviews, on platforms like YouTube and Vimeo for a self-explanatory presentation of the project and its main results and outcomes. Furthermore, this task will monitor the project's achievements in terms of scientific publications in journals and presentations at conferences and will arrange for contributions to technical blogs and articles. Contributions to open-source software repositories on GitHub and Bitbucket will be closely monitored and orchestrated, fostering an open-source culture. Q&A sessions on platforms like Stack Overflow will be established for knowledge exchange. Workshops, training sessions, webinars, and participation in policy fora and debate hubs, will be organized with the support of all partners, to contribute to discussions related to and making recommendations for sustainable development. KPI based monitoring will be employed to continuously assess the impact and effectiveness of communication and dissemination channels. Output: D11.1 Task 11.2 Policy recommendations and barriers M1-M60 (LiU, Imperial, All WP partners): Policy research has shown that different circular economy policies affect different lifecycle stages . H4C focuses primarily on the end-of-life stage and thus is affected by EU policies such as producer responsibility and packaging rules, standards and upcoming legal definitions of, for example, remanufacturing or the Ecodesign Regulation. It is also affected by national, regional and local policies such as public procurement rules, options provided at recycling stations etc (ibid.). Moreover, research has shown that different product groups are affected by different policies and since H4C processes several material flows there is a wide variety of policies to consider. Therefore, to make policy recommendations the following steps will be taken: 1)Map key policies on the EU, national, regional and local level that affect each of the key material flows; 2)Interview stakeholders identified in T2.2 from the project and other regions in Europe on whether the identified policies from step 1 are barriers or drivers, and identify policy gaps and improvement opportunities; 3)Analyse findings from steps 1 and 2 to identify policy recommendations for each material flow; 4)Use focus groups of diverse stakeholders to conduct a cost-benefit analysis of the identified policy recommendations from step 3 with the aim of finding low hanging fruit and high impact recommendations, such as, policies that will positively impact several material flows or are expected to yield high impacts. The findings will be documented in a report and two policy briefs: one targeting the EU policy level and one targeting the national, regional, and local policy levels. Output: D11.2 Task 11.3 Exploitation, Business models and IPR Management M6-M60 (HSRT, All WP partners): It focuses on aligning project outcomes with market demands, by developing a comprehensive exploitation strategy, considering societal, commercial, and scientific impacts. Emphasis is on cross-sectoral collaboration and industrialization of solutions for broad impact. Task will perform detailed market analysis to identify target markets, key actors, and monetization opportunities, including potential financial partnerships for post-project financing. The objective is financial viability and market alignment of project outcomes. Development of exploitation roadmaps, detailing roles, responsibilities, milestones, and IPR ownership, is crucial. These roadmaps will guide the commercialization process, assessing innovation potential in market terms. Analytical tools like SWOT and Porter's Five Forces will support strategy development. Managing IPR is vital, with strategies for protection and post-project exploitation. The final goal is a comprehensive exploitation and commercialization strategy, outlining viable products and initial business models for successful market entry, adhering to industry best practices. Ensuring Theseus not only achieves groundbreaking research but translates it into market-ready products, impacting the market Output: D11.3. Task 11.4 Commercialization of solutions and trading opportunities M21-M60 (EMVIS, HSRT, All WP partners): It is dedicated to laying foundational steps for the future commercialization of project's outcomes. This task entails a strategic approach towards understanding market feasibility and potential commercial pathways post-project. It includes initiating dialogues with prospective commercial partners and investors, crucial for gauging market interest and potential collaborations. Market feasibility studies form a key component, assessing the practicality of market entry. It includes refining and finalizing go-to-market strategies. While T11.3 develops the initial business models, T11.4 tailors these models to specific market segments and customer needs, based on the insights gained from industry engagement. This involves fine-tuning marketing, distribution, and sales strategies to align with the unique characteristics of each target market. It also includes refining and finalizing go-to-market strategies. Output: D11.4 Task 11.5 Standardization of novel technologies and activities M6-M60 (DS, SN, All WP partners): It will develop 1) an

analysis of the applicable standardization landscape and the contribution to the ongoing and future standard developments and 2) a comprehensive analysis of the existing landscape in I-US, Circularity and Secondary Materials, including examination of standards and standardization activities. Key relevant regulations and EU standardization requests will be taken into consideration to create a synergy with EU-strategies. The mapping will subsequently be updated and used by other WPs to ensure compatibility and interoperability, and link with existing standards in the market. Collaboration with the most relevant standardization Technical Committee(s) in CEN and ISO will be set up, e.g. CEN/TC 465 Sustainable Cities and Communities and SABE workgroup mapping standards influencing CE, to ensure that results and knowledge is integrated into ongoing and future developments, to interact with market stakeholders, and use the standardization system as a tool for dissemination. Also, based on input from WPs3-8, activities will be performed to facilitate the acceptance and utilisation of the developed solutions by the market. Standardization acts as a catalyst and builds trust in new technologies and innovations as it is a formal and transparent process. Task will contribute to ongoing and future developments in the specific topics, promoting the inclusion of the findings of the project in new or revised standards that can be used by the European or international industry. Depending on the maturity of the technologies and the sector, this can involve providing information to standardization Technical Committees, mutual knowledge sharing, submitting technical proposals and even promoting the elaboration of new standard documents. Output: D11.5

Task 11.6 Stakeholder engagement and Theseus bubbles M6-M60 (Imperial, LiU, RISE): Based on T2.2 results, it establishes a collaborative innovation model for the Theseus solutions and fosters knowledge sharing by the active involvement of diverse groups of stakeholders. Stakeholders, standardization bodies, local authorities and decisionmakers, experts, companies and associated EU projects as well as local communities will be brought together to establish three core Bubbles. The Bubbles will serve as an instrumental platform to collectively address barriers and identify opportunities for circularity considering the local context. Bubbles include 1) Policy, Local Authorities, and Standards Harmonization Hub, 2) Community Engagement Hub, and 3) Market Accelerator Hub. It facilitates innovation dissemination by overseeing and optimizing communication channels between WP11 Tasks to maximize the exploitation of results. Activities of Theseus Bubbles include collaborative working sessions (every 6-8 months - online) and the organization of workshops (can be open to wider stakeholders) to disseminate important findings and receive feedback from relevant groups. On-site demonstrations and presentations will be used to engage stakeholders and demonstrate practical applications of project results. These activities will support the exploitation of project results and market acceptance and help accelerate the uptake/scale-up of the supply chains, technologies and products. Output: D11.6

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Theseus PM, QA & Ethics v1	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	6
D1.2	Theseus PM, QA & Ethics v2	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	24
D1.3	Theseus PM, QA & Ethics v3	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	36
D1.4	Theseus PM, QA & Ethics v4	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	48
D1.5	Theseus PM, QA & Ethics final	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	60
D1.6	Data Management Plan v1	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	6
D1.7	Data Management Plan v2	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	24
D1.8	Data Management Plan v3	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	36
D1.9	Data Management Plan v4	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	48

D1.10	Data Management Plan final	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	60
D1.11	Short Interim Management Report v1	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.12	Short Interim Management Report v2	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	36
D1.13	Short Interim Management Report v3	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	48
D2.1	Governance Model v1	WP2	16 - LiU	R — Document, report	PU - Public	9
D2.2	Governance Model v2	WP2	16 - LiU	R — Document, report	PU - Public	24
D2.3	Governance Model final	WP2	16 - LiU	R — Document, report	PU - Public	36
D2.4	Participatory approaches v1	WP2	16 - LiU	R — Document, report	PU - Public	9
D2.5	Participatory approaches v2	WP2	16 - LiU	R — Document, report	PU - Public	24
D2.6	Participatory approaches final	WP2	16 - LiU	R — Document, report	PU - Public	36
D2.7	Material flow analysis v1	WP2	1 - ICCS	R — Document, report	PU - Public	9

D2.8	Material flow analysis v2	WP2	1 - ICCS	R — Document, report	PU - Public	24
D2.9	Material flow analysis final	WP2	1 - ICCS	R — Document, report	PU - Public	36
D2.10	Regional/interregional pathways v1	WP2	13 - UAEGEAN	R — Document, report	PU - Public	9
D2.11	Regional/interregional pathways v2	WP2	13 - UAEGEAN	R — Document, report	PU - Public	24
D2.12	Regional/interregional pathways final	WP2	13 - UAEGEAN	R — Document, report	PU - Public	36
D2.13	Characterization of value chains v1	WP2	37 - RISE	R — Document, report	PU - Public	9
D2.14	Characterization of value chains v2	WP2	37 - RISE	R — Document, report	PU - Public	24
D2.15	Characterization of value chains final	WP2	37 - RISE	R — Document, report	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D2.16	Digital interactions and planning v1	WP2	18 - CORE	R — Document, report	SEN - Sensitive	9
D2.17	Digital interactions and planning v2	WP2	18 - CORE	R — Document, report	SEN - Sensitive	24
D2.18	Digital interactions and planning final	WP2	18 - CORE	R — Document, report	SEN - Sensitive	36
D3.1	GIS tool v1	WP3	21 - TLNV	OTHER	SEN - Sensitive	18

D3.2	GIS tool v2	WP3	21 - TLNV	OTHER	SEN - Sensitive	36
D3.3	GIS tool final	WP3	21 - TLNV	OTHER	SEN - Sensitive	48
D3.4	Symbiotic models v1	WP3	1 - ICCS	OTHER	PU - Public	18
D3.5	Symbiotic models v2	WP3	1 - ICCS	OTHER	PU - Public	36
D3.6	Symbiotic models final	WP3	1 - ICCS	OTHER	PU - Public	48
D3.7	Dynamic DPP and MR app v1	WP3	1 - ICCS	OTHER	SEN - Sensitive	18
D3.8	Dynamic DPP and MR app v2	WP3	1 - ICCS	OTHER	SEN - Sensitive	36
D3.9	Dynamic DPP and MR app final	WP3	1 - ICCS	OTHER	SEN - Sensitive	42
D3.10	Prediction logistics module v1	WP3	18 - CORE	OTHER	SEN - Sensitive	18
D3.11	Prediction logistics module v2	WP3	18 - CORE	OTHER	SEN - Sensitive	36
D3.12	Prediction logistics module final	WP3	18 - CORE	OTHER	SEN - Sensitive	48
D3.13	Data Exchange Channels v1	WP3	1 - ICCS	OTHER	SEN - Sensitive	18
D3.14	Data Exchange Channels v2	WP3	1 - ICCS	OTHER	SEN - Sensitive	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.15	Data Exchange Channels final	WP3	1 - ICCS	OTHER	SEN - Sensitive	48
D3.16	Theseus H4C Markeplace v1	WP3	22 - IRIS	OTHER	SEN - Sensitive	18
D3.17	Theseus H4C Markeplace v2	WP3	22 - IRIS	OTHER	SEN - Sensitive	36
D3.18	Theseus H4C Markeplace final	WP3	22 - IRIS	OTHER	SEN - Sensitive	48
D4.1	Technologies requirements	WP4	3 - NTUA	R — Document, report	SEN - Sensitive	9
D4.2	Sorting lines (TRL 5-ready)	WP4	1 - ICCS	OTHER	PU - Public	18
D4.3	Sorting lines (TRL 6-ready)	WP4	1 - ICCS	OTHER	PU - Public	36
D4.4	Sorting lines (TRL 7/8-ready)	WP4	1 - ICCS	OTHER	PU - Public	48
D4.5	Autonomous Platform v1	WP4	12 - DUTH	OTHER	PU - Public	18
D4.6	Autonomous Platform v2	WP4	12 - DUTH	OTHER	PU - Public	36
D4.7	Autonomous Platform final	WP4	12 - DUTH	OTHER	PU - Public	48
D4.8	Smart bins for Attica v1	WP4	1 - ICCS	R — Document, report	PU - Public	18
D4.9	Smart bins for Attica v2	WP4	1 - ICCS	R — Document, report	PU - Public	36
D4.10	Smart bins for Attica final	WP4	1 - ICCS	R — Document, report	PU - Public	48
D4.11	XR for human-robot interaction alpha version	WP4	1 - ICCS	OTHER	PU - Public	36
D4.12	XR for human-robot interaction beta version	WP4	1 - ICCS	OTHER	PU - Public	48

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.1	Biological recycling for textiles v1	WP5	23 - CIRCE	R — Document, report	SEN - Sensitive	18
D5.2	Biological recycling for textiles v2	WP5	23 - CIRCE	R — Document, report	SEN - Sensitive	36
D5.3	Biological recycling for textiles final	WP5	23 - CIRCE	R — Document, report	SEN - Sensitive	48
D5.4	Biodegradable masterbatches v1	WP5	11 - HMU	R — Document, report	SEN - Sensitive	18
D5.5	Biodegradable masterbatches v2	WP5	11 - HMU	R — Document, report	SEN - Sensitive	36
D5.6	Biodegradable masterbatches final	WP5	11 - HMU	R — Document, report	SEN - Sensitive	48
D5.7	Construction materials v1	WP5	40 - SINTEF	R — Document, report	PU - Public	18
D5.8	Construction materials v2	WP5	40 - SINTEF	R — Document, report	PU - Public	36
D5.9	Construction materials final	WP5	40 - SINTEF	R — Document, report	PU - Public	48
D5.10	Biobased thin films v1	WP5	11 - HMU	R — Document, report	PU - Public	18
D5.11	Biobased thin films v2	WP5	11 - HMU	R — Document, report	PU - Public	36
D5.12	Biobased thin films final	WP5	11 - HMU	R — Document, report	PU - Public	48

D5.13	Thin-film materials v1	WP5	3 - NTUA	R — Document, report	PU - Public	18
D5.14	Thin-film materials v2	WP5	3 - NTUA	R — Document, report	PU - Public	36
D5.15	Thin-film materials final	WP5	3 - NTUA	R — Document, report	PU - Public	48
D6.1	RES technologies	WP6	9 - CERTH	R — Document, report	PU - Public	9
D6.2	Solar thermal technology v1	WP6	9 - CERTH	R — Document, report	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D6.3	Solar thermal technology v2	WP6	9 - CERTH	R — Document, report	PU - Public	36
D6.4	Solar thermal technology final	WP6	9 - CERTH	R — Document, report	PU - Public	48
D6.5	CHP fuel cell system v1	WP6	9 - CERTH	R — Document, report	PU - Public	18
D6.6	CHP fuel cell system v2	WP6	9 - CERTH	R — Document, report	PU - Public	36
D6.7	CHP fuel cell system final	WP6	9 - CERTH	R — Document, report	PU - Public	48
D6.8	DH smart technologies v1	WP6	9 - CERTH	OTHER	PU - Public	18
D6.9	DH smart technologies v2	WP6	9 - CERTH	OTHER	PU - Public	36

D6.10	DH smart technologies final	WP6	9 - CERTH	OTHER	PU - Public	48
D7.1	Analysis for Attica water needs	WP7	3 - NTUA	R — Document, report	PU - Public	9
D7.2	Decentralized Water Reuse v1	WP7	3 - NTUA	R — Document, report	PU - Public	18
D7.3	Decentralized Water Reuse v2	WP7	3 - NTUA	R — Document, report	PU - Public	36
D7.4	Decentralized Water Reuse final	WP7	3 - NTUA	R — Document, report	PU - Public	48
D7.5	NBS for Water Reuse v1	WP7	3 - NTUA	R — Document, report	PU - Public	18
D7.6	NBS for Water Reuse v2	WP7	3 - NTUA	R — Document, report	PU - Public	36
D7.7	NBS for Water Reuse final	WP7	3 - NTUA	R — Document, report	PU - Public	48
D7.8	Fertigation of reclaimed water v1	WP7	3 - NTUA	R — Document, report	PU - Public	18
D7.9	Fertigation of reclaimed water v2	WP7	3 - NTUA	R — Document, report	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D7.10	Fertigation of reclaimed water final	WP7	3 - NTUA	R — Document, report	PU - Public	48
D7.11	Valorization of sewage sludge v1	WP7	3 - NTUA	R — Document, report	PU - Public	18

D7.12	Valorization of sewage sludge v2	WP7	3 - NTUA	R — Document, report	PU - Public	36
D7.13	Valorization of sewage sludge final	WP7	3 - NTUA	R — Document, report	PU - Public	48
D7.14	Master Plan for CE in Water v1	WP7	3 - NTUA	R — Document, report	PU - Public	38
D7.15	Master Plan for CE in Water final	WP7	3 - NTUA	R — Document, report	PU - Public	54
D8.1	Biobased materials and Packaging flow initial report	WP8	11 - HMU	R — Document, report	PU - Public	36
D8.2	Textiles flow initial report	WP8	14 - TUC	R — Document, report	PU - Public	36
D8.3	CDW & Glass flow initial report	WP8	1 - ICCS	R — Document, report	PU - Public	36
D8.4	Energy flow initial report	WP8	9 - CERTH	R — Document, report	PU - Public	36
D8.5	Water flow initial report	WP8	3 - NTUA	R — Document, report	PU - Public	36
D8.6	Biobased materials and Packaging flow report v2	WP8	11 - HMU	R — Document, report	PU - Public	48
D8.7	Textiles flow initial report v2	WP8	14 - TUC	R — Document, report	PU - Public	48
D8.8	CDW & Glass flow report v2	WP8	1 - ICCS	R — Document, report	PU - Public	48
D8.9	Energy flow report v2	WP8	9 - CERTH	R — Document, report	PU - Public	48

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D8.10	Water flow report v2	WP8	3 - NTUA	R — Document, report	PU - Public	48
D8.11	Biobased materials and Packaging flow final report	WP8	11 - HMU	R — Document, report	PU - Public	56
D8.12	Textiles flow final report	WP8	14 - TUC	R — Document, report	PU - Public	56
D8.13	CDW & Glass flow final report	WP8	1 - ICCS	R — Document, report	PU - Public	56
D8.14	Energy flow final report	WP8	9 - CERTH	R — Document, report	PU - Public	56
D8.15	Water flow final report	WP8	3 - NTUA	R — Document, report	PU - Public	56
D8.16	Validation of Frameworks V1	WP8	1 - ICCS	R — Document, report	PU - Public	40
D8.17	Validation of Frameworks final report	WP8	1 - ICCS	R — Document, report	PU - Public	56
D8.18	Impact assessment V1	WP8	23 - CIRCE	R — Document, report	PU - Public	40
D8.19	Impact assessment final report	WP8	23 - CIRCE	R — Document, report	PU - Public	56
D9.1	Synergies v1	WP9	17 - A.SPIRE	R — Document, report	PU - Public	36
D9.2	Synergies v2	WP9	17 - A.SPIRE	R — Document, report	PU - Public	48
D9.3	Synergies final	WP9	17 - A.SPIRE	R — Document, report	PU - Public	60
D9.4	Replication areas v1	WP9	37 - RISE	R — Document, report	PU - Public	36
D9.5	Replication areas v2	WP9	37 - RISE	R — Document, report	PU - Public	48
D9.6	Replication areas final	WP9	37 - RISE	R — Document, report	PU - Public	60

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D9.7	Replicability Roadmap v1	WP9	1 - ICCS	R — Document, report	PU - Public	36
D9.8	Replicability Roadmap v2	WP9	1 - ICCS	R — Document, report	PU - Public	48
D9.9	Replicability Roadmap final	WP9	1 - ICCS	R — Document, report	PU - Public	60
D10.1	H4C models initial	WP10	1 - ICCS	R — Document, report	PU - Public	30
D10.2	H4C models final	WP10	1 - ICCS	R — Document, report	PU - Public	60
D10.3	TM-WEMW nexus tool initial	WP10	45 - ICL	OTHER	PU - Public	30
D10.4	TM-WEMW nexus tool final	WP10	45 - ICL	OTHER	PU - Public	60
D10.5	Cities climate modelling v1	WP10	3 - NTUA	R — Document, report	PU - Public	36
D10.6	Cities climate modelling final	WP10	3 - NTUA	R — Document, report	PU - Public	48
D10.7	AR app for awareness raising alpha version	WP10	1 - ICCS	OTHER	PU - Public	36
D10.8	AR app for awareness raising beta version	WP10	1 - ICCS	OTHER	PU - Public	60
D10.9	Training toolkit v1	WP10	45 - ICL	R — Document, report	PU - Public	24

D10.10	Training toolkit v2	WP10	45 - ICL	R — Document, report	PU - Public	48
D10.11	Training toolkit final	WP10	45 - ICL	R — Document, report	PU - Public	60
D10.12	Impact on health v1	WP10	20 - RISA	R — Document, report	PU - Public	36
D10.13	Impact on health final	WP10	20 - RISA	R — Document, report	PU - Public	54
D10.14	De-risking assessment v1	WP10	24 - HSRT	R — Document, report	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D10.15	De-risking assessment final	WP10	24 - HSRT	R — Document, report	PU - Public	54
D11.1	Communication Dissemination v1	WP11	24 - HSRT	DEC —Websites, patent filings, videos, etc	PU - Public	6
D11.2	Communication Dissemination v2	WP11	24 - HSRT	DEC —Websites, patent filings, videos, etc	PU - Public	24
D11.3	Communication Dissemination v3	WP11	24 - HSRT	DEC —Websites, patent filings, videos, etc	PU - Public	48
D11.4	Communication Dissemination final	WP11	24 - HSRT	DEC —Websites, patent filings, videos, etc	PU - Public	60

D11.5	Policy Recommendations v1	WP11	16 - LiU	R — Document, report	PU - Public	36
D11.6	Policy Recommendations final	WP11	16 - LiU	R — Document, report	PU - Public	54
D11.7	Exploitation Strategy v1	WP11	1 - ICCS	R — Document, report	PU - Public	6
D11.8	Exploitation Strategy v2	WP11	24 - HSRT	R — Document, report	PU - Public	24
D11.9	Exploitation Strategy v3	WP11	24 - HSRT	R — Document, report	PU - Public	36
D11.10	Exploitation Strategy final	WP11	24 - HSRT	R — Document, report	PU - Public	60
D11.11	Market analysis v1	WP11	19 - EMVIS	R — Document, report	PU - Public	36
D11.12	Market analysis final	WP11	19 - EMVIS	R — Document, report	PU - Public	60
D11.13	Standardization activities v1	WP11	43 - DS	R — Document, report	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D11.14	Standardization activities final	WP11	43 - DS	R — Document, report	PU - Public	54
D11.15	Stakeholder Engagement v1	WP11	45 - ICL	R — Document, report	PU - Public	24
D11.16	Stakeholder Engagement v2	WP11	45 - ICL	R — Document, report	PU - Public	48

D11.17	Stakeholder Engagement final	WP11	45 - ICL	R — Document, report	PU - Public	60

Deliverable D1.1 – Theseus PM, QA & Ethics v1

Deliverable Number	D1.1	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Theseus PM, QA & Ethics v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
PM, reporting , gender ethics issues report as a reference point during project duration. Reflects the work of T1.1, T1.2 and T1.3.			

Deliverable Number	D1.4	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Theseus PM, QA & Ethics v4		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP1
Description			

PM, reporting , gender ethics issues report as a reference point during project duration. Reflects the work of T1.1, T1.2 and T1.3.

Deliverable D1.2 – Theseus PM, QA & Ethics v2

Deliverable Number	D1.2	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Theseus PM, QA & Ethics v2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP1
Description			
PM, reporting , gender ethics issues report as a reference point during project duration. Reflects the work of T1.1, T1.2 and T1.3..			

Deliverable D1.3 – Theseus PM, QA & Ethics v3

Deliverable Number	D1.3	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Theseus PM, QA & Ethics v3		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1
Description			
PM, reporting , gender ethics issues report as a reference point during project duration. Reflects the work of T1.1, T1.2 and T1.3.			

Deliverable D1.4 – Theseus PM, QA & Ethics v4 Deliverable D1.5 – Theseus PM, QA & Ethics final

Deliverable Number	D1.5	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Theseus PM, QA & Ethics final		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	60	**Work Package No**	WP1
Description			
PM, reporting , gender ethics issues report as a reference point during project duration. Reflects the work of T1.1, T1.2 and T1.3..			

Deliverable Number	D1.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Management Plan v3		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1

Description

Data of value chains & technologies; Standards metadata; maintenance & storage procedures. Reflects the work of T1.2 .

Deliverable D1.6 – Data Management Plan v1

Deliverable Number	D1.6	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Management Plan v1		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
Data of value chains & technologies; Standards metadata; maintenance & storage procedures. Reflects the work of T1.2 .			

Deliverable D1.7 – Data Management Plan v2

Deliverable Number	D1.7	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Management Plan v2		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP1
Description			
Data of value chains & technologies; Standards metadata; maintenance & storage procedures. Reflects the work of T1.2			

Deliverable D1.8 – Data Management Plan v3 Deliverable D1.9 – Data Management Plan v4

Deliverable Number	D1.9	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Management Plan v4		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP1
Description			
Data of value chains & technologies; Standards metadata; maintenance & storage procedures. Reflects the work of T1.2 .			

Deliverable Number	D1.12	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Short Interim Management Report v2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1
Description			

Short Interim Management Report: objective for the period, progress obtained and outcome , issues encounter and challenges to address forward. Reflects the work of T1.1

Deliverable D1.10 – Data Management Plan final

Deliverable Number	D1.10	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Management Plan final		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	60	**Work Package No**	WP1
Description			
Data of value chains & technologies; Standards metadata; maintenance & storage procedures. Reflects the work of T1.2 .			

Deliverable D1.11 – Short Interim Management Report v1

Deliverable Number	D1.11	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Short Interim Management Report v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP1
Description			
Short Interim Management Report: objective for the period, progress obtained and outcome , issues encounter and challenges to address forward. Reflects the work of T1.1			

Deliverable D1.12 – Short Interim Management Report v2 Deliverable D1.13 – Short Interim Management Report v3

Deliverable Number	D1.13	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Short Interim Management Report v3		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP1
Description			
Short Interim Management Report: objective for the period, progress obtained and outcome , issues encounter and challenges to address forward. Reflects the work of T1.1			

Deliverable D2.1 – Governance Model v1

Deliverable Number	D2.1	**Lead Beneficiary**	16 - LiU

Description
Governance model for Regional and Intra Regional I-US. Reflects the work of T2.1.

Deliverable Name	Governance Model v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2
Description			
Governance model for Regional and Intra Regional I-US. Reflects the work of T2.1.			

Deliverable D2.2 – Governance Model v2

Deliverable Number	D2.2	**Lead Beneficiary**	16 - LiU
Deliverable Name	Governance Model v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Governance model for Regional and Intra Regional I-US. Reflects the work of T2.1			

Deliverable D2.3 – Governance Model final

Deliverable Number	D2.3	**Lead Beneficiary**	16 - LiU
Deliverable Name	Governance Model final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP2

Deliverable D2.4 – Participatory approaches v1

Deliverable Number	D2.4	**Lead Beneficiary**	16 - LiU
Deliverable Name	Participatory approaches v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2
Description			
Participatory approaches and co-creation processes. Reflects the work of T2.2.			

Deliverable D2.5 – Participatory approaches v2

Description			
Mapping of materials, water and energy flows and 2050 Projections. Reflects the work of T2.3.			
Deliverable Number	D2.5	**Lead Beneficiary**	16 - LiU

Deliverable Name	Participatory approaches v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Participatory approaches and co-creation processes. Reflects the work of T2.2			

Deliverable D2.6 – Participatory approaches final

Deliverable Number	D2.6	**Lead Beneficiary**	16 - LiU
Deliverable Name	Participatory approaches final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP2
Description			
Participatory approaches and co-creation processes. Reflects the work of T2.2.			

Deliverable D2.7 – Material flow analysis v1

Deliverable Number	D2.7	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Material flow analysis v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2

Deliverable D2.8 – Material flow analysis v2

Deliverable Number	D2.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Material flow analysis v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Mapping of materials, water and energy flows and 2050 Projections. Reflects the work of T2.3.			

Deliverable D2.9 – Material flow analysis final

Description			
Regional & interregional pathways for symbiotic flows linked to EU & local regional plans. Reflects the work of T2.4.			
Deliverable Number	D2.9	**Lead Beneficiary**	1 - ICCS

Deliverable Name	Material flow analysis final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP2
Description			
Mapping of materials, water and energy flows and 2050 Projections. Reflects the work of T2.3.			

Deliverable D2.10 – Regional/interregional pathways v1

Deliverable Number	D2.10	**Lead Beneficiary**	13 - UAEGEAN
Deliverable Name	Regional/interregional pathways v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2
Description			
Regional & interregional pathways for symbiotic flows linked to EU & local regional plans. Reflects the work of T2.4.			

Deliverable D2.11 – Regional/interregional pathways v2

Deliverable Number	D2.11	**Lead Beneficiary**	13 - UAEGEAN
Deliverable Name	Regional/interregional pathways v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2

Deliverable D2.12 – Regional/interregional pathways final

Deliverable Number	D2.12	**Lead Beneficiary**	13 - UAEGEAN
Deliverable Name	Regional/interregional pathways final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP2
Description			
Regional & interregional pathways for symbiotic flows linked to EU & local regional plans. Reflects the work of T2.4.			

Deliverable D2.13 – Characterization of value chains v1

Description			
Characterization of value chains and KPIs definition and workshops results. Reflects the work of T2.5.			
Deliverable Number	D2.13	**Lead Beneficiary**	37 - RISE

Deliverable Name	Characterization of value chains v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2
Description			
Characterization of value chains and KPIs definition and workshops results. Reflects the work of T2.5.			

Deliverable D2.14 – Characterization of value chains v2

Deliverable Number	D2.14	**Lead Beneficiary**	37 - RISE
Deliverable Name	Characterization of value chains v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Characterization of value chains and KPIs definition and workshops results. Reflects the work of T2.5.			

Deliverable D2.15 – Characterization of value chains final

Deliverable Number	D2.15	**Lead Beneficiary**	37 - RISE
Deliverable Name	Characterization of value chains final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP2

Deliverable D2.16 – Digital interactions and planning v1

Deliverable Number	D2.16	**Lead Beneficiary**	18 - CORE
Deliverable Name	Digital interactions and planning v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP2
Description			
Digital plans for each pilot/demonstration site mapping data requirements. Reflects the work of T2.6.			

Deliverable D2.17 – Digital interactions and planning v2

Description			
GIS tool for dynamic symbiotic maps and reverse logistics optimization module. Reflects the work of T3.1			
Deliverable Number	D2.17	**Lead Beneficiary**	18 - CORE

Deliverable Name	Digital interactions and planning v2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP2
Description			
Digital plans for each pilot/demonstration site mapping data requirements. Reflects the work of T2.6.			

Deliverable D2.18 – Digital interactions and planning final

Deliverable Number	D2.18	**Lead Beneficiary**	18 - CORE
Deliverable Name	Digital interactions and planning final		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2
Description			
Digital plans for each pilot/demonstration site mapping data requirements. Reflects the work of T2.6.			

Deliverable D3.1 – GIS tool v1

Deliverable Number	D3.1	**Lead Beneficiary**	21 - TLNV
Deliverable Name	GIS tool v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3

Deliverable D3.2 – GIS tool v2

Deliverable Number	D3.2	**Lead Beneficiary**	21 - TLNV
Deliverable Name	GIS tool v2		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3
Description			
GIS tool for dynamic symbiotic maps and reverse logistics optimization module. Reflects the work of T3.1			

Deliverable D3.3 – GIS tool final

Description			
Matching frameworks and development of symbiotic models module. Reflects the work of T3.2.			
Deliverable Number	D3.3	**Lead Beneficiary**	21 - TLNV

Deliverable Name	GIS tool final		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP3
Description			
GIS tool for dynamic symbiotic maps and reverse logistics optimization module. Reflects the work of T3.1.			

Deliverable D3.4 – Symbiotic models v1

Deliverable Number	D3.4	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Symbiotic models v1		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP3
Description			
Matching frameworks and development of symbiotic models module. Reflects the work of T3.2.			

Deliverable D3.5 – Symbiotic models v2

Deliverable Number	D3.5	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Symbiotic models v2		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP3

Deliverable D3.6 – Symbiotic models final

Deliverable Number	D3.6	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Symbiotic models final		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP3
Description			
Matching frameworks and development of symbiotic models module. Reflects the work of T3.2.			

Deliverable D3.7 – Dynamic DPP and MR app v1

Description			
DPPs and end-users interactions though AR module. Reflects the work of T3.3.			
Deliverable Number	D3.7	**Lead Beneficiary**	1 - ICCS

Deliverable Name	Dynamic DPP and MR app v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
DPPs and end-users interactions though AR module. Reflects the work of T3.3.			

Deliverable D3.8 – Dynamic DPP and MR app v2

Deliverable Number	D3.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Dynamic DPP and MR app v2		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3
Description			
DPPs and end-users interactions though AR module. Reflects the work of T3.3.			

Deliverable D3.9 – Dynamic DPP and MR app final

Deliverable Number	D3.9	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Dynamic DPP and MR app final		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP3

Deliverable D3.10 – Prediction logistics module v1

Deliverable Number	D3.10	**Lead Beneficiary**	18 - CORE
Deliverable Name	Prediction logistics module v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
Predictive intelligence module for material and water flows among actors of the H4C. Reflects the work of T3.4.			

Deliverable D3.11 – Prediction logistics module v2

Description			
Data Harmonization and Secure data exchange channels and platform. Reflects the work of T3.5.			
Deliverable Number	D3.11	**Lead Beneficiary**	18 - CORE

Deliverable Name	Prediction logistics module v2		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3
Description			
Predictive intelligence module for material and water flows among actors of the H4C. Reflects the work of T3.4.			

Deliverable D3.12 – Prediction logistics module final

Deliverable Number	D3.12	**Lead Beneficiary**	18 - CORE
Deliverable Name	Prediction logistics module final		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP3
Description			
Predictive intelligence module for material and water flows among actors of the H4C. Reflects the work of T3.4.			

Deliverable D3.13 – Data Exchange Channels v1

Deliverable Number	D3.13	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Exchange Channels v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3

Deliverable D3.14 – Data Exchange Channels v2

Deliverable Number	D3.14	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Data Exchange Channels v2		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3
Description			
Data Harmonization and Secure data exchange channels and platform. Reflects the work of T3.5.			

Deliverable D3.15 – Data Exchange Channels final

Description			
Marketplace and modules integration across digital H4C. Reflects the work of T3.6.			
Deliverable Number	D3.15	**Lead Beneficiary**	1 - ICCS

Deliverable Name	Data Exchange Channels final		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP3
Description			
Data Exchange Channels final. Reflects the work of T3.5.			

Deliverable D3.16 – Theseus H4C Markeplace v1

Deliverable Number	D3.16	**Lead Beneficiary**	22 - IRIS
Deliverable Name	Theseus H4C Markeplace v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
Marketplace and modules integration across digital H4C. Reflects the work of T3.6.			

Deliverable D3.17 – Theseus H4C Markeplace v2

Deliverable Number	D3.17	**Lead Beneficiary**	22 - IRIS
Deliverable Name	Theseus H4C Markeplace v2		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3

Deliverable D3.18 – Theseus H4C Markeplace final

Deliverable Number	D3.18	**Lead Beneficiary**	22 - IRIS
Deliverable Name	Theseus H4C Markeplace final		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP3
Description			
Marketplace and modules integration across digital H4C. Reflects the work of T3.6.			

Deliverable D4.1 – Technologies requirements

Deliverable Number	D4.1	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Technologies requirements		
Description			
Deployment aspects for the sorting lines (Mixed urban wastes, Textile sorting, Plastic recovery, Metal sorting/ Glass Sorting) . Reflects the work of T4,2, T4,3, T4,4. T4,5.			

Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP4
Description			
Requirements analysis, concept design and optimization for material and sorting solutions. Reflects the work of T4.1.			

Deliverable D4.2 – Sorting lines (TRL 5-ready)

Deliverable Number	D4.2	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Sorting lines (TRL 5-ready)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP4
Description			
Preliminary developments for the sorting lines (Mixed urban wastes, Textile sorting, Plastic recovery, Metal sorting/ Glass Sorting) . Reflects the work of T4,2, T4,3, T4,4. T4,5.			

Deliverable D4.3 – Sorting lines (TRL 6-ready)

Deliverable Number	D4.3	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Sorting lines (TRL 6-ready)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4

Deliverable D4.4 – Sorting lines (TRL 7/8-ready)

Deliverable Number	D4.4	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Sorting lines (TRL 7/8-ready)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP4
Description			
Evaluation for the sorting lines (Mixed urban wastes, Textile sorting, Plastic recovery, Metal sorting/ Glass Sorting) . Reflects the work of T4,2, T4,3, T4,4. T4,5.			

Deliverable D4.5 – Autonomous Platform v1

Deliverable Number	D4.5	**Lead Beneficiary**	12 - DUTH

Description
Platform Integration and Navigation Strategies for Optimized Plant Material Flows. Reflects the work of T4.6.

Deliverable Name	Autonomous Platform v1		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP4
Description			
Platform Integration and Navigation Strategies for Optimized Plant Material Flows. Reflects the work of T4.6.			

Deliverable D4.6 – Autonomous Platform v2

Deliverable Number	D4.6	**Lead Beneficiary**	12 - DUTH
Deliverable Name	Autonomous Platform v2		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4
Description			
Platform Integration and Navigation Strategies for Optimized Plant Material Flows. Reflects the work of T4.6.			

Deliverable D4.7 – Autonomous Platform final

Deliverable Number	D4.7	**Lead Beneficiary**	12 - DUTH
Deliverable Name	Autonomous Platform final		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP4

Deliverable D4.8 – Smart bins for Attica v1

Deliverable Number	D4.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Smart bins for Attica v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP4
Description			
Placement optimization Analysis for Smart Bins in West Attica Region. Reflects the work of T4.7.			

Deliverable D4.9 – Smart bins for Attica v2

Description			
Alpha and Beta version of the XR application. Reflects the work of T4.8.			
Deliverable Number	D4.9	**Lead Beneficiary**	1 - ICCS

Deliverable Name	Smart bins for Attica v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4
Description			
Placement optimization Analysis for Smart Bins in West Attica Region. Reflects the work of T4.7.			

Deliverable D4.10 – Smart bins for Attica final

Deliverable Number	D4.10	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Smart bins for Attica final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP4
Description			
Placement optimization Analysis for Smart Bins in West Attica Region. Reflects the work of T4.7.			

Deliverable D4.11 – XR for human-robot interaction alpha version

Deliverable Number	D4.11	**Lead Beneficiary**	1 - ICCS
Deliverable Name	XR for human-robot interaction alpha version		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4

Deliverable D4.12 – XR for human-robot interaction beta version

Deliverable Number	D4.12	**Lead Beneficiary**	1 - ICCS
Deliverable Name	XR for human-robot interaction beta version		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP4
Description			
Alpha and Beta version of the XR application. Reflects the work of T4.8.			

Deliverable D5.1 – Biological recycling for textiles v1

Description			
Conclusions on chemical recycling of textiles and optimization though IA of parameters. Reflects the work of T5.1.			
Deliverable Number	D5.1	**Lead Beneficiary**	23 - CIRCE

Deliverable Name	Biological recycling for textiles v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP5
Description			
Conclusions on chemical recycling of textiles and optimization though IA of parameters. Reflects the work of T5.1.			

Deliverable D5.2 – Biological recycling for textiles v2

Deliverable Number	D5.2	**Lead Beneficiary**	23 - CIRCE
Deliverable Name	Biological recycling for textiles v2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
Conclusions on chemical recycling of textiles and optimization though IA of parameters. Reflects the work of T5.1.			

Deliverable D5.3 – Biological recycling for textiles final

Deliverable Number	D5.3	**Lead Beneficiary**	23 - CIRCE
Deliverable Name	Biological recycling for textiles final		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP5

Deliverable D5.4 – Biodegradable masterbatches v1

Deliverable Number	D5.4	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biodegradable masterbatches v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP5
Description			
Development of biodegradable masterbatches & characterisation of industrial scale. Reflects the work of T5.2.			

Deliverable D5.5 – Biodegradable masterbatches v2

Description			
Raw materials for fired brick production and concrete recipes & Leaching impact. Reflects the work of T5.3.			
Deliverable Number	D5.5	**Lead Beneficiary**	11 - HMU

Deliverable Name	Biodegradable masterbatches v2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
Development of biodegradable masterbatches & characterisation of industrial scale. Reflects the work of T5.2.			

Deliverable D5.6 – Biodegradable masterbatches final

Deliverable Number	D5.6	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biodegradable masterbatches final		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP5
Description			
Development of biodegradable masterbatches & characterisation of industrial scale. Reflects the work of T5.2.			

Deliverable D5.7 – Construction materials v1

Deliverable Number	D5.7	**Lead Beneficiary**	40 - SINTEF
Deliverable Name	Construction materials v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5

Deliverable D5.8 – Construction materials v2

Deliverable Number	D5.8	**Lead Beneficiary**	40 - SINTEF
Deliverable Name	Construction materials v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP5
Description			
Raw materials for fired brick production and concrete recipes & Leaching impact. Reflects the work of T5.3.			

Deliverable D5.9 – Construction materials final

Description			
Biobased thin film for packaging using bioplastics extracted. Reflects the work of T5.4			
Deliverable Number	D5.9	**Lead Beneficiary**	40 - SINTEF

Deliverable Name	Construction materials final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Raw materials for fired brick production and concrete recipes & Leaching impact. Reflects the work of T5.3.			

Deliverable D5.10 – Biobased thin films v1

Deliverable Number	D5.10	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biobased thin films v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Biobased thin film for packaging using bioplastics extracted. Reflects the work of T5.4			

Deliverable D5.11 – Biobased thin films v2

Deliverable Number	D5.11	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biobased thin films v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP5

Deliverable D5.12 – Biobased thin films final

Deliverable Number	D5.12	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biobased thin films final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5
Description			
Biobased thin film for packaging using bioplastics extracted. Reflects the work of T5.4			

Deliverable D5.13 – Thin-film materials v1

Description			
Verified thin-film materials from electrospinning process. Reflects the work of T5.5.			
Deliverable Number	D5.13	**Lead Beneficiary**	3 - NTUA

Deliverable Name	Thin-film materials v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Verified thin-film materials from electrospinning process. Reflects the work of T5.5.			

Deliverable D5.14 – Thin-film materials v2

Deliverable Number	D5.14	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Thin-film materials v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP5
Description			
Verified thin-film materials from electrospinning process. Reflects the work of T5.5.			

Deliverable D5.15 – Thin-film materials final

Deliverable Number	D5.15	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Thin-film materials final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP5

Deliverable D6.1 – RES technologies

Deliverable Number	D6.1	**Lead Beneficiary**	9 - CERTH
Deliverable Name	RES technologies		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP6
Description			
Introduction of innovative RES technologies to District heating (DH) systems . Reflects the work of T6.1.			

Deliverable D6.2 – Solar thermal technology v1

Description			
Design and specifications for Innovative solar thermal into DH and cooling of buildings. Reflects the work of T6.2.			
Deliverable Number	D6.2	**Lead Beneficiary**	9 - CERTH

Deliverable Name	Solar thermal technology v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP6
Description			
Design and specifications for Innovative solar thermal into DH and cooling of buildings. Reflects the work of T6.2.			

Deliverable D6.3 – Solar thermal technology v2

Deliverable Number	D6.3	**Lead Beneficiary**	9 - CERTH
Deliverable Name	Solar thermal technology v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
Design and specifications for Innovative solar thermal into DH and cooling of buildings. Reflects the work of T6.2.			

Deliverable D6.4 – Solar thermal technology final

Deliverable Number	D6.4	**Lead Beneficiary**	9 - CERTH
Deliverable Name	Solar thermal technology final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6

Deliverable D6.5 – CHP fuel cell system v1

Deliverable Number	D6.5	**Lead Beneficiary**	9 - CERTH
Deliverable Name	CHP fuel cell system v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP6
Description			
Utilization of green H2 for heat and electricity through CHP fuel cell system adaptation . Reflects the work of T6.3.			

Deliverable D6.6 – CHP fuel cell system v2

Description			
Digital smart technologies for DH towards improved energy management and control. . Reflects the work of T6.4.			
Deliverable Number	D6.6	**Lead Beneficiary**	9 - CERTH

Deliverable Name	CHP fuel cell system v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
Utilization of green H2 for heat and electricity through CHP fuel cell system adaptation . Reflects the work of T6.3..			

Deliverable D6.7 – CHP fuel cell system final

Deliverable Number	D6.7	**Lead Beneficiary**	9 - CERTH
Deliverable Name	CHP fuel cell system final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6
Description			
Utilization of green H2 for heat and electricity through CHP fuel cell system adaptation . Reflects the work of T6.3.			

Deliverable D6.8 – DH smart technologies v1

Deliverable Number	D6.8	**Lead Beneficiary**	9 - CERTH
Deliverable Name	DH smart technologies v1		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP6

Deliverable D6.9 – DH smart technologies v2

Deliverable Number	D6.9	**Lead Beneficiary**	9 - CERTH
Deliverable Name	DH smart technologies v2		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
Digital smart technologies for DH towards improved energy management and control. . Reflects the work of T6.4.			

Deliverable D6.10 – DH smart technologies final

Description			
Results from demonstration of decentralized systems, sewer mining and rainwater harvesting . Reflects the work of T7.2.			
Deliverable Number	D6.10	**Lead Beneficiary**	9 - CERTH

Deliverable Name	DH smart technologies final		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6
Description			
Digital smart technologies for DH towards improved energy management and control. . Reflects the work of T6.4.			

Deliverable D7.1 – Analysis for Attica water needs

Deliverable Number	D7.1	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Analysis for Attica water needs		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP7
Description			
A comprehensive water flow analysis will be performed for the entire region of Attica. Reflects the work of T7.1.			

Deliverable D7.2 – Decentralized Water Reuse v1

Deliverable Number	D7.2	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Decentralized Water Reuse v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7

Deliverable D7.3 – Decentralized Water Reuse v2

Deliverable Number	D7.3	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Decentralized Water Reuse v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7
Description			
Results from demonstration of decentralized systems, sewer mining and rainwater harvesting Reflects the work of T7.2.			

Deliverable D7.4 – Decentralized Water Reuse final

Description			
A blueprint for use of NBS for rainwater/stormwater harvesting in urban and peri-urban areas Reflects the work of T7.3.			
Deliverable Number	D7.4	**Lead Beneficiary**	3 - NTUA

Deliverable Name	Decentralized Water Reuse final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
Results from demonstration of decentralized systems, sewer mining and rainwater harvesting Reflects the work of T7.2.			

Deliverable D7.5 – NBS for Water Reuse v1

Deliverable Number	D7.5	**Lead Beneficiary**	3 - NTUA
Deliverable Name	NBS for Water Reuse v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7
Description			
A blueprint for use of NBS for rainwater/stormwater harvesting in urban and peri-urban areas Reflects the work of T7.3.			

Deliverable D7.6 – NBS for Water Reuse v2

Deliverable Number	D7.6	**Lead Beneficiary**	3 - NTUA
Deliverable Name	NBS for Water Reuse v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Deliverable D7.7 – NBS for Water Reuse final

Deliverable Number	D7.7	**Lead Beneficiary**	3 - NTUA
Deliverable Name	NBS for Water Reuse final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
A blueprint for use of NBS for rainwater/stormwater harvesting in urban and peri-urban areas Reflects the work of T7.3.			

Deliverable D7.8 – Fertigation of reclaimed water v1

Description			
Benefits and risks arising from the large-scale fertigation of crops with reclaimed water Reflects the work of T7.4.			
Deliverable Number	D7.8	**Lead Beneficiary**	3 - NTUA

Deliverable Name	Fertigation of reclaimed water v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7
Description			
Benefits and risks arising from the large-scale fertigation of crops with reclaimed water Reflects the work of T7.4.			

Deliverable D7.9 – Fertigation of reclaimed water v2

Deliverable Number	D7.9	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Fertigation of reclaimed water v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7
Description			
Benefits and risks arising from the large-scale fertigation of crops with reclaimed water Reflects the work of T7.4.			

Deliverable D7.10 – Fertigation of reclaimed water final

Deliverable Number	D7.10	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Fertigation of reclaimed water final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7

Deliverable D7.11 – Valorization of sewage sludge v1

Deliverable Number	D7.11	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Valorization of sewage sludge v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7
Description			
Results from fast pyrolysis and torrefaction for sewage sludge valorization. Reflects the work of T7.5.			

Deliverable D7.12 – Valorization of sewage sludge v2

Description			
A master plan for integrating CE into water and wastewater services. Reflects the work of T7.6.			
Deliverable Number	D7.12	**Lead Beneficiary**	3 - NTUA

Deliverable Name	Valorization of sewage sludge v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7
Description			
Results from fast pyrolysis and torrefaction for sewage sludge valorization . Reflects the work of T7.5.			

Deliverable D7.13 – Valorization of sewage sludge final

Deliverable Number	D7.13	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Valorization of sewage sludge final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
Results from fast pyrolysis and torrefaction for sewage sludge valorization . Reflects the work of T7.5.			

Deliverable D7.14 – Master Plan for CE in Water v1

Deliverable Number	D7.14	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Master Plan for CE in Water v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	38	**Work Package No**	WP7

Deliverable D7.15 – Master Plan for CE in Water final

Deliverable Number	D7.15	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Master Plan for CE in Water final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	54	**Work Package No**	WP7
Description			
A master plan for integrating CE into water and wastewater services. Reflects the work of T7.6.			

Deliverable D8.1 – Biobased materials and Packaging flow initial report

Description			
CDW & Glass flow: Initial Report in pilots and demonstration activities. Reflects the work of T8.3.			
Deliverable Number	D8.1	**Lead Beneficiary**	11 - HMU

Deliverable Name	Biobased materials and Packaging flow initial report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8
Description			
Biobased materials and Packaging flow: Initial Report in pilots and demonstration activities . Reflects the work of T8.1.			

Deliverable D8.2 – Textiles flow initial report

Deliverable Number	D8.2	**Lead Beneficiary**	14 - TUC
Deliverable Name	Textiles flow initial report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8
Description			
Textiles flow: Initial Report in pilots and demonstration activities. Reflects the work of T8.2.			

Deliverable D8.3 – CDW & Glass flow initial report

Deliverable Number	D8.3	**Lead Beneficiary**	1 - ICCS
Deliverable Name	CDW & Glass flow initial report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8

Deliverable D8.4 – Energy flow initial report

Deliverable Number	D8.4	**Lead Beneficiary**	9 - CERTH
Deliverable Name	Energy flow initial report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8
Description			
Energy flow: Initial Report in pilots and demonstration activities. Reflects the work of T8.4.			

Deliverable D8.5 – Water flow initial report

Deliverable Number	D8.5	**Lead Beneficiary**	3 - NTUA

Description
Textiles flow: Report in pilots and demonstration activities v2. Reflects the work of T8.2.

Deliverable Name	Water flow initial report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8
Description			
Water flow: Initial Report in pilots and demonstration activities. Reflects the work of T8.5.			

Deliverable D8.6 – Biobased materials and Packaging flow report v2

Deliverable Number	D8.6	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biobased materials and Packaging flow report v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP8
Description			
Biobased materials and Packaging flow: Report in pilots and demonstration activities, results and valuable insights v2. Reflects the work of T8.1.			

Deliverable D8.7 – Textiles flow initial report v2

Deliverable Number	D8.7	**Lead Beneficiary**	14 - TUC
Deliverable Name	Textiles flow initial report v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP8

Deliverable D8.8 – CDW & Glass flow report v2

Deliverable Number	D8.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	CDW & Glass flow report v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP8
Description			
CDW & Glass flow: Report in pilots and demonstration activities v2. Reflects the work of T8.3.			

Deliverable D8.9 – Energy flow report v2

Description
Biobased materials and Packaging flow: Final Report in pilots and demonstration activities, results and valuable insights. Reflects the work of T8.1.

Deliverable Number	D8.9	**Lead Beneficiary**	9 - CERTH
Deliverable Name	Energy flow report v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP8
Description			
Energy flow: Report in pilots and demonstration activities v2. Reflects the work of T8.4.			

Deliverable D8.10 – Water flow report v2

Deliverable Number	D8.10	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Water flow report v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP8
Description			
Water flow: Report in pilots and demonstration activities v2. Reflects the work of T8.5.			

Deliverable D8.11 – Biobased materials and Packaging flow final report

Deliverable Number	D8.11	**Lead Beneficiary**	11 - HMU
Deliverable Name	Biobased materials and Packaging flow final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8

Deliverable D8.12 – Textiles flow final report

Deliverable Number	D8.12	**Lead Beneficiary**	14 - TUC
Deliverable Name	Textiles flow final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8
Description			
Textiles flow: Final Report in pilots and demonstration activities, results and valuable insights. Reflects the work of T8.2.			

Description
Water flow: Final Report in pilots and demonstration activities, results and valuable insights. Reflects the work of T8.5.

Deliverable D8.13 – CDW & Glass flow final report

Deliverable Number	D8.13	**Lead Beneficiary**	1 - ICCS
Deliverable Name	CDW & Glass flow final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8
Description			
CDW & Glass flow: Final Report in pilots and demonstration activities, results and valuable insights. Reflects the work of T8.3.			

Deliverable D8.14 – Energy flow final report

Deliverable Number	D8.14	**Lead Beneficiary**	9 - CERTH
Deliverable Name	Energy flow final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8
Description			
Energy flow: Final Report in pilots and demonstration activities, results and valuable insights. Reflects the work of T8.4.			

Deliverable D8.15 – Water flow final report

Deliverable Number	D8.15	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Water flow final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8

Deliverable D8.16 – Validation of Frameworks V1

Deliverable Number	D8.16	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Validation of Frameworks V1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	40	**Work Package No**	WP8
Description			
Validation of digital governance and collaboration schemes. Reflects the work of T8.6			

Description
Circularity and sustainability performance and impact of all demonstrators. Reflects the work of T8.7.

Deliverable D8.17 – Validation of Frameworks final report

Deliverable Number	D8.17	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Validation of Frameworks final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8
Description			
Validation of digital governance and collaboration schemes. Reflects the work of T8.6.			

Deliverable D8.18 – Impact assessment V1

Deliverable Number	D8.18	**Lead Beneficiary**	23 - CIRCE
Deliverable Name	Impact assessment V1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	40	**Work Package No**	WP8
Description			
Circularity and sustainability performance and impact of all demonstrators. Reflects the work of T8.7.			

Deliverable D8.19 – Impact assessment final report

Deliverable Number	D8.19	**Lead Beneficiary**	23 - CIRCE
Deliverable Name	Impact assessment final report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	56	**Work Package No**	WP8

Deliverable D9.1 – Synergies v1

Deliverable Number	D9.1	**Lead Beneficiary**	17 - A.SPIRE
Deliverable Name	Synergies v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP9
Description			
Report on synergies, common activities with H4C, CCRI etc – T9.1 and T9.2			

Description
Replication to Malmö and MO areas, common developments and experience. Reflects the work of T9.3.

Deliverable D9.2 – Synergies v2

Deliverable Number	D9.2	**Lead Beneficiary**	17 - A.SPIRE
Deliverable Name	Synergies v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP9
Description			
Report on synergies, common activities with H4C, CCRI etc – T9.1 and T9.2			

Deliverable D9.3 – Synergies final

Deliverable Number	D9.3	**Lead Beneficiary**	17 - A.SPIRE
Deliverable Name	Synergies final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP9
Description			
Report on synergies, common activities with H4C, CCRI etc – T9.1 and T9.2			

Deliverable D9.4 – Replication areas v1

Deliverable Number	D9.4	**Lead Beneficiary**	37 - RISE
Deliverable Name	Replication areas v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP9

Deliverable D9.5 – Replication areas v2

Deliverable Number	D9.5	**Lead Beneficiary**	37 - RISE
Deliverable Name	Replication areas v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP9
Description			
Replication to Malmö and MO areas, common developments and experience. Reflects the work of T9.3.			

Deliverable D9.6 – Replication areas final

Deliverable Number	D9.6	**Lead Beneficiary**	37 - RISE
Deliverable Name	Replication areas final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP9
Description			
Replication to Malmö and MO areas, common developments and experience. Reflects the work of T9.3.			

Deliverable D9.7 – Replicability Roadmap v1

Deliverable Number	D9.7	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Replicability Roadmap v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP9
Description			
Roadmap for replicability to additional sites in Greece and around EU. Reflects the work of T9.4.			

Deliverable D9.8 – Replicability Roadmap v2

Deliverable Number	D9.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Replicability Roadmap v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP9

Deliverable D9.9 – Replicability Roadmap final

Deliverable Number	D9.9	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Replicability Roadmap final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP9
Description			
Roadmap for replicability to additional sites in Greece and around EU. Reflects the work of T9.4.			

Description			
Roadmap for replicability to additional sites in Greece and around EU. Reflects the work of T9.4.			
Deliverable Number	D10.3	**Lead Beneficiary**	45 - ICL

Deliverable Name	TM-WEMW nexus tool initial		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP10

Deliverable D10.1 – H4C models initial

Deliverable Number	D10.1	**Lead Beneficiary**	1 - ICCS
Deliverable Name	H4C models initial		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP10
Description			
A report documenting the developed database on circularity solutions and policies that can be modelled in the two IAMs, as well as the open science guidelines and ontologies for documenting model outputs. Reflects the work of T10.1.			

Deliverable D10.2 – H4C models final

Deliverable Number	D10.2	**Lead Beneficiary**	1 - ICCS
Deliverable Name	H4C models final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP10
Description			
A report documenting the developed database on circularity solutions and policies that can be modelled in the two IAMs, as well as the open science guidelines and ontologies for documenting model outputs. Reflects the work of T10.1.			

Deliverable D10.3 – TM-WEMW nexus tool initial Deliverable D10.4 – TM-WEMW nexus tool final

Deliverable Number	D10.4	**Lead Beneficiary**	45 - ICL
Deliverable Name	TM-WEMW nexus tool final		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP10
Description			
Territorial Metabolism and Integrated TM-WEMW nexus tool initial version. Reflects the work of T10.2.			

Description
Territorial Metabolism and Integrated TM-WEMW nexus tool initial version. Reflects the work of T10.2.

Deliverable D10.5 – Cities climate modelling v1

Deliverable Number	D10.5	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Cities climate modelling v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10
Description			
Report on environmental and economic assessment of Theseus interventions. . Reflects the work of T10.3.			

Deliverable D10.6 – Cities climate modelling final

Deliverable Number	D10.6	**Lead Beneficiary**	3 - NTUA
Deliverable Name	Cities climate modelling final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP10
Description			
Report on environmental and economic assessment of Theseus interventions. . Reflects the work of T10.3.			

Deliverable D10.7 – AR app for awareness raising alpha version

Deliverable Number	D10.7	**Lead Beneficiary**	1 - ICCS
Deliverable Name	AR app for awareness raising alpha version		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10

Deliverable D10.8 – AR app for awareness raising beta version

Deliverable Number	D10.8	**Lead Beneficiary**	1 - ICCS
Deliverable Name	AR app for awareness raising beta version		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP10
Description			
Social Awareness increase through AR – app development and activities. Reflects the work of T10.4.			

Description
Social Awareness increase through AR – app development and activities. Reflects the work of T10.4.
Description

Upskilling & Training activities in region cities and universities. Reflects the work of T10.5.

Deliverable D10.9 – Training toolkit v1

Deliverable Number	D10.9	**Lead Beneficiary**	45 - ICL
Deliverable Name	Training toolkit v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP10
Description			
Upskilling & Training activities in region cities and universities. Reflects the work of T10.5.			

Deliverable D10.10 – Training toolkit v2

Deliverable Number	D10.10	**Lead Beneficiary**	45 - ICL
Deliverable Name	Training toolkit v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP10
Description			
Upskilling & Training activities in region cities and universities. Reflects the work of T10.5.			

Deliverable D10.11 – Training toolkit final

Deliverable Number	D10.11	**Lead Beneficiary**	45 - ICL
Deliverable Name	Training toolkit final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP10

Deliverable D10.12 – Impact on health v1

Deliverable Number	D10.12	**Lead Beneficiary**	20 - RISA
Deliverable Name	Impact on health v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10
Description			
Report on the impact of the H4C implementation on the health of citizens. Reflects the work of T10.6.			

Deliverable D10.13 – Impact on health final

Deliverable Number	D10.13	**Lead Beneficiary**	20 - RISA
Deliverable Name	Impact on health final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	54	**Work Package No**	WP10
Description			
Report on the impact of the H4C implementation on the health of citizens. Reflects the work of T10.6.			

Deliverable D10.14 – De-risking assessment v1

Deliverable Number	D10.14	**Lead Beneficiary**	24 - HSRT
Deliverable Name	De-risking assessment v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10
Description			
Findings, insights, and strategic recommendations on successful de-risking strategies. Reflects the work of T10.7.			

Deliverable D10.15 – De-risking assessment final

Deliverable Number	D10.15	**Lead Beneficiary**	24 - HSRT
Deliverable Name	De-risking assessment final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	54	**Work Package No**	WP10

Deliverable D11.1 – Communication Dissemination v1

Description			
Findings, insights, and strategic recommendations on successful de-risking strategies. Reflects the work of T10.7.			
Deliverable Number	D11.1	**Lead Beneficiary**	24 - HSRT

Deliverable Name	Communication Dissemination v1		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP11
Description			
Communication and Dissemination plan and activities. Reflects the work of T11.1.			

Deliverable Number	D11.4	**Lead Beneficiary**	24 - HSRT
Deliverable Name	Communication Dissemination final		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP11

Description

Communication and Dissemination plan and activities. Reflects the work of T11.1.

Deliverable D11.2 – Communication Dissemination v2

Deliverable Number	D11.2	**Lead Beneficiary**	24 - HSRT
Deliverable Name	Communication Dissemination v2		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP11
Description			
Communication and Dissemination plan and activities. Reflects the work of T11.1.			

Deliverable D11.3 – Communication Dissemination v3

Deliverable Number	D11.3	**Lead Beneficiary**	24 - HSRT
Deliverable Name	Communication Dissemination v3		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP11
Description			
Communication and Dissemination plan and activities. Reflects the work of T11.1.			

Deliverable D11.4 – Communication Dissemination final Deliverable

D11.5 – Policy Recommendations v1

Deliverable Number	D11.5	**Lead Beneficiary**	16 - LiU
Deliverable Name	Policy Recommendations v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP11
Description			
Policy recommendations and barriers. Reflects the work of T11.2.			

Deliverable D11.6 – Policy Recommendations final

Description			
Exploitation Strategy and Business Models. Reflects the work of T11.3.			
Deliverable Number	D11.6	**Lead Beneficiary**	16 - LiU

Deliverable Name	Policy Recommendations final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	54	**Work Package No**	WP11
Description			
Policy recommendations and barriers. Reflects the work of T11.2.			

Deliverable D11.7 – Exploitation Strategy v1

Deliverable Number	D11.7	**Lead Beneficiary**	1 - ICCS
Deliverable Name	Exploitation Strategy v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP11
Description			
Exploitation Strategy and Business Models. Reflects the work of T11.3.			

Deliverable D11.8 – Exploitation Strategy v2

Deliverable Number	D11.8	**Lead Beneficiary**	24 - HSRT
Deliverable Name	Exploitation Strategy v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP11

Deliverable D11.9 – Exploitation Strategy v3

Deliverable Number	D11.9	**Lead Beneficiary**	24 - HSRT
Deliverable Name	Exploitation Strategy v3		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP11
Description			
Exploitation Strategy and Business Models. Reflects the work of T11.3.			

Deliverable D11.10 – Exploitation Strategy final

Description			
Commercialization of solutions and trading opportunities. Reflects the work of T11.4.			
Deliverable Number	D11.10	**Lead Beneficiary**	24 - HSRT

Deliverable Name	Exploitation Strategy final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP11
Description			
Exploitation Strategy and Business Models. Reflects the work of T11.3.			

Deliverable D11.11 – Market analysis v1

Deliverable Number	D11.11	**Lead Beneficiary**	19 - EMVIS
Deliverable Name	Market analysis v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP11
Description			
Commercialization of solutions and trading opportunities. Reflects the work of T11.4.s			

Deliverable D11.12 – Market analysis final

Deliverable Number	D11.12	**Lead Beneficiary**	19 - EMVIS
Deliverable Name	Market analysis final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP11

Deliverable D11.13 – Standardization activities v1

Deliverable Number	D11.13	**Lead Beneficiary**	43 - DS
Deliverable Name	Standardization activities v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP11
Description			
Standardization of novel technologies and activities. Reflects the work of T11.5.			

Deliverable D11.14 – Standardization activities final

Description			
Stakeholder engagement and Theseus bubbles. Reflects the work of T11.6.			
Deliverable Number	D11.14	**Lead Beneficiary**	43 - DS

Deliverable Name	Standardization activities final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	54	**Work Package No**	WP11
Description			
Standardization of novel technologies and activities. Reflects the work of T11.5.			

Deliverable D11.15 – Stakeholder Engagement v1

Deliverable Number	D11.15	**Lead Beneficiary**	45 - ICL
Deliverable Name	Stakeholder Engagement v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP11
Description			
Stakeholder engagement and Theseus bubbles. Reflects the work of T11.6.			

Deliverable D11.16 – Stakeholder Engagement v2

Deliverable Number	D11.16	**Lead Beneficiary**	45 - ICL
Deliverable Name	Stakeholder Engagement v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP11

Deliverable D11.17 – Stakeholder Engagement final

Deliverable Number	D11.17	**Lead Beneficiary**	45 - ICL
Deliverable Name	Stakeholder Engagement final		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	60	**Work Package No**	WP11
Description			
Stakeholder engagement and Theseus bubbles. Reflects the work of T11.6.			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Collaboration tool, visual id & Theseus handbook	WP1	1 - ICCS	Website and social tools, consortium collaboration tool are available	4
2	DM, D&C and Exploitation plans establishment	WP11	24 - HSRT	D11.1, D11.3 (first versions) are submitted	6
3	Completion of baseline phase	WP2	1 - ICCS	Baseline and the frameworks are set up (WP2 deliverables first versions submitted)	8
4	External Advisory Board established	WP1	1 - ICCS	EAB participation in a session, feedback documentation available	12
5	Development phase completed	WP2, WP1, WP9, WP3, WP8, WP11, WP7, WP5, WP6, WP10, WP4	3 - NTUA	Technologies designed, developed and ready for implementation/integration in the pilots.	24
6	Implementation phase completed	WP1, WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	3 - NTUA	Technologies implemented in the value chains (D5.1-5.5, D6.2-6.4, D7.2-7.5 updated, D8.1-8.5 first version submitted)	36
7	Validation phase completed	WP1, WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	3 - NTUA	Technologies are validated (D5.1-5.5, D6.2-6.4, D7.2-7.5 completed, D8.1-8.5 updated)	48

8	Replication & Upscaling phase	WP9	37 - RISE	Replication completed in SE and NO (D9.2 completed)	60
9	Final event & successful end of project	WP1	1 - ICCS	Final event takes place	60

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Milestones or deliverables delayed	WP1, WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	Continuously monitoring activities, detailed analysis in WP/Task implementation levels to ensured delays any are recognised early, with timely and effective corrective actions. In crucial cases, the Project Officer could be involved (M,M).
2	Underestimated or overspending resources	WP1, WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	Situations will be detected by WP1. ExB will analyse: i) re-arranging resources among partners; ii) committing further internal resources; iii) re-planning activities; iv) mobilise more skilled and efficient HR & Ad-hoc recovery plans (L,M).
3	One or more partners leaving consortium	WP1, WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	Given the good reputation of partners, with records in H2020/HE, possibility is unlikely. General Assembly will decide whether uncovered activities will carried-out by existing or new partners (L,M).
4	One of the partners (SME) goes bankrupt	WP1, WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	The economic uncertainty driven by the Covid-19 pandemic. To overcome a potential bankrupt, advanced allocation of funding will be carefully organised to minimise this risk, and the ExB will identify existing or new partners to fill the gap (L,M).

5	Missing/delayed equipment	WP5, WP6, WP7, WP4	Potential delays (due to unavailability or transport) may occur in the delivery/availability of equipment. Partners involved will monitor the supply chain to mitigate delays and provide alternative solutions (M,H).
6	Reluctance to provide or insufficient data	WP2, WP3, WP8, WP5, WP6, WP7, WP4	The CA will include and define the terms regarding access to data and existing knowledge. Secure mechanisms will be deployed for data sovereignty and sharing for data characterised as “sensitive” and “confidential” (M,M).
7	Deviations from expectation, invalid specifications	WP3, WP8, WP5, WP6, WP7, WP4	Iterative development will guarantee early detection of deviations and invalid specs. The integration of mature technologies minimises these chances. Technical Manager will monitor the design and development with technical partners, to avoid misalignment, ensure detailed

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			specifications of components, identify ineffective designs and take proactive measures (M,M).
8	Interoperability problems	WP3, WP4	Best practices will be used upon design of components, aligned with open and interoperability standards. Consortium possess deep knowledge of interoperability with technical partners to agree on decisions that may impact interoperability (L,L).
9	Technologies upscaling not expected performance	WP5	MA/CP: The GA will decide on alternative designs which can achieve a valid compromise between performance and costs targets. While exploring synergies among stakeholders, alternatives will be considered in case of need (M,M).
10	Security of pilots’ equipment/facilities	WP8, WP7	Security measures will be used to avoid vandalism, thefts, such as protection fences, alarm and cameras where necessary as well as selection of places that are guarded or closed during the night hours (M,M)

11	Optimising processes causing delays	WP3, WP5, WP6, WP7, WP4	The project will conduct research and analysis on processes and potential challenges during optimisation, and implement a robust management plan, including regular reviews, identify delays or setbacks early with involvement of experts (L,M).
12	Difficulties in using tools developed by the project	WP3	To facilitate the adoption, the tool developers we will ensure friendly user interface, accompanied by instructions and training according to the needs (L,H).
13	Disputes over ownership of IPR	WP2, WP3, WP9, WP8, WP5, WP6, WP7, WP11, WP10, WP4	IPR and access rights clauses will be included in CA which will be signed before the project starts. Further support to be provided by the legal experts involved in the project for IP conflict resolution. Ownership will be decided according to CA (M,L).
14	Competition and low users' acceptance	WP9, WP8, WP11, WP10	Market intelligence activities upon exploitation planning will ensure continuous monitoring of competition landscape to ensure the quality of results. Well defined cost-benefit validation. Co-creation with end-users to enhance acceptance (L,L).
15	New superior methodologies arise	WP3, WP5, WP6, WP7, WP4	Trends within research areas will be constantly monitored, and strong liaisons with relevant projects and initiatives will be established. The scientific choices will be reoriented, considering the latest advances to avoid approaches being obsolete (M,M).

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
16	Coordination with relevant activities	WP10	WP10 will support cooperation with relevant activities and projects, ensure proper links with different Working Groups and the rest H4C (L,M).

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	**Timing** (month)	**Location**	**Comments**
RV1	18	ICCS premises/hybrid	
RV2	36	ICCS premises/hybrid	
RV3	48	ICCS premises/hybrid	
RV4	60	ICCS premises/hybrid	

Version Number	Date	Change
V1.0	18/06/2024	First DoA Add figure 1 again - Add table of Contexts and proper corrections of headings within the whole document - Add Milestones' Leaders - Deliverables arrangement (split to different versions for different deadlines) – task 10.1 description has been minor revised in line with the ESR comments to better clarify flows as well as the Deliverable 10.1 – Deliverables 4.2-4.6 adjusted from material level to maturity level per version, to be more efficient - Deliverables 7.7 (previously 7.9) and 10.4 adjusted to have two versions instead of three. – Pilot Deliverables adjusted., Section 4 Added, Short Interim Management Reports added to Deliverables, Section 3.2.1 Added with the Associated Partners
V2.0	03/07/2024	Updates corresponding on the comments from UC for GAP provided at 14/6/2024– SO1 and SO2 updated page 2,3- End users/industries highlighted in Section 1.1.2 and IP5. Page 11 – Clarification added for Sludge valorization. Page 9. Table was created in section 1.1.1 as a reference to the methodologies to be used. Page 5- Figure 8 Overall Methodology in Scientific Methodology was updated to display the integration of Innovation Pillars into methodology. Page 13- 2.1.4 Potential barriers and Mitigation Measures were enhanced. Page 25-26 - Dissemination & Communication Measures were enhanced to refer specific entities page 27-28 – Task 3.1 was updated to highlight the data sources Part A - Figure 12 Theseus PERT chart and text for WP interactions were inserted to display interlinkages as well as Interdependencies between work packages and activities Page 35 - 2.1.3 Quantification of potential long-term impact Theseus table was updated to highlight the unfolding of impacts to EU. Page 25 – Consortium table was updated to justify the role of associate partners. Page 37 – Deliverable table was updated to include the relevant tasks per deliverable. Part A. – Footnote added in assumption for clarification. Page 25. – Soa was updated to link with previous and ongoing projects to enhance the innovation to be developed. Pages 6-10
V3.0	19/07/2024	Task 9.3 has been updated to include the APs (Part A). Updated section 3.1h for NTUA.
V4.0	23/07/2024	Section 3.2.1, PAGE 38-39, was updated including: 1)subcontracting of TVP clarifications 2)APs role clarifications

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TITLE: A First of a kind Hub for circularity demonstrator for Attica and peripheral regions; Topic addressed: HORIZON-CL4-2024-TWIN-TRANSITION-01-38 Hubs for circularity for industrialised urban peripheral areas (Processes4Planet partnership) #@APP-FORM-HERIAIA@#

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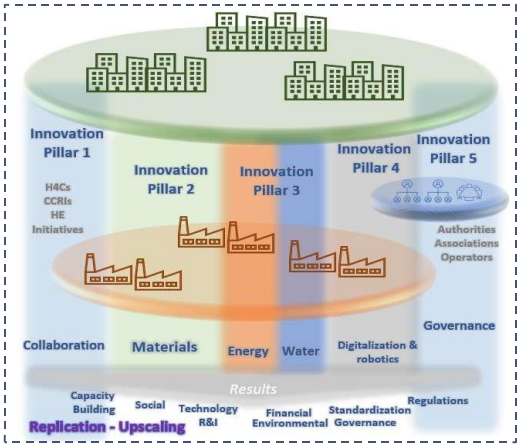

1 Excellence #@REL-EVA-RE@#

1.1 Objectives and ambition #@PRJ-OBJ-PO@#

Industrial-Urban symbiosis (I-US) involves municipalities and regions cooperating with industries on the management of resources, waste, energy, water, infrastructures, services, transportation and networks. I-US harnesses circularity potential through sector integration. It offers a novel approach for regional eco-industrial development[1], supporting urban transition towards sustainability, offering effective ways to reduce resource consumption and emissions in the city[2] with tangible results[3], addressing many challenges and offering future opportunities[4,5]. That is why I-US has been recognized by the process industries as a key strategy for achieving carbon neutrality by 2050 and is one of 4 key pillars in the roadmap by ASPIRE[6]. To accelerate circular transition a wide range of related actions should be implemented. Thus, cooperation of stakeholders at all levels is required and setting up First-of-a-kind (FOAK) large scale plants integrating I-US cases is a key step for proving and scaling up I-US solutions in regional and interregional level. This inter-disciplinary and cross-sectoral character is implemented by Hubs4Circularity (H4C) in a FOAK approach, as key instruments for delivering territorial I-US solutions in Europe by Processes4Planed partnership[7]. Theseus will develop a systemic I-US solution, a FOAK H4C demonstrating solutions in semiindustrial scale and involving authorities, research operators, companies, and industries from Attica (incl. Athens metropolitan area[7] -Greek capital, 68 municipalities, 2.928 km^2, ~3,7 million inhabitants ~35% of national- several industrial and semi-urban areas) and surrounding regions. Theseus will validate new or existing technologies and develop solutions for water, energy and materials based on regional needs (Innovation Pillar (IP) 2 , 3) and digital/robotic technologies (IP4). The hub will operate under innovative governance and cooperation models of involved stakeholders, to enable pilots, upscale solutions, exploit outcomes, attract further companies and expand existing ecosystem (IP1,5). Water, energy and material flows were targeted selected, being widely spread, having high potential all over Europe and included in EU/P4P strategies. The overall systemic, based on regional science approach and IP1,IP5 activities will ensure strong insights to smoothly replicate Theseus solutions to other EU regions. Several existing innovations from H4C, I-US and Circular Cities and Regions Initiative (CCRI) projects will be used, retrofitted and expanded. All the activities will be supported transversely by digital solutions (IP4).

The systemic solution of Theseus is expected to have significant impact on the region, while metaphorically recalling the mythical figure of Theseus, king and great reformer of Athens.

1.1.1 Vision and Special Objectives

SO1 To initiate, boost and fast track systemic transformation in Attica and peripheral regions by deploying circular solutions with minimal

environmental, social and financial impact through implementation of I-US (including flows of materials, energy, water) at FOAK plant scale towards the first real hub for circularity (H4C) in Greece, contributing to deploying climate neutral solutions, closing circular loops and achieving global leadership to related solutions (P4P objectives).

[1] 2020, Yuli Bian et al., A Sectoral Eco-Eciency Analysis on Urban-Industrial Symbiosis
[2] 2020, Sun Lu et al., Energy-saving and carbon emission reduction effect of urban-industrial symbiosis implementation with feasibility analysis in the city
[3] 2020, Chunyang Lu et al., Uncovering the benefits of integrating industrial symbiosis and urban symbiosis targeting a resource-dependent city: A case study of Yongcheng, China
[4] 2019, M.A. Butturi et al., Renewable energy in eco-industrial parks and urban-industrial symbiosis: A literature review and a conceptual synthesis
[5] 2018, Luca Fraccascia, Industrial symbiosis and urban areas: a systematic literature review and future research directions
[6] Processes4Planet 2050 roadmap https://www.spire2030.eu/sites/default/files/pressoffice/publication/processes4planet_2050_roadmap_jan2021.pdf [7] https://www.aspire2050.eu/p4planet/about-p4planet
[7] https://www.emta.com/network/athens/identification-of-the-metropolitan-area/

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Theseus will embrace multiple aspects of ecosystem to foster circular transition within Attica and other regions, based on five innovation pillars (IPs) assuring the transformation is addressed by a systemic holistic approach. IPs embrace technical innovation and process-based approach, focusing on symbiotic flows and closing the loops of water, energy (IP3), materials (IP2), and on digitalization, robotics (IP4), embedded by governance (IP5) where key role is played by Attica Region and Municipalities, enhanced through collaboration (IP1). IP1 will enhance cross-sectoral collaboration, replication and collaboration with HE projects e.g. W2W, DigInTraCE, H4Cs and ECoP e.g. IS2H4C, REDOL, and CCRIs e.g. CSSBoost. IP1 aims to raise social citizens awareness using novel engaging technologies like AR. The systemic transformation within Attica will be complemented by assessing the impact at flow level and hub overall in terms of potential benefits on the health of the citizens and roll-out risks analysis. The approach proposed will contribute to EU Cities Mission, by helping Athens, one of mission's cities, becoming climate neutral and paving H4Cs, enabling other cities to follow.
Relation to the call: "A systemic solution for a H4C including technological and non-technological as well as regional and interregional aspects: closing circularity loops for mixed/combined materials streams based on upcycling and process-based approach to a range of recycling solutions including conversion and downstream, complex multi-material streams, valorisation of waste streams (urban mining); Citizens living in cities will benefit from a healthier environment through industrial/urban symbiosis by lowering emissions through circular and renewable energy sources and waste reduction; … to support the goals of the smart cities mission by contributing to a healthier urban industrial symbiosis through waste reduction; Use established reporting methodologies… interact with the ECoP for support… on technological and non-technological issues"
KPI1.1 At least 5 examples of technological innovations on products, processes, or robotics implemented; KPI 1.2 Establishment of at least 6 cross-sectoral collaborations, replication related collaborations, collaborations with other projects; KPI 1.3 Health benefits of hub implementation identified and impact assessed at least 6 flows; KPI 1.4 At least 4 de-risking strategies presented. KERs: ALL

SO2 To align the Strategic Masterplan on Circularity of resource management with P4P objectives, thus deploying climate neutral solutions, closing circular loops and achieving global leadership to related solutions, following H4C principles and through the implementation of the 9 proposed circular flows.
Theseus will design and implement closing of circular loops involving 9 material flows connecting different manufacturing sectors and P4P industries (cement, chemicals, non-ferrous metals, water). It will propose full technological schemes from secondary resource collection up to the validation by the final user. Secondary resources coming from combined and mixed sources: textiles, Municipal Solid Waste (MSW), glass, Construction & demolition waste (CDW) and marbles are collected & separated/sorted. This is followed by processing, to be used as feedstock for creating circular products to be tested by final users from construction, packaging, glass, metals sectors. It will evaluate the circularity of resource/material flows and value (intrinsic circularity and consequential sustainability) creation. This novel methodology will enable the investigation of virgin materials substitution and waste reduction, degree of circularity, quality changes (e.g. downcycling, upcycling) and environmental, economic, and social impacts (e.g. climate neutrality). Theseus will ensure its solutions avoid false redirection of secondary resources (displace virgin material in one sector but hinder another) to abide by EU recycling targets or create negative effects. Material Flow Analysis (MFA) will support to identify flows and stocks of materials in TM-WEMW[8] model, validating trade-offs and avoid burden shifting between nexus and value chains. Those activities will be enhanced through the Strategic Master Plan on CE, which will include pricing, systems design, and regulatory framework for all the Hub sectors and will be developed by local authorities and scientific partners. The activities will support the implementation of the Regional Plan on Waste management which has been developed by the Attika Region.

[8] Territorial Metabolism Water-Energy-Materials-Waste

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Relation to the call: "Proposals can address either materials, water or energy, or a combination of those. Management and processing of secondary resource streams through e.g., collection, disassembly, sorting, purification, concentration, recycling (including but not limited to chemical, metallurgical, or bio-based resources), logistics and trading for their valorisation for the use as feedstock for other plants and companies across sectors and/or across value chains. Connections with manufacturing industries are expected…"

KPI2.1 At least 6 technological schemes from collection to use established; KPI2.2 At least 5 circular products tested by end users; KERs: 4-9

SO3 To Demonstrate the valorisation of reclaimed water reuse at various scales and rainwater harvesting leading to the reduction of freshwater consumption. To implement decarbonization strategies to outdated district heating (DH) systems contributing closing energy loops. To demonstrate innovative sewage sludge thermal treatment for energy and materials recovery through fast pyrolysis.

Theseus expanding tested solutions from previous projects (HYRDOUSA, TRINEFLEX, IMPETUS) will implement centralized & decentralized solutions for treatment and valorisation of non-conventional water sources (wastewater, rain/storm). Novel decentralised MBR sewer mining technologies will be deployed in the sewer network of Athens to irrigate green spaces, reducing freshwater demand and heat stress especially during heatwaves. Centralised reclaimed reuse of agricultural irrigation will be integrated with fertigation process leading to increased soil biodiversity and good plant growth with minimal chemical fertilizers. Sewage sludge produced from centralized WWTPs will be transformed via innovative fast pyrolysis process to high calorific value gas and biochar to be valorised. Also, nature-based solutions (NBS - bioswales) for rain/stormwater valorisation for urban and industrial use within Athens, and standard rainwater harvesting solutions to support urban farming practices in households and schools will be applied. Different decarbonization strategies of DH will be tested on peripheral scale

and studied theoretically to reach conclusions for future implementation in Attica. An innovative solar thermal technology for DH and cooling of 2 buildings in Ptolemaida town will be demonstrated and the potential connection to city's central DH network will be assessed. Green H2 use for producing heat via fuel cell system will also be demonstrated in the local CERTH building. The introduction of other renewable energy sources (locally sourced bio-waste/ biomass) will be studied for city network, currently exploiting waste heat from fossil fuel (decarbonization til 2028). All approaches will be combined with replicators to produce strategy for other EU regions and upscaling in Attica.

Relation to the call: "Consider when applicable the co-development of industrial decarbonization strategies with urban district heating networks, i.e., based on a socio-economic optimum in the cascading re-use of waste heat and using the district heating network to supply low temperature process heat…; Proposals can address either materials, water or energy, or a combination of those…; Reduce the freshwater consumption of the urban area by 50%, and re-use 90% of the solid waste generated by the water treatment".

KPI 3.1 Decentralized reclaimed water reuse solutions for irrigation of urban green spaces (sewer mining); KPI 3.2 Centralized reclaimed water reuse solution in the Eastern Attika serviced by 3 medium scale WWTPs for agricultural irrigation (~3.6 million p.e.); KPI 3.3: Rainwater harvesting solutions including NBS to valorise rainwater/stormwater through urban farming activities and urban use; KPI 3.4 Sewage sludge valorisation through i)fast pyrolysis to produce biochar, biooil and gas and through ii) torrefaction to achieve energy densification KPI 3.4 Co-development of 3 decarbonization strategies for Ptolemaida DH operator KPI 3.5 Real application of renewable

sources for heating/cooling needs of 2 buildings considering CE aspects (recyclable materials for solar thermal systems etc.). KERs: 10-14

SO4 To facilitate the activities within the Hub and foster circularity through digitalization and robotics. To further enhance the transition and foster implementation of H4C concept through diverse digital technologies at practical and theoretical level.

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Theseus will develop advanced sorting & separation technologies with digital tools and robotics to enhance flows' quality and purity and reuse potential. Sorting will be applied on textiles and advanced separation in urban wastes, plastics, metals, glass. Smart bin optimization will be applied focusing on space limitations, typical in Attica, improving the separation capabilities for biowaste/organic from MSW. Theseus will introduce multi-sensor classifiers and multiple-robot setups to sorting lines. Human-robot collaboration will be fostered through MR techniques. I-US Attica digital Marketplace will be developed, a digital community hub, engaging companies and industries from Attica and surrounding regions, supporting secure data exchange and features for traceability, DPPs, MFA, matchmaking, LCA, AI prediction, optimization, improved logistics (and reverse), scheduling and financial tools. Dynamic DPPs (T3.3) for novel circular products will be developed, adjustable and scalable to other sectors, for informed decision making and opportunities identified by matchmaking features (T3.2). A territorial visualization GIS tool (T3.1) will be developed for dynamic symbiotic maps in the region while PRLM[10] (T3.4) will predict the material flows providing insights into bottlenecks and opportunities for decision-making on synergies. TM-WEMW nexus modelling tool (WP9) will also be integrated to analyze territorial resources interactions, providing valuable insights to planners and decision makers by integrating scenario planning capabilities considering socio-demographic, economic, policy, population, housing, and behavioral factors influencing Theseus solution impacts. Theseus will develop a Replication Roadmap Tool , which will improve identifying opportunities and decision-making for replicators. Also, computation of pertinent financial metrics and suitable economic indicators will facilitate the estimation and mitigation of economic impacts and associated risks on specific material flow processes with potential for improved economic efficiency.
Relation to the call: "Digital tool, recycling and sensor-based waste sorting, modelling tools, including material passport and information on material streams, as basis for resource management towards fully integrated LCA and Material Flow Analysis MFA (on diverse levels) and for creating transparency and matchmaking opportunities across hubs. Prepare for tagging/matrix for complex consumer products and innovative approach to end-of life materials. Establish IT infrastructures and tools that provide a secure basis for the integrated management and the preservation of confidentiality of sensitive data, it might not be in the same location as the demonstrator and serve the needs of multiple hubs".
KPI 4.1 To develop H4C Marketplace with advanced matchmaking features, secure data exchange and at least 50 registered users by the end of project; KPI 4.2 At least 5 advanced sorting lines developed, using digital tools, human/robot collaboration via Extended Reality (XR) and data analysis; KPI 4.3 At least 5 material parameters added in DPP continuously updated along value chain; KPI 4.4 To validate the sustainability of sorting mechanisms based on techno-economic, LCA, MFA; KPI 4.5 To develop a predictive resource logistics module for identifying potential business opportunities, risks or bottlenecks KPI 4.6 At least 3 smart bin placement optimization proposals produced that applicable to the region of Attica by the end of the project. KERs: 15-21

SO5 To collaborate with local actors to create conditions for successful replication. Local authorities' representatives of replicators will be closely interacting with Theseus and following project activities, preparing the ground for implementation of the activities in their regions.
Theseus will establish replicators in Sweden, Norway: 1)Malmö, IS experienced, with special focus on topics related to the local Resource Hub to foster regional cooperation and innovation through technologies and social processes, applying a novel collaborative system modelling approach by WECOS[9] project within same region, adapted for Malmö; 2)MO, to establish a regional I-US reference group, enabling a systemic approach and policy innovation during the project and after. The replications will enable mutual learning between the three locations in an iterative process, focused on promoting H4C through main hub's innovations and policies. T9.3, T9.4 of replicators and collaboration with CCRI, will provide conclusions and recommendations.
Relation to the call: "Plan in detail the replication and adaption of the concept, including the simulation and the business case and exploitation strategy of the First of a Kind hubs, in two to three alternative locations in close cooperation with the relevant local actors. The replication cases should be part

[9] https://mesam.se/projekt/wecos-waste-to-energy-wte-in-swedens-circular-economy-collaborative-system-dynamics-modelling

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[10] Predictive resource logistics module

of the proposal. Include local and regional authorities in an active collaboration to create favourable and coherent place-based framework conditions".
KPI 5.1 Establish 1 replication case in Sweden and 1 in Norway; KPI 5.2 Establishment of a regional I-US policy arena, hosting 3 policy workshops; KPI 5.3 Trial study of at least 3 innovations from the main Attica hub to replicators; KPI 5.4 Replication Roadmap Tool tested by at least 3 new potential replication areas. KERs: 1-3

SO6 To facilitate implementation of the hub in the region and beyond, and to assure better acceptance of Theseus solutions, to enhance collaboration, to assure participative approach and knowledge exchange as well as future launching to the market.
Theseus will include activities to enhance acceptance of its concepts, concerning active participation, education and awareness raising. Approaches for collaboration between diverse stakeholders for effective co-creation (T2.2, T11.6) to address barriers and identify opportunities within the local context. Theseus will provide tailored training and mentoring for upskilling decision-makers, employees, and citizens including workshops for co-creating and validating educational materials (T10.5). Special attention will be given to increasing social awareness towards sustainability and acceptance., thus AR technology will be used with gamification and educational functionalities (T10.4). Next, activities related to a set of guidelines and recommendations will be developed based on experiences and standardization. Common hub circularity guidelines based on activities performed by A.SPIRE to align ongoing P4P projects in terms of technologies proposed by H4Cs operating across Europe (T9.1), will be supported by Regional and Intraregional development recommendations and action plans related to integration of the Theseus H4C actions in EU, National, Regional and Municipal Plans (T2.4), Theseus will also provide policy guidelines based on MFA, health impact (T10.6) and de-risking (T11.2). Standardization activities (T11.5) will increase acceptance and trust for future launching on the market linking with the final group of SO6 activities, which are related to market analysis. In this context outcomes will be aligned with market demands to propose exploitation and commercialization strategy for feasible products and business models based on best practices (T11.3). Market feasibility and potential pathways for commercialization of solutions beyond project duration will be studied (T11.4). The gender perspective and inclusiveness is considered throughout all project activities, ensured by WP1 and as described in 1.2.5.
Relation to the call: "Favour participative management with the local community and study the evolution of the social impact of the hub, whilst also considering gender perspective and inclusiveness; Implement a social innovation action involving at least one of the local community actors and, additional actions to facilitate relations and to involve the local community actors e.g., exchanging knowledge with the educational establishments and developing flexible learning resources; Include a plan to extend the hub to additional players parties…"
KPI 6.1 Diss & Comm and Exploitation KPIs fulfilled as per 2.2; KP1 6.2 Minimum 2 co-creation activities; KPI 6.3 Minimum 4 trainings performed; KPI 6.4 AR application shared with > 400 end-users to validate >4 targeted campaigns with immersive content achieving>75% acceptance of solutions assessed by >200 assessment forms; KPI 6.5 Minimum 3 standardization, activities performed. KERs: 1-3, 22-24

In summary, the below concepts and methodologies, grouped by their common action field, will be developed to serve the 6 Theseus SOs:

SO	Methodologies	WPs / Tasks
1,5,6	Collaboration, Co-creative Activities, Stakeholder Engagement & Participation (related to social Innovations)	T2.2, T11.6
1,5,6	Social Awareness via AR app and gamification, XR for human-robot interaction (related to social Innovations)	T10.4, T4.8

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1,6	Training and mentoring for upskilling (related to social Innovations)	T10.5
1,6	Governance scheme & capacity / RIC building (related to social Innovations)	T2.1, T2.2
1,6	Regional Science principles (related to social Innovations)	T2.4
1,6	Replication and mutual learning between main hub and replicators (related to social Innovations)	WP9
2,4	Material Flow Analysis & characterization (and related more focused sub-analysis)	T2.3, T2.5 (T4.1, T4.6, T4.7, T7.1)
2,4	Digital modules supporting the implementation and optimization of the symbiotic flows: GIS map, DPP, Matchmaking, Predictive resource logistics module	WP3
2,4,6	Territorial Metabolism Water-Energy-Materials-Waste Nexus Tool	T10.2
2,4, 6	Climate Change integrated assessment models	T10.1
2,4, 6	Cities Climate models and evaluation of environmental and economic impacts via LCA, Dynamic LCA , LCC	T10.3
2,4, 6	Impact Assessments of Flows	T8.7
1,6	Commercialization strategy, business models and market analysis	T11.3, T11.4
1,6	Health Benefits and Impact Analysis (social Innovations)	T10.6
1,6	Risk analysis & De-Risking activities	T10.7, T11.2
1,6	Standardization activities, Policy recommendations	T11.5, T11.2

1.1.2 Ambition

Theseus ambition is summarized in the below 5 Innovation Pillars (IPs). The related Innovations to be developed per IP so the project will achieve the main and the special objectives, are described below.

2.1.2.1. Innovations per THESEUS Innovation Pillars & relevant Tasks/KERs:

A)Collaboration between different sites, replicators, regions, existing initiatives/projects, stakeholders & society

SoA: H4C Community of Practice (CoP) offers network, knowledge and assessment tools for H4C & I-US to cover the gap between different sites and initiatives. Results are not systematically exploited by next projects resulting in separated or duplicated efforts. Regions successful in I-US are characterized by ambitious policy, alignment, proximity between diverse organizations, but also by internal drivers: economic benefits, environmental responsibility, stakeholder pressure for sustainable practices and regulatory restrictions e.g. Kalundborg. A known challenge is public and private sectors alignment, crucial for policy development and I-US innovations. MO (replicator) has seen innovative successful collaboration between public, private and research sectors based on smart specialisation and industrial cluster models e.g. ACT (host KPH, 106orgs) facilitating interactions of energy, water, and material between. Helsingborg (Malmo replicator area) has a history of I-US participatory processes for Waste/Energy in WECOS[12].

Beyond: H4C-CoP ideas enhanced by A.SPIRE into a methodology for knowledge exchange, a digital I-US community hub will be developed, including marketplace and repository for knowledge exchange, to support effective collaboration of all the stakeholders and different areas and integrate, enhance and validate solutions (e.g. digital or physical testbeds). As Theseus baseline, I-US solutions from existing H4Cs (REDOL, IS2H4C) and CCRIs (CSSBoost) and upcycling technologies from W2W, DigInTraCE projects will be enhanced (by the common partners) and used. Theseus will provide proper tools to overcome the gap between different sites, public and private sectors, using experience gained from replication areas practices, from ACT and WECOS project. Theseus will adapt these existing strategies and align them to Attica needs, creating natural and organic symbiosis (I-US). ACT practices will also support reaching final users and stakeholders through activities in replicator and main hub. Based on this mutual learning between main hub and replicators, a systemic related approach and a social innovation will

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be formed during the project, enabling an iterative interactive process, to transfer and promote innovations, practices and policies between the 3 different areas. T2.2 participation, T10.4 AR and T11.6 engagement will focus in reaching final users and engage stakeholders. Implemented in T2.2, WP9, WP10, WP11, KERs 1-3. Synergies with DigInTraCE, W2W, IS2H4C, REDOL, CSSBoost projects

B)Governance Model towards sustainable Regional, Intraregional and Interregional development

Several Governance Models and management approaches are developed at regional, municipal or project level to facilitate symbiosis and related operational and business models. Synergies are developed at regional level, but they are not long term, neither coordinated to multiply their impact, or aligned with other areas, national actions and regional, national or EU Plans. Systems dynamics (SD) models have been widely applied in modelling multi-dimensional socio-economic and environmental issues of regional/urban complex systems[10]. Also, Regional Institutional Capacity (RIC) is recognized as a dynamic force for I-US enabling diverse regional actors (businesses, public authorities, civil society, financiers) to build collective (relational, knowledge, mobilization) capacity to address issues of common concern[11].

Theseus will develop a Hybrid Governance Model integrating both regional development approaches and project management participatory and co-creation aspects, in T2.1, T2.2. An analysis of local /national strategies, along with regional science principles, will ensure that there is interregional interaction and cooperation, multiplying the impacts of interventions, intra and interregional. Circular economy roadmaps (CERMs)[12] will be developed, as strategic blueprints including actions, initiatives, timelines and KPIs to measure impacts for CE transition towards sustainability (using innovative ML approaches to connect I-US performance with overall regional resilience[13]) focused on resource optimization, business value, exploitation, standardization and policy recommendation. RIC will be enabled, while SD models approaches from WECOS project will be also exploited. Implemented in WP1, WP2, WP10, WP11 KERs 22-24. Synergies with DigInTraCE, W2W, IS2H4C, REDOL, CSSBoost projects

Collaboration Framework Guidelines, AR app and Replication Tool KERs 1-3 SRL[17] 5 SRL 7

Governance & Roadmaps Governance Model, CERMs, Policy and Standards KERs 22-23 SRL 5 SRL 7

A) Marketplace, Matchmaking & Prediction Modules, DPPs & Secure data exchange channels

I-US involves resource-sharing to enhance sustainability and efficiency. Digital platforms are pivotal in this by connecting businesses, identifying synergies, and managing resource exchanges efficiently. They are often cloud based ecosystems where industries interact and share information in real time, supporting traceability, product and material passports. Advanced analytics and AI on large datasets can identify opportunities, with ML predict symbiotic synergies. Blockchain is explored for transparent and trusted transactions. IoT systems collect real-

[12] https://mesam.se/projekt/wecos-waste-to-energy-wte-in-swedens-circular-economy-collaborative-system-dynamics-modelling/

time data to identify surplus resources and monitor environmental impacts. Beyond: Theseus will develop innovative modules: sophisticated matchmaking for I-US based on existing efforts (Accelwater, Coralis, Sharebox, IS2H4C), dynamic DPPs (DigInTraCE), prediction, improved logistics, PRLM, scheduling, financial tools, GIS on MFA, LCA and TM-WEMW approaches. The modules will be integrated in the I-US Attica digital Marketplace. The platform (by IRIS) will include Natural Language Processing (NLP) focused on interaction of computers with human language. NLP systems often use a combination of linguistic rules and ML algorithms to process and understand language. Recent advances in Deep Learning (DL), neural networks, and large-scale language models have significantly improved their performance and accuracy. Infrastructures for secure data exchange will be developed. Implemented in WP3 KERs 18-21. Synergies with Sharebox, Coralis, Accelwater, IS2H4C projects

B) Sorting and Separation Mechanisms for waste upcycling (combined with P5 innovations)

Current waste sorting advancements integrate AI sensing and robotics for efficient automation. Sensor combinations of Red-Green-Blue-Depth (RGBD), Near-Infrared (NIR), and hyperspectral imaging (HSI) cameras, achieve categorization accuracy nearing 99.95%[14]. Robots excel in material

[10] https://www.sciencedirect.com/science/article/abs/pii/S0195925523003220
[11] https://www.diva-portal.org/smash/get/diva2:1513638/FULLTEXT02.pdf
[12] https://www.sciencedirect.com/science/article/pii/S0959652623042245
[13] https://www.researchgate.net/publication/369452237_Prediction_models_and_testing_of_resilience_in_regions [17] https://data.europa.eu/doi/10.2777/381211
[14] https://doi.org/10.1007/s10311-023-01604-3 https://doi.org/10.1016/j.wasman.2019.06.035

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handling with commercial solutions achieving over 3000 moves per hour per robot[15]. However, challenges remain e.g. high costs due to absence of fully commercialized for automated operations, research focused on single-robot instead of wide scalability[20].

Beyond: Theseus will introduce multi-sensor classifiers and multiple-robot setups to sorting lines to increase adaptability and automation. By developing tailored sub-systems for material recovery and seamlessly integrating them into existing lines, their capacity and efficiency will be amplified in a flexible manner. ICCS will develop, train and test new robust datasets in real conditions, to produce digital tools for mass flow analysis of urban waste, identify and quantify each distinct valorisation chain.

B2) Sorting of Textiles (Textiles T8.2) SoA: 15-20% of textiles are sorted within EU giving textile waste significant role in pollution and climate change. Sorted textiles have diverse pathways such as upcycling, reclaiming for reuse in production or repurposing for e.g. furniture, agriculture. Current sorting practices mainly involve manual labour, lacking automation for efficient categorization by fibre composition, sizing, and pre-processing on recyclers' specifications. Challenges persist due to related needs (e.g. type, quality, condition) in a single process. Post-sorting, textile waste requires mechanical, chemical, or biological processing. The optimization of these processes, currently done ad hoc, can be challenging for continuous processes with varied waste compositions.

Theseus aims to automated AI classification and robotic material handling line to accommodate the diverse range of disposed garments for efficient categorization. This will accelerate sorting procedures and enable the recovery of textiles by fibre type, differentiating between thermoplastic, non-thermoplastic, and mixed. Using an XR application, a human operator will collaborate with robots by optimizing the handling of complex textiles (such as unfolding). Theseus will also develop AI algorithms to optimize recycling and degradation by relating critical process parameters to material properties.

Current practices for unsorted MSW leverage machine vision to effectively sort materials such as plastics, paper, metals and glass with accuracy rates often exceeding 90%[16]. Nonetheless, due to materials diversity and inherent lack of industry-oriented datasets, difficulties persist in

Beyond: Theseus will develop extensive datasets for precise waste identification in mixed streams, distinguishing contaminants and multi material items. Modular, multi-sensory classification systems powered by AI, adjusted to the individual needs of existing lines will be deployed to recover plastics, metal, paper, and textiles from MSW, extending to distinguishing specific plastics, such as PET, PE, and bio-based polymers. Additionally, multi-robot cells will facilitate automated handling. A mass flow analysis of circulated waste will identify optimal placement locations for smart waste bins in West Attica, enhancing material recovery rates using tailor-made integrated separation. The project will collaborate with W2W & DigInTraCE projects for shared advancements. Human operator training during sorting will be optimized with (XR) improving accuracy and speed of separation for more efficient and effective management of MSW

B4) Recovery of copper scrap (CWD/Industrial Wastes T8.3) SoA: metals from CDW involves automated processes accurately in cases of contamination or complex packaging.

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[15] https://packagingeurope.com/news/ai-robotic-picking-technology-aspires-to-optimise-recycling-operations-and-address-labour-shortages/10184.article [20] https://doi.org/10.1007/s12008-023-01320-w

[16] https://doi.org/10.1016/j.wasman.2022.02.009.

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that employ magnetic and eddy current configurations. Latest advances demonstrate use of visual recognition systems coupled with robotics, with instances exhibiting capacity to handle waste up to 30kg in up to 6000 picks per hour[17]. However, difficulties remain in accurately sorting metals by grade and detecting impurities within the stream and differentiating between similar metals or identifying hybrid materials.

Theseus will develop an adjustable dedicated sorting line, expanding W2W project pilot to extract metals from mixed CDW streams, prioritizing efficient scrap retrieval by precise grade-based categorization. Integrating adaptable AI-based classification and specialized robot manipulators aims to boost sorting accuracy and enhance metal recovery efficiency within CDW streams in a modular fashion without hindering the functionality of current processing facilities.

Glass bottles are approximately 18.5% of Europe's annual packaging waste. In 2020, recycling rates for glass have been promising, over 75% to undergo recycling[18]. Optical imaging sorting technologies such as vision, and spectrometry-based like XRF exhibit good accuracy rates in glass, effectively exceeding 85%, for bottle sorting lines, but a notable challenge lies in identifying broken or cracked glass, crucial for negative sorting[19, 20].

Beyond: Theseus will establish a tailor-made, automated

glass sorting line to manage glass waste originating from Athenian Brewery. The modular sorter will support the current bottle handling line without negative impacts and its function will centre around recycling used bottles and broken glass using a multi-sensing approach that employs data fusion from machine vision and hyperspectral imaging. This initiative also holds potential synergies with W2W, within the context of CDW management. B1)-B5) Implemented in Task WP4 KERs 15-18. Synergies with DigInTraCE, REDOL, W2W, ACT UK projects

Developments	Key Advantage & Progress	Start TRL	Foreseen End TRL
Urban sorters	Automated textiles separation and sorting for mixed waste & plastics and XR app	KERs 14-18	TRL5 7
Industrial sorters	Glass/copper & metals sorting system	KERs 17	TRL5 7

innovations) contribute

A)

Solar thermal technology for District Heating (DH) has been evolving to provide efficient and sustainable solutions for community needs. Integration of solar thermal systems with existing DH allows a gradual transition to renewable energy (decarbonisation). The High vacuum Flat Panels (HVFP) solar thermal technology is a highly innovative non-concentrated technology which exhibits minimal thermal losses, no degradation over time, and exceptional lifetime. TVP has developed, patented and SolarKeyMark certified first generation of HVFP (branded as 'MT-Power'). Several technological innovations like the metal-to-glass seal and self-regenerative getter pump ensure that high vacuum is retained over 25 years, and make panels nearly 100% recyclable. HVFPs, capturing direct and diffuse solar irradiation, outperform (can exceed 50%) any other collector in efficiency at low-to-medium temperature range.[21] HVFPs can deliver relatively high temperature (80-110C) during winter. TVP has developed the next generation of HVFPs (branded as "HT-Power", up to 250C, stagnation >350C)[22] thanks

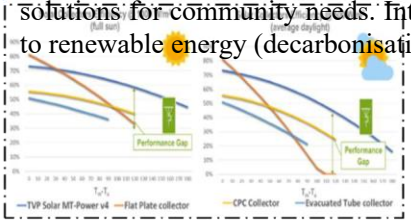

Figure 1 Certified data comparison

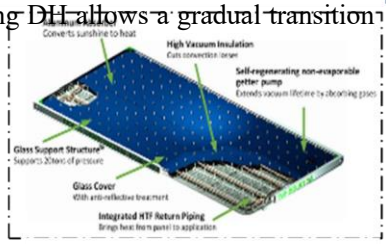

Figure 2 TVP's HVFP solar thermal collector

[17] Zen Robotics, https://www.terex.com/zenrobotics/
[18] https://www.statista.com/statistics/1258851/glass-recycling-rate-in-europe/
[19] https://doi.org/10.3389/frsus.2022.901885
[20] Redwave X-Stream Glass https://redwave.com/en/loesungen/recycling/glass
[21] https://doi.org/10.1364/OE.26.00A480
[22] Overall absorber performances: efficiency 60% at 250°C (current value 45% at 250°C); 25 years of operation with <5% degradation. Relevant publications: [28] https://www.sciencedirect.com/science/article/pii/S0927024822001702?via%3Dihub

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to a new Selective Solar Absorber (SSA)[28]. Related RTD has been carried out under ESSteam[23] (60m^2 pilot, TRL6 by 6/24). The new panel will provide highly efficient solar cooling coupled with double-effect absorption chillers and can generate low-pressure steam for industrial processes. Solar systems with 1st gen TVP's are used in DHs in Switzerland (800m^2 2020), Netherlands (48.000m^2 2024); Germany (6.000m^2 2023). Industrial applications are running in Brazil, France, Italy, while solar thermal cooling in Spain and Kuwait.

Beyond: Theseus will demonstrate a SDHC solution of 2nd gen HVFPs in dual mode: heating during winter (Oct-Apr) and cooling during summer for carbon-free tertiary, industrial / residential applications in line with local decarbonisation strategies. TVP will improve SDHC 2nd generation panels: (i) Model, simulate and assess potential configurations (incl. cold storage) for optimal heating & cooling. (ii) Selection of the suitable (double effect) absorption chiller; (iii) Develop appropriate control systems for the dual operation, dynamically based on weather and performance; (iv) Study replication cases for cascade scenarios with or without waste heat; (v) Elaborate on Heat-as-a-Service (HaaS) and Cold-as-a-Service (CaaS) business models towards replication and scale-up to maximize socio-economic and environmental impact; achieving a Levelised Cost of Heat 40-60 €/MWh.[24][25]. Implemented in WP6, Task 8.4 producing KERs 13-14. Synergies with EVELIXIA Coralis projects

B) Centralized and decentralized management and valorization of non-conventional water sources

Significant strides have been made over the last few years towards the sustainable management and valorisation of non-conventional water sources which include among others wastewater, rainwater stormwater. Notable progress includes the upgrade of central WWTPs to produce adequate quality reclaimed water in alignment with the recent Water Reuse Regulation (2020/741), the development of innovative decentralized reclaimed water reuse systems, like those based on sewer mining concepts for treating wastewater as evidenced in projects like H2020 NextGen and IMPETUS. Additionally, the emergence of decentralized Nature-Based Solutions (NBS) to manage and valorise non-conventional water sources (such as those developed in HYDROUSA, CARDIMED), particularly in urban settings, exemplifies an innovative approach to circular water flows, simultaneously contributing to climate change adaptation (flood/drought) and augmenting water availability for various uses. Furthermore, thermal treatment processes (drying, incineration) for the management of sewage sludge have been applied at large scale in the two largest WWTPs of Greece (Psyttalia and Syndos) However, technical limitations for these circular systems, socioeconomic constraints, insufficient investments and regulatory compliance often block the initial setup costs, while recent increases in energy prices challenge their operational and maintenance costs. The integration of the decentralised systems with existing centralized infrastructure poses several technical, administrative, logistical, and financial challenges. Scaling up these solutions to serve larger populations remains an obstacle, especially in densely populated urban areas. Maintenance and sustainability issues are still not fully addressed, and there is a lack of well-defined business models and financial frameworks for water service providers. Centralised wastewater systems can also play a key role as sources of reused water in bulk but consume more energy than conventional water supply systems. Moreover, the risk associated with the consumption of crops irrigated with reclaimed water needs to be evaluated on a case-by-case basis. The processes of dewatered sewage sludge drying and incineration suffer from high CAPEX and OPEX, while land application can be problematic or even impossible in urban settings. Consequently, finding alternative pathways for sewage sludge valorisation is necessary.

Beyond: Theseus will go beyond the current SOTA by integrating and valorising non-conventional water flows into circular and symbiotic value chains by coordinating the adoption of innovative centralized and decentralized solutions at city and regional scales also considering the process by-products. This will be achieved by analysing non-domestic water flows in Attica, focusing on sectors with high water demand and by aligning these demands with EU standards, we ensure both compliance and efficient water use. The benefits of centralized reclaimed water reuse in agriculture will be

[23] http://essteam-project.eu/
[24] https://solarheateurope.eu/2023/02/28/solar-district-heating-plant-in-geneva-230-tons-of-co2-saved-and-over-1-gwh-thermal-energy-produced-in-2-years/
[25] https://solarthermalworld.org/news/37-mw-solar-district-heating-plant-in-the-netherlands-with-outstanding-features/

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scientifically assessed focusing on the effect on soil (soil health – biodiversity) and crops (benefits of fertigation) and the risk of crops consumption will be evaluated focusing on the fate of emerging contaminants other chemicals and pathogens. A principa innovation is the use of decentralized water reuse practices, employing sewer mining technology to repurpose reclaimed water for irrigation which is both operationally efficient and adaptable to various urban contexts. Additionally, we will assess the cost-effectiveness and sustainability of these systems, in line with regulatory requirements. The project also integrates NBS for water recovery in urban areas enhancing water availability and reducing flood risks and supporting urban farming practices. In parallel, we will implement the nove processes of fast pyrolysis and torrefaction to recover both energy and high added value products from sewage sludge, fostering significan improvements in energy efficiency and circularity for centralized WWTPs. Importantly the project will bridge the 'gap' between demonstration and large-scale adoption by co-developing with the Region of Attica, a Strategic Regional Master Plan for Circularity in Attica synthesizing insights from demonstrations to guide investment and align with regulatory frameworks. This plan will catalyse regiona

upscaling of circular water management practices and solutions

Implemented in WP7, Task 8.5 KERs 10-12. Synergies with HYDROUSA

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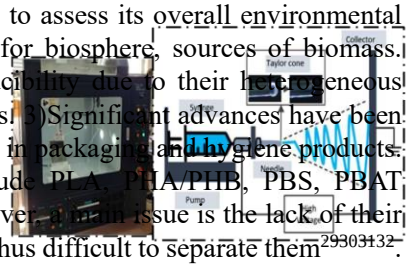
Accelwater, AquaSPICE, CARDIMED, IMPETUS projects

Developments	Key Advantage & Progress – Start TRL Foreseen End TRL
CHP system H2	for production of heat and electricity through a CHP KERs 13 TRL5 7
DH system New	generation of HVFP solar thermal technology for greening DH KERs 14 TRL5 7
Water Sewer Mining System for Wastewater Valorisation KER 10 TRL5 7; Centralized reclaimed water fertigation KER 12 management TRL5 7; NBS for rain/stormwater harvesting and valorisation KERs 10,12 TRL5 7; Fast pyrolysis and torrefaction systems for sewage sludge valorisation KER 11 TRL4 6[26]	

IP5 Symbiotic flows (materials) and upcycling (A, B, C sub-innovations) contribute to SO2

A) Packaging upcycling from MSW

SoA 1)OECD[27] reveals globally a significant surge in plastic production (2m to 460m tons 1950-2019), only 9% of plastic waste is recycled, with nearly 50% disposed in EU, while COVID-19 has exacerbated situation. Smart packaging (SP) is designed to prevent or minimize food waste, having challenges in commercial viability, regulatory standards, waste preservation, process optimization, recycling etc. to assess its overall environmental contribution[28]. 2) Bioplastics are renewable, abundant, nutritious and safe for biosphere, sources of biomass. However, there are challenges such as high cost due, performance reproducibility due to their heterogeneous feedstock, intractable and exhaustive exploration to form sustainable solutions. 3)Significant advances have been made in biodegradable and compostable polymeric materials with applications in packaging and hygiene products. Widely used bio-based and petroleum-based biodegradable polymers include PLA, PHA/PHB, PBS, PBAT certified according to latest regulations (e.g. EN13432, ASTM D6400). However, a main issue is the lack of their recycling as they are often mixed with other materials e.g. food scraps, paper, thus difficult to separate them[29][30]. New recycling streams processing approaches e.g. addition of additives, blending with neat/ fresh bio-degradable polymers should be developed, both addressed, boosting further the reduction of carbon footprint and cost of existing biodegradable plastic masterbatches (now at 2-3 €/Kg).



Beyond:

[26] Techno-economic assessment of upscaling and operating a multistage pyrolysis reactor with a nominal input of > 1 ton of sludge/ day, along with the analysis of the mass and energy balances, thus a technology demonstrated in relevant environments.
[27] Organization for Economic Co-operation and Development, https://www.oecd.org/environment/plastic-pollution-is-growing-relentlessly-as-waste-management-and-recycling-fall-short.htm
[28] https://doi.org/10.3390/polym1506148
[29] https://www.sciencedirect.com/science/article/pii/S004896972302065X
[30] https://doi.org/10.1007/s12221-012-0339-x https://doi.org/10.1177/0040517515603808
[31] https://doi.org/10.1088/2043-6262/1/4/043002 https://doi.org/10.1007/s10965-013-0105-9 https://doi.org/10.1021/acs.chemrev.8b00593
[32] https://doi.org/10.1007/s12221-012-0339-x https://doi.org/10.1016/S0032-3861(00)00250-0

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1 1)Nanofiber films and micro/nano-composite powder with enhanced properties will be developed from recycled polymers. In the
:, the nanofiber films with encapsulated agents will be coated on conventional polymers for the formation ilayer films that can be used for packaging applications. In 2nd case, the produced micro/nano-composite will be co- extruded with conventional polymeric pellets to produce batches of enhanced pellets with al uses in packaging, agricultural films etc. For this purpose, the electrohydrodynamic process of spinning will be used, a cost-effective and promising technique for fabrication of polymeric micro- or nano-ed fibres, micro and nano-films that can act as coatings in several materials. It is an efficient and simple ulation technique in terms of versatility, flexibility and ease[36], with green and safe processing under normal atures, without harmful chemicals. During electrospinning the polymer solution is charged with high to be ejected to the collector. Towards its route, the solvent evaporates and the jet solidifies forming e fibres, deposited on the collector[37]. Fibres' have unique properties for exploitation in packaging, textiles) Theseus will develop a robotic materials system/platform automating synthesis and characterization of tic formulations (100% biobased/ biodegradable) to develop a single-use biobased thin film and/or recycled waste extracted from biomass. AI planning software will be used for analysis in development cycles, ely learning from prior cycles and design and implement new material recipes based on user-

specified performance goals. 3)The following biodegradable recycled materials/ formulations will be developed and investigated: (1) neat polymers, (2) mixed polymer powders, towards biodegradable polymer blends, and (3) polymer powders mixed in solid state or via a twinscrew melt compounding process with different (nano-)additives e.g. CaCO3, ZnO, SiO2 nanoparticles, etc., exploiting research by HMU in thermoplastics processing via melt-mixing, thermomechanical recycling processes and physicochemical characterisation[33].The cost of additives on the above formulations will be considered as a function of the potential increase of properties e.g. mechanical, thermomechanical, etc. to be achieved. Namely, the recycled based thermoplastic polymeric materials / compounds that will be developed by HMU will be tested by melt-rheology, DMA, DSC, FT-IR, GPC and tensile testing on dog-bone as well as on melt-blown film samples. Implemented in T5.2,5.4,5.5 KERs 5,6,8. Synergies with REDOL project

B)Textiles upcycling

SoA: Less than 1% of materials used in clothing is recycled into clothing at a global scale[40]. Mechanical recycling is the most mature option, with 20 % of collected and shredded textiles in Europe used as industry wipes, padding and filling for automotive industry and as insulation[34]. However, textiles cannot be recycled mechanically without reducing properties and quality. Chemical recycling is mature for thermoplastic fibres but not for natural fibres while mixed fibres is a challenge in recycling. The presence of dyes and certain chemicals in fabrics can also prevent recycling. Moreover, LCA show that chemical recycling has greater costs and contributes to an overall negative impact in energy consumption and environmental effects.

Theseus will develop a fibre-spinning process (melt spinning for thermoplastic or solvent/wet spinning for natural fibres) combined with encapsulation of active compounds within fibres. This will lead to novel circular high-value products, with desired properties (water repellent, oil repellent, antimicrobial, flame retardancy). In addition, to make chemical recycling feasible at industrial scale, Theseus will develop a glycolysis reactor with ionic liquids and supported by microwave technology, followed by post-condensation and spinning steps, represents a sustainable depolymerisation-polymerisation process able to integrate renewable energy sources more efficiently. All the above will be combined with the sorting activities described. Implemented in T5.1 KERs 4,7. Synergies with EXTENDED, ACT UK, REDOL, Textile waste refinery projects

C)Innovative Construction Materials from CDW and other Wastes

SoA 1)Constructions is the largest resources consumer (35–45%) and 40% of global GHG emissions from material use. CDW is the 3040% of total solid waste. CE approaches in cement and concrete industries offer economically opportunities. Tailoring composition and feedstock materials offers significant potential for reducing embodied carbon and energy of cement production. Concrete production is flexible utilizing an array of materials as binders and aggregates, thus a recommended option for repurposing post-use materials e.g. CDW, glass, fibrereinforced polymers, mixed plastic, rubber[35]. Brick production from CDW is also important because of the detrimental effects

[33] https://doi.org/10.3390/ma14020466 https://doi.org/10.1016/j.polymertesting.2022.107545 [40] Ellen MacArthur Foundation, A new textiles economy: Redesigning fashion's future (2017).
[34] https://www.europarl.europa.eu/RegData/etudes/BRIE/2022/729405/EPRS_BRI(2022)729405_EN.pdf
[35] https://doi.org/10.1016/j.conbuildmat.2019.117869 https://doi.org/10.1016/j.conbuildmat.2012.02.066 https://doi.org/10.1016/j.resconrec.2017.08.005 https://doi.org/10.1016/j.jclepro.2021.127180 https://doi.org/10.3390/su13084113

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of bricks and ceramics in recycled aggregates from CDW used in concrete. Replacing up to 25% of the raw materials have been demonstrated at laboratory scale[36] whereas up to 5% ceramic waste are being used in some factories. CDW recycling is challenging due to variation in brick, ceramic and waste glass content, pollution level, chemistry and mineralogy. 2)Supplementary cementitious materials (SCMs), e.g. siliceous, aluminosiliceous, or calcium powders, are used as partial replacements of Portland in modern concrete, or as by-products of industries, e.g., coal fly ash and slag, or natural minerals with much lower energy input in processing, e.g., clay[37]. Alkali-activated binders and related geopolymers have been promoted as a component of the current and future toolkit of "full cement replacement systems". The primary binding phases are derived by the polycondensation of aluminosilicates sourced from clays, fly ash, or slags, with an alkaline activator that accelerates the reaction process. Ongoing research emphasizes optimizing precursor utilization and tailoring and mixture formulations, exploring novel/recycled materials, assessing their long-term behaviour and integrating them into mainstream construction practices[38].

1) Theseus will use CDW in brick production, utilizing the whole fraction, combinations and waste glass as binder in concrete products. Environmental acceptability will be assessed for the new materials[39]. The leaching performance along with typical scenario will determine the impact (e.g. leaching to soil, surface water body and groundwater) for products in granular and monolithic forms for concrete and brick products. The aim is to quantify the leaching over lifetime. Ceramic, brick, and fines contents constrain the recycling potential. Theseus will seek novel insights into the cementitious activity of entire fractions of recycled concrete, glass wastes, etc., for their utilization as SCMs in blended cement and alkali-activated binders or related geopolymers for producing concrete. The performance of selected mortar mixtures, tailored for pre-cast elements and 3D-printability, will be validated in lab. Also, concrete recipes including glass will be tested at the ready-mixed concrete producer. The objective is to mitigate the environmental impact of cement and concrete manufacturing without compromising product's economic and technical viability, minimizing raw materials and reducing emissions. Theseus will demonstrate a viable route for pulverised mixed CDW as raw material for brick production replacing local materials where feedstock contains min 10% of CDW and 6-8% of ceramic waste. 2) The incorporation of high-quality and uniformity of recycled mineral wools from project partners promotes the development of high-performance construction panels with enhanced mechanical properties, durability, and sustainability. Implementing sophisticated methods (thermal or chemical treatment) can enhance the quality of recycled mineral Wool, optimizing reactivity and compatibility with cementitious binders and reliability in performance of the final concrete product. Implemented in T5.3, KER 4. Synergies with W2W, REDOL, DigInTraCE projects

Developm ents	Key Advantage & Progress – Start TRL Foreseen End TRL
Packaging and fibres circular products	Thin films, masterbatches and fibres from MSW and textiles (used in ThreacePoly and Panchart, as end users as per Pilots KERs 5-8 ons
Circular materials for constructio n	Circular constructi on material from CDW and textiles (used in Kl user, as per description 1.2.1) KERs 4 TRL5 7

#§PRJ-OBJ-PO§#

1.2. Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

[36] Dubale, M.; Vasi´c, M.V.; Goel, G.; Kalamdhad, A.; Singh, L.B. Utilization of Construction and Demolition Mix Waste in the Fired Brick Production: The Impact on Mechanical Properties. Materials 2023, 16, 262.
[37] https://doi.org/10.1016/J.CEMCONRES.2010.12.001 https://doi.org/10.1016/j.cemconres.2019.05.008 https://doi.org/10.1016/J.CEMCONCOMP.2022.104885
[38] https://doi.org/10.1146/annurev-matsci-070813-113515 https://doi.org/10.1016/J.CEMCONCOMP.2022.104885 https://doi.org/10.1016/j.cemconres.2015.04.013 https://doi.org/10.1016/j.cscm.2023.e02772
[39] Engelsen, C.J., van der Sloot, H.A., Kosson, D.A., Leaching assessment methodologies and regulations for evaluating the use of secondary and recycled materials in concrete and mortars

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1.2.1 Overall Concept and Scientific Methodology

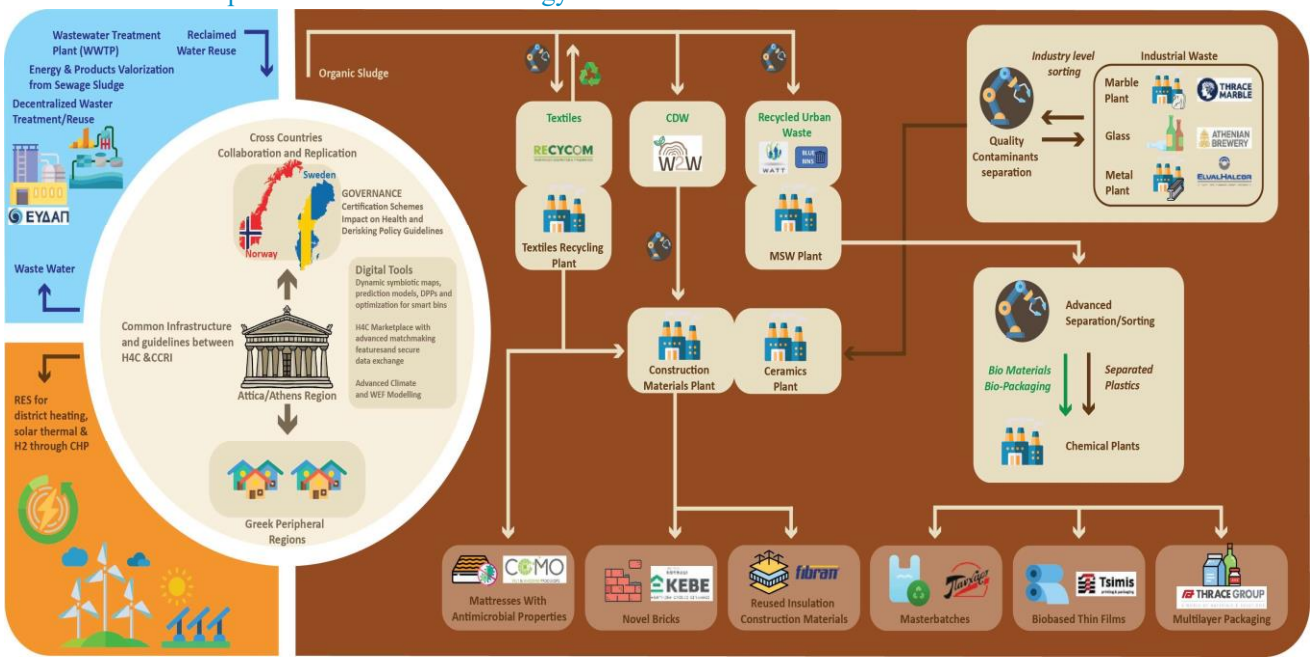

Figure 3 Overall Concept & Overview of Water Energy Material Flows and Pillars

2.1.2.2. Overall Concept and Systemic Approach

Using and combining the innovations described above, a story line per IP is summarized below:

IP1 Collaboration (new & enhancement of existing) & IP5 Governance (regional & interregional development)

A set of existing I-US practices, methodologies, technologies and tools from existing H4Cs, CCRI and IS projects and H4C CoP (connection already established thought common partners and A.SPIRE, CIRCE in T9.1, 9.2) will be exploited as baseline, adjusted and/or enhanced in several WPs, to fit and operate in Theseus. Successful cases and approaches will be transferred via mutual learning between the main hub and replicators. In WP2, LiU will design and facilitate the establishment of the I-US governance model, and collaboration structures between involved stakeholders, different sites, private and public sector and end-users of Theseus and of the intra-regional ecosystem of Attica. Detailed analysis of regional and national plans and materials flows mapping and characterization, along with territorial analysis of interregional interactions with peripheral regions, will support the setting up of projections and pathways to symbiotic flows and upscale to interregional and national level, thus multiplying the impact of interventions. Theseus will provide new insights to local, regional and national plans and develop among others Circular economy roadmaps (CERMs), setting up also strong basis for optimization, business value, exploitation, standardization and policy recommendations, regional institutional capacity and overall sustainable regional development.

IP2 Symbiotic flows (materials) and upcycling T8.1, T8.2, T8.3

Packaging upcycling from MSW: Theseus' IP5 focuses on improving mutually beneficial material flows and upcycling initiative textiles, biobased materials, packaging, and CDW are all given special consideration. These material flows were deliberately chosen for their major environmental effect and the ability to successfully incorporate the main principles of the circular economy. This concept is well-aligned with both Greek and European environmental objectives. In Greece, there is a significant gap and scope for improvement in the management of such material flows. Biobased materials will also be extracted. Theseus investigates cutting-edge recycling processes aimed to convert MSW streams into useful secondary materials (Pilot #2: Biobased Materials and Packaging Symbiotic flows in T8.1).

Textiles: The current textile recycling infrastructure is still in its early phases, resulting in low rates of recovery and reuse, although this is projected to change soon. This project presents automated sorting methods for textiles that promise to improve material segregation efficiency, resulting in improved recycling rates and less reliance on landfills (Pilot #1: Textiles flow in T8.2).

Innovative Construction Materials from CDW and other Industrial Wastes: While advances have been achieved in recycling packaging waste and CDW, there is still work to be done in upcycling these materials into higher value

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products that can revolutionize entire value chains and pave the way for sustainability. This gap underscores the need for novel waste management systems, which Theseus intends to address using cutting-edge technology and methodologies. When it comes to CDW the project focuses on developing sustainable building materials that integrate recycled content, so encouraging environmentally friendly construction practices (Pilot #3: CDW flow in T8.3).

IP3 Symbiotic flows (water / energy) T8.5, T8.4

Water: Theseus, we analyse potential non-domestic water flows across Attica, industrial water needs per sector and quality, to ensure that demand, fluctuations, and uncertainties from domestic to irrigation meet EU and Greek Water Reuse Regulations and industrial requirements. Decentralized water reuse strategies will be explored, for urban gardening and farming activities. Innovative sewer mining systems will be deployed, to use treated wastewater for irrigation and addressing urban heat stress in Athens, aiming operational efficiency and sustainability. The long-term benefits of centralized reclaimed water reuse for agricultural irrigation will be investigated considering the fate and removal of emerging contaminants and the impact of the fertigation practice to soil health and plants growth. Also, an extensive risk assessment will be carried out to showcase potential risks arising from human consumption of these crops. Theseus will also strategically deploy decentralised NBS for rain/stormwater valorisation within Athens. The approach extends to redesigning urban parks, incorporating rainwater harvesting, and pro-actively modifying runoff patterns, thus alleviating urban flooding and reducing stress on potable water resources. In parallel, valorisation of sewage sludge produced in WWTPs will be addressed, by implementing advanced thermal treatment processes (fast pyrolysis, torrefaction) to produce energy and added value products. These will be integrated into Strategic Regional Master Plan in collaboration with Attica Region, bringing together insights, focusing on investment opportunities, maturing system designs, and aligning with regulations to fasttrack upscaling at regional level.

Energy: The application of solar thermal technology to buildings of tertiary sector will be tested on site. Solar thermal panels will be installed in the ground beside CERTHs building in Ptolemaida to support to current area's DH across seasons, for both heating and cooling purposes, by also installing absorption chiller replacing current electricity powered cooling circuit. The nearby Mpodosakeio Hospital will also benefit from the construction of a small-scale DH to cover heating but mainly cooling needs all over the year. Hot water will be transferred through an underground pipe to feed Hospitals current absorption chiller and provide extra cooling for surgery rooms, new medical sections, lowering carbon footprint (currently big energy consumption). The potential of connecting this pipe with the central DH system of Ptolemaida will also be examined. The hospital is currently using cooling with roof solar thermal panels, trying to engage renewables and be transformed in the future to a green Hospital (zero emissions healthcare unit). Also, using green H2 (from RES) for supporting heating of CERTHs building will be demonstrated. H2 will be produced in a container outside of the building (from EVELIXIA). A study being conducted by CERTH (regional funding) for transferring green Hydrogen to Hospital for production of heat and electricity through a fuel cell system. Based on the above gradual city decarbonization could be achieved in the future.

IP4 Digitalization & robotics (incl. sorting)

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Theseus will incorporate 5 distinct sorting lines for T8.1-T8.3: Textiles: concentrates on automating textile sorting processes carried out by RECYCOM, aiming to streamline and enhance the sorting accuracy. Textiles from purple bins will feed a conveyor line. An image-based recognition module comprising of RGB or HSI equipment will categorize textiles into natural, thermoplastic, and mixed fibre. Next, a multirobot setup will handle and deposit each item into its corresponding container. Metals: will expand the capabilities of ICCS's wood sorter in W2W. Setup will be based on Halcor premises and will retrieve scrap metal from mixed CDW, through precise grading. Combined machine vision and X-Ray-based classification modules will detect specific materials, impurities and contaminants. Robot arms equipped with specialized grippers will sort classified items in individual containers for further processing. The pilot will also explore pre- and post-treatment (e.g. shredding, balling, briquetting) to seamlessly integrate with current workflows. MSW: will be implemented at TERNA facility in Arkadia region and will undergo targeted modular upgrades to refine the sorting of plastic, metal, paper and textiles, to optimize operational line's sorting capabilities through AI-driven machine vision and robotization, elevating both its efficiency and processing capacity, to sort and manage diverse waste components more effectively. Plastics: advanced recycling via negative sorting within a generic material stream from WATT's Blue Bins. Enhancements aim for precise differentiation among various thermoplastics such as PET, PE, and bio-based polymers through HSI and DL. Potential synergies will be explored with W2W and DigInTraCE, for shared advancements. Glass: will recycle used glass bottles originating from AB. Used bottles will traverse along a conveyor line and will undergo a 2-stage categorization i)an optical sensor will detect defects and damage in the bottles ii)each bottle will be characterized by size and brand. Intact bottles will be grouped via a robot and forwarded for sterilization and rebottling. Damaged bottles will be recycled. Theseus will develop a Marketplace H4C platform: facilitating circularity, management, matchmaking, modelling and optimization and decision making, under secure data exchange. Several tools will be integrated:

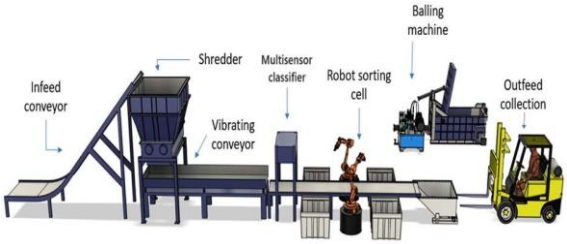


Figure 4: Indicative proposed concept for scrap metal sorting

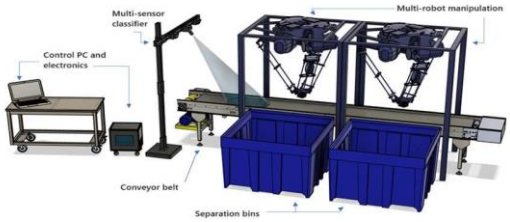


Figure 5: Indicative proposed concept for mixed waste sorting

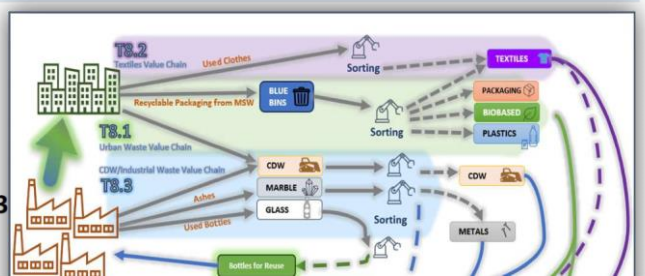
GIS tool for dynamic symbiotic maps: mapping and visualization of dynamic symbiotic loops, spatial allocation of activities, informed decisions; Matchmaking and symbiotic models: advanced matchmaking functionalities and recommenders expanding existing work and research. DPPs: adjustable for each innovative circular product with dynamically updated information. Informed decision taking within the supply chain. A mixed reality application creating value for the end user. Prediction of resource logistics: PRLM to predict material and water, to enable users identify business opportunities, risks or bottlenecks, while exploring scalability, replication opportunities.

2.1.2.3. Scientific Methodology

The main methodology is displayed below, indicating how existing knowledge, expertise and tools will be used from existing or ongoing

2.1.2.4. Value Chains of the 5 Pilots Descriptions

	Partners



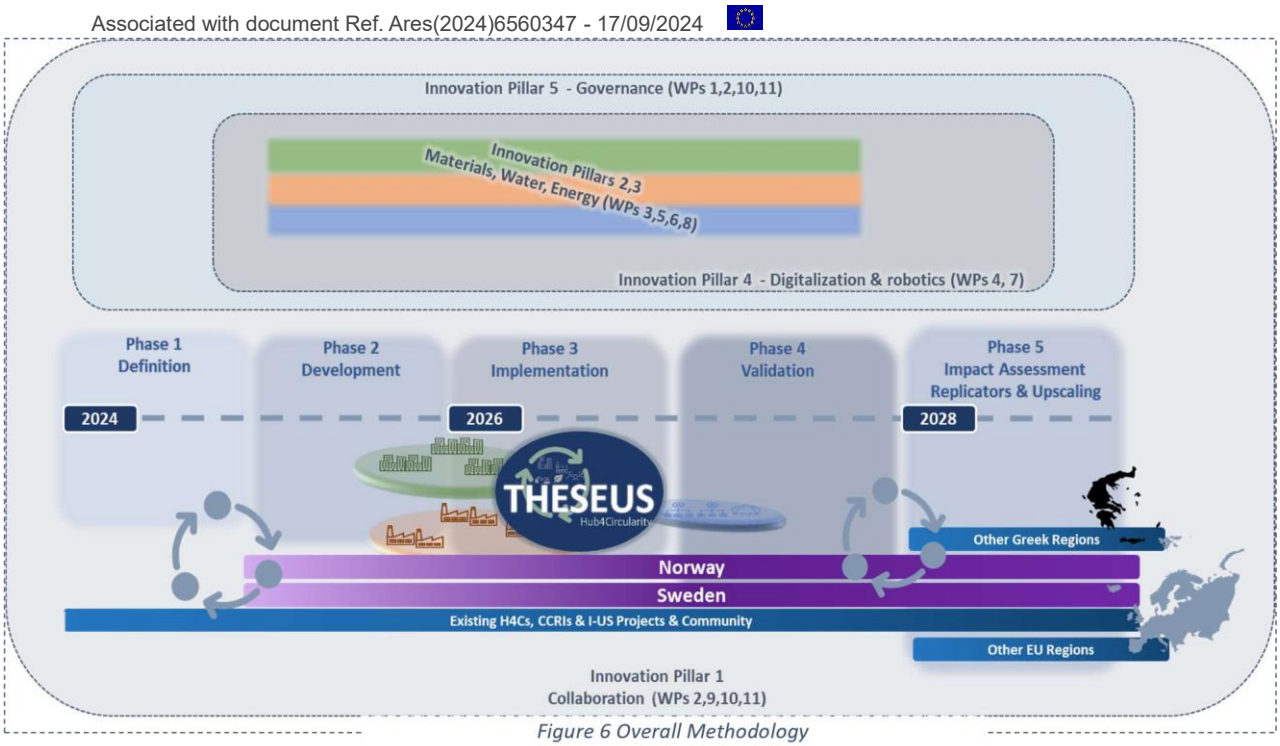


Figure 6 Overall Methodology

H4Cs, CCRIs and I-US Projects, to Establish the Attica Theseus H4C, develop and validate the innovations and the replicators and prepare for upscaling. PM, Communication activities will be performed throughout the project. The phases are Definition M1-M8: This Phase will set up the baseline and the frameworks under which Theseus consortium, project and the H4C will be developed and operate. The synergies, knowledge, expertise, tools and practices to be transferred from H4Cs, other projects and replication areas, into Theseus will be defined and prepared for integration. Preliminary studies will be performed in waste, energy and materials in the territory of Theseus. It is implemented mainly in WP2 and WP1. Development M9-M24: It will design and develop the proposed technologies and innovations in terms of materials waste, energy and water, as well as the sorting and digital systems, based on the existing expertise and tools defined in P1 and they will be enhanced and adjusted accordingly and integrated. The pilot's setup will be prepared. It is implemented mainly in WP3-7. Implementation M25-M36: It will finalize all the technologies and innovations, sorting and digital systems. They will be combined and integrated into value chains, forming the Theseus pilots will be developed and completed. In Parallel replicators will be prepared based on knowledge from main H4C activities. It is implemented mainly in WP3-7, WP8. Validation M37-M48: This Phase will validate Theseus pilots and complete the value chains with the testing and validation of the circular products and solutions. The replication areas will be also up and running. Proper refining will be done based on feedback from the value chains as well as the replicators. Assessments, policy recommendations and standardisation activities will be fed with insights from the pilot's demonstration. It is implemented mainly in WP8. Replication and Upscaling M49-M60: It will finalize the replication areas, transfer to and from the main H4C knowledge and insights. Impact assessments will be developed for solutions and the pilots, along with final policy recommendations and standardization. The conditions for upscaling to other EU regions, replication and full exploitation towards commercialisation will be analysed and project outcomes will be summarized. It is implemented mainly in PW9, WP10, WP11. An iterative knowledge exchange process between the 3 locations (main HUB and replication areas) will be active, though all phases.

Materials Pilots Value Chains, Circular Products (CP), End Users (Pilots #4, #5, #6; via Innovation Pillars 2 & 4 story lines) 2.1.2.5. Pilot #1: Textiles Symbiotic Flows

Textiles

Coordinated & demonstrated in T8.2

Used clothes/textiles collected from RECYCOM collection points (purple bins) within communities to ensure raw materials separation and higher quality of collected

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textiles. An extra source of textiles is from the sorting processes in Pilot #2, on which textiles are identified and separated from MSW. Separation in T4.4 so textiles collected will undergo advanced sorting & separation to categorize them based on fibre type, color, and condition (e.g. thermoplastic, non-thermoplastic, mixed fibres), to determine suitability for recycling or upcycling.

Based on the properties of the textile waste recovered, the feedstock material will undergo an appropriate fibre-spinning process paired with encapsulation of active compounds, leading to the production of novel circular high-value products, with desired fibre properties based on the application. Also, novel linker bio-chemistry will be tested to attach the actives with the fibrous materials, while the functionalization will be achieved using plasma treatment. 2 novel methods will be utilized to produce new fibres: (a) a glycolysis reactor with ionic liquids supported by microwave technology followed by post-condensation and spinning; (b) a fibre-spinning process (melt spinning/electrospinning processes for thermoplastic or solvent/wet spinning for natural fibres) combined with encapsulation of active compounds.

Figure 8: Indicative spinning process for thermoplastic textile waste stream

CP#1 Recycled Fiber will be processed to extract fibers suitable for crafting into new textile materials, focusing on creating a sustainable source of textile fibers, reducing the need for virgin materials. The new spun fibres with antimicrobial COMO properties will be mixed with existing packing material to produce mattresses.

CP#2/3 Thin Films & Multi Layers will be created from recycles textiles and used as raw materials in various sectors, Tsimis including fashion, home decor, and industrial uses, showcasing the versatility of recycled textiles, in ThracePoly

CP#5/6: Bricks and cement from recycled textiles will be created and used as ICM. Recycled textiles will be incorporated into construction materials, contributing to sustainable building practices and reducing the environmental footprint of KEBE constructions. The new flame retardancy spun fibres will be used as secondary construction material in

KPI 1 - Recycling Rate: >10% Increase in the percentage of textile waste that is successfully recycled and diverted from landfills; KPI 2 - Reduction in virgin materials 'use: >15% decrease in virgin materials 'use due to the substitution of them with recycled textiles in product manufacturing; KPI 3 – GHGs reduction: >10% in GHGs due to reduced raw materials use; KPI 4 - Innovative Products: At least three new products or applications developed from recycled textiles, indicating the success in creating value-added products from waste.

2.1.2.6. Pilot #2: Biobased materials and Packaging Symbiotic Flows

MSW; Packaging, Biobased, Plastics

Coordinated & demonstrated in T8.1 Partners

Mixed recycled materials will be collected from Blue Bins (mainly packaging recyclables) managed by WATT. Recyclable materials in Blue Bins are a compilation of different materials with separation challenges. Indicatively, WATT general Greece's MSW: paper 28.5%, biowaste (organic) 39.2%, plastic 12%, glass 4%, textiles 2%.

Mixed recycled materials will be separated to with the sorter developed in T4.4 producing Packaging, Biobased and

ICCS

Plastics flows, as well as Textiles in 4 qualities to feed the Source of .

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Prep. Sources

CP#2: Biobased Thin Films of various biobased targeted properties will be developed using bioplastics extracted. NCSR/Tsimis

CP#3: Multilayer Packaging: An electrospun coating film with incorporated functional materials (antimicrobial and antioxidant compounds) that acts as the carrier of active compounds that will be controllably released can be incorporated in a multilayer conventional packaging. Thus, the multilayer system will have enhanced physicochemical properties and incorporated functional compounds. NTUA pilot line is capable to prepare nanocomposites and fibers from carbon neutral materials with different functionalities using electrospinning process with capability of 5 kg coated film with electrospun nanofibers. NTUA ThracePoly

CP#4: Masterbatches: Recycled biodegradable thermoplastic polymer formulations with the optimum performance and decreased cost set by KPIs in T5.2 will be shredded by HMU and delivered to PanChart for the film-blowing process at industrial environment, ending up "recycled & recyclable biodegradable polymeric films" for packaging, bags, etc. PanChart will need for each batch of extrusion/ film blowing 100-120Kgr of pellet material with melt rheological properties to apply the proper industrial-scale extruder speed and film blowing process parameters. optimise them towards the best quality of the blown-film as well as trying to minimise the energy consumption of the overall process. The film produced will be tested by PanChart for its printability (e.g. for supermarket bag applications) and sent to HMU for mechanical (tensile), thermomechanical (DMA), DSC and TGA analyses.

KPI 1 20% reduction time of material recipes testing for bio based thin films; KPI 2 10% faster creation of multilayer packages with specific attributes; KPI 3 at least 50% reduced cost of the recycled compounds, for masterbatches; KPI 4 10% increased tensile modulus and 25% strength for masterbatches

2.1.2.7. Pilot #3: CDW & Industrial Waste Symbiotic Flows

CDW & industrial waste; Metals, Glass, Marble

Coordinated & demonstrated in T8.3 Partners

CDW consist of dirt, sand, gravel, concrete, stones, bricks, wood, metals, glass, plastics, paper, and fabric, etc. They will be collected from civil engineering works (roads, bridges, sewers, networks, sidewalks, and space renovations etc.), as well as from AB their demolition, restoration and/or collapse; The collection of pulverised waste glass will be performed during CDW fromThraceM windows, glass doors, glass partitions, and other glass elements used in buildings; Collecting waste marble during CDW will arble be performed in collaboration with AB and ThraceMarble through effective segregation and disposal.

CDW will be sorted dependent on particle sizes for an optimal reactivity environment to produce hydraulic blended cement or

alkali-activated materials (AAMs)/Geopolymers (GP) as alternative precursors. Coarse particle fractions will be sorted and

ICC

S applied as aggregates. Considering optimal packing density, reactive binder materials will be thoroughly mixed with different recycled aggregates and water/seawater to produce concrete blends.

Freshly produced concrete blends will be combined with differently recycled fibres or waste glass enhancing circularity aspects

TUD

and improving the engineering and functional properties of the prototype of resulting insulation panels. Following intensive

SINTEF mixing, pre-casting and additive manufacturing will be performed to produce insulation panels. On site separation on the premises of the plants aiming to increase the circularity in resources within the plants , improve the

Halco

r usage of secondary materials and allow a total increase on recycled materials both for glass and copper. This is achieved through

AB advanced separation/sorting and dedicated shredding for the copper.

CP #5: Insulation Panels of (Circular Product #2) The development of insulating panels based on selected binder precursors and aggregates and particular waste glass and wools offers an innovative system for high energy efficiency in buildings, used for both pre-cast and 3D printed elements. Based on the experimental results, the proposed panel will be investigated and compared to conventional insulating building solutions

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(e.g., rock wool, glass wool, etc.) considering not TUD only the thermal performance, but also considering other important characteristics, including acoustic insulation, fire Fibran resistance, mechanical strength, durability, cost, in situ installation, and eco-compatibility. To develop an optimal design of fiber-based concrete mixture for application in existing walls, different combinations of recycled component materials will be considered in the analysis.

CP #6 Bricks with increased recycled content: CDW fines (and glass as a binder in some cases) will be demonstrated in the brick production at KEBE: i) Apply pre-homogenisation strategy to ensure stable chemistry of the raw materials. This includes laboratory mixing strategies of refined CDW to assess chemical and mineralogical properties. ii) Based on laboratory-made bricks, refined CDW will replace 10-50% of the clay based natural raw materials in the pilot trials. iii) SINTEF Each replacement level will be assigned to the product quality (e.g. strength, colour, water absorption, linear shrinkage, KEBE bulk density etc.) in order to document and classify the bricks containing refined CDW. iiii) Establish the maximum replacement level of CDW without compromising the quality parameters for most relevant markets (loadbearing and nonloadbearing applications).

KPI 1 up to 50% of recycled materials in the end products; KPI 2 80% recyclability/repair ability of insulation materials in CDW

Water & Energy Value Chains (Pilots #1, #2, #3, via Innovation Pillar 3 Study line)

2.1.2.8. Pilot #4: Energy Symbiotic Flows

Management	Energy Coordinated & demonstrated in T8.4	Partners Innovative
	technology will be tested to a block of 2 buildings (CERTHs and Mpodosakeio Hospital). The TVP ,DHCP CLEAN application of solar thermal technology to buildings of tertiary sector will be tested on site. Around 1600m^2 of solar thermal panels (HVFPs) of 1 MW, will be installed in the ground ~3000m^2 (beside CERTHs building in Ptolemaida) to cover heating (support to current area's DH) and cooling purposes (solar cooling). Hot water from solar thermal panels will feed the heat exchanger of the building (October-May), whereas solar cooling will	
	be applied (June-September) through the installation of an absorption chiller (~500-800kW) to replace current electricity powered cooling circuit. The nearby Mpodosakeio Hospital will also benefit from the construction of a small-scale DH to cover heating but mainly cooling needs all year. Hot water will be transferred through an underground insulated pipe to feed Hospitals current absorption chiller (distance 500m) and provide extra cooling for surgery rooms, new medical sections, lowering the carbon footprint (Hospital has big energy consumption throughout the year). The potential of connecting this pipe with the central DH system of Ptolemaida will also be examined to minimize heating losses. The hospital is currently using cooling with roof solar thermal panels. Using green H2 (from RES) for supporting heating of CERTHs building will be demonstrated, H2 to be produced in a container outside of building (deployed through EVELIXIA project). There is also a study being conducted from CERTH (regional funding) for transferring green Hydrogen to Hospital to be utilized for production of heat and electricity through a fuel cell system (mobile). Greening of DH of Ptolemaida will be gradually achieved (currently supported from operation of lignite thermal power stationsuse of waste heat).	
USE #1:	Hot water of high temperature (produced by HVFP) will be utilized for heating and cooling purposes of 2 buildings (CERTHs building and nearby Mpodosakeio Hospital).	TVP,DHCP CLEAN
USE #2:	Utilization of green H2 (produced locally in CERTH via EVELIXIA horizon project) for production of heat and electricity through a small fuel cell system~5KW (small scale implementation) for CERTHs building.	CERTH

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USE #3:	Implementation of a study from CERTH towards introduction of innovative technologies through RES for covering the heating load of DH System of Ptolemaida city.	CERTH
USE #4:	Implementation of study from CERTH towards Digitalization of district heating systems in terms of improved energy management and control and optimized energy sources utilization.	CERTH
KPI 1 annual heating energy production from RES (solar thermal) > 1500 MWh; KPI 2 annual heating consumption from RES in CERTHs building (for DH and solar cooling: 200-250 MWh, KPI 3 annual heating consumption from RES in Mpodosakeio Hospital building (for DH and solar cooling): 1200-1250 MWh		

2.1.2.9. Pilot #5: Water Symbiotic Flows – Centralized / Decentralized

Value Chain:	Water	
	Coordinated & demonstrated in T8.5	Partners
Sources	Wastewater will be treated at sewer mining facilities for irrigation purposes to increase water availability and resilience of urban parks in the City of Athens. Reclaimed water from the centralized WWTP of Eastern Attica will be applied at a 0.2-0.3 ha agricultural field to assess benefits and risks of fertigation practice. Rainwater will be collected from rooftops and/or from surface runoff through NBS and other engineered solutions, both as a flood mitigation measure and for recovery and use for irrigation purposes in the urban and peri-urban environments at the appropriate time (dry period) to support farming practices. Sewage sludge from the Psyttallia and Metamorfosis WWTPs will be tested as feedstock to recover energy and materials.	EYDAP, NTUA ADDMA, SusCITY
Treat	Sewage Sludge will be treated by fast pyrolysis and torrefaction for energy and materials valorisation (biochar, biooil, gas). Decentralized system of Water will be developed in T7.4 so to decrease the pressures from water availability or for artificial recharge to protect the coastal aquifers from sea water intrusion and improve groundwater quality in general	EYDAP, NTUA ADDMA, SusCITY
USE#1:	Production of biochar, biooil and gas from Sewage Sludge (T7.5)	
USE#2:	Reclaimed Water used for irrigation of crops and urban uses (T7.2 and T7.4)	
USE#3:	Harvested Rainwater and Stormwater used for green spaces and urban farming (T7.2 and T7.3)	
KPI 1 Volume of available water produced or collected for irrigation from decentralised systems > 80m^3; KPI2 Rainfall runoff volume decreasing from NBS in urban environments > 15%		

1.2.2 Replicators

A) S : Malmö is the 3rd largest city in Sweden, its region covering 1,347 km^2 , 271,573 people. Öresund Region (Malmö, Copenhagen) is home to 4 million people. Malmö was one of the earliest and most-industrialised cities in Scandinavia, with ambitious targets for climate neutrality and one of the EU Cities Mission incl NetZeroCities. Malmö have signed climate contracts with 200 companies, the present environmental programme was adopted by the Council in 2021. One important goal is increased resource efficiency. Increased consumption is causing the Earth's resources to be consumed at an ever-increasing rate, and Malmö must undergo a transition to a more circular economy where spillage and waste is minimized, and energy efficiency is facilitated. Products, materials, and resources must be fully utilized, re-used where possible and finally recycled. Malmö also has a Waste and eco-cycle plan with Prevention of waste, Efficient re-use, and Reduced waste flows. Sources and Value Chains: Industrial and urban residual flows and DH; residual heat from DH production and industry, waste and water, textiles (Siptex, automated sorting), plastics – better separation from the waste streams, food etc. Main Goals to be achieved through project: Establish resource collaboration between actors in Malmö region including both private business sector and public sector and value chains. The goal is in Theseus to support the 200+ companies with climate contracts. Knowledge transfer and exchange between Attica hub and Malmö regarding technology, policy, business and innovation. Co-development with stakeholders leading to integrated scenario analysis and valorisation of Theseus innovations through a system dynamics model (WECOS). Co-develop decision support tool for value chain improvement.

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Norway: Mo i Rana (MO) is the second largest industrial center in Norway, located in Nordland County, just below the artic circle. The industry in MO stems from the 19th century and today over 100 businesses with extensive IS (SRL level 6), employing over 2900 people, and with a turnover of over 1 billion €/year are present in MIP. At the core are 3 smelters, including a steel smelter recycling steel from the urban sector. The smelters provide products and by-products utilized by the urban sector: rebars, asphalts, concrete fillers and aggregates. Further, the district heating network is mainly powered by surplus energy and cooling water is utilized for fish farming. Through work in clusters, such as ACT (> 100 businesses from process industries and related value chains) and Concrete Innovation Cluster (CIC, >80 businesses), particularly IS, but also some I-US has been promoted. Objectives related to the replication: 1) Testing and adaption of methodology for systematic promotion of I-US from Attica hub, 2) adaption of Attica hub innovations to develop foundation (business model and plans) for > 2 I-US value chains and 3) adapt Attica hub methodologies for geographically expanded areas. Sources and Value Chains: CDW: Management and re-utilization of CDW from urban and industrial building stocks. Plastics: Plastics sorted from household mixed waste and industrial waste collection points. Textiles: Textiles sorted from household mixed waste and end-of-life textiles from collection points. Water sludge: Urban sludge and sludge from on land fish farming. Main Goals to be achieved through project: Co-develop the CDW value chain with partners of ACT and CIC, including identifying logistic solutions, formulating business models, studying the effect of incentives and impact on sustainability; facilitate establishment of the other described value chains based on: adaptation studies, sustainability assessments and business model formulations for replicating main hub innovations in 1) MO based hub for circularity and 2) geographically expanded hub in Nordland. Main goals will be achieved under the guidance and uptake of a regional reference group to also expand on current collaboration models and policy for circularity (IS and I-US).

1.2.3 Synergies with other national and international projects

Relevant Objectives	Added Value in Theseus
TEXTENDED (Horizon Europe), Dec 2022-Nov 2026, RISE	
An innovative I-US approach of discarded textiles with the development of a Blueprint, a knowledge-based masterplan for optimized cycling. Under same topic of REDOL H4C.	RISE leads on the Sustainability section of the project, involving LCA, sLCA and LCC.
W2W (Horizon Europe), Jan 2024-Dec 2027– ICCS (Project Coordinator)	
A comprehensive framework for multi-dimensional cascade valorisation of wood from C&D and furniture waste including sorting, upcycling and digital tools.	DPP expertise, sorting and separation technologies will be used and enhanced by ICCS. The CDW flow of W2W will be used in parallel in THESEUS project.
REDOL (Horizon Europe), Dec 2022-Nov 2026 – CIRCE (Project Coordinator), ICCS, IRIS	
Establishment of H4Cs in urban areas with zero waste policies by optimising value chains with digital tools.	CIRCE will bring expertise from packaging, plastics, construction waste, and textiles. ICCS, IRIS will use knowledge on H4C/I-US digital tools.
IS2H4C (Horizon Europe), Jan 2024 – Jan 2028 H4C (IRIS, ICCS)	
An innovation workplan of H4C development in diverse areas of process industry surrounded by rural and/or urban settings.	IRIS. ICCS will use knowledge on H4C/I-US digital tools, marketplace, recommenders and indicators measuring nontechnological hub impact.
CORALIS (H2020), Oct 2020-Sep 2024 – CIRCE (Project Coordinator), ICCS, RISE	
Pathways creation for resource decarbonisation in energy intensive sector through viable IS approaches.	CIRCE and ICCS will use knowledge for circular resources, monitoring and IS data sharing & management. RISE will transfer and adapt SRLM.
SHAREBOX (H2020), Sep 2015-Aug 2019 – IRIS	
Secure Management for Shared Process for flexible energy and material flows.	IRIS will transfer practices/solutions from digital platform
ACCELWATER (H2020), Nov 2020-Oct 2024 – ICCS, NTUA	
WaterEnergyFood nexus in food industries for effective use of resources.	Water reuse practices(NTUA);AI & IS platform practices(ICCS)
HYDROUSA (H2020) July 2018 – June 2023 – NTUA (Project Coordinator)	

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Non-conventional water sources NBS valorisation, WEFE Nexus	NBS in urban peri-urban environment; Fertigation using reclaimed water.
DigInTrace (Horizon Europe), Jan 2023-Dec 2026 – ICCS (Project Coordinator), NTUA, CIRCE, IRIS	
A decentralised traceability platform for tracking, sorting, quality certification schemes, AI-based decision-making mechanisms.	Separation / sorting systems and AI digital tools for circular resource valorisation. Expertise in materials flows.
Autosort for Circular Textiles Demonstrator ACT UK (Innovate UK), Oct 2023-Sep 2025 - HUD	
A pioneering pilot-scale fully integrated, automated sorting and pre-	Lessons learnt from this pilot facility will inform the textile case
processing demonstrator for post-consumer waste textiles (ATSP).	study.
Textile waste refinery to produce recycled plastic, cellulose and dye (EPSRC), Oct 2023-Sep 2026 - HUD (PC), ICL	
A novel AI green recovery of polyester & cellulose from mixed textile waste.	The outcomes in terms of optimized reuse of textiles.
AquaSPICE (H2020), Nov 2020-Jan 2025 – TUC, ICCS, NTUA, HUD	
Materializing circular water use in process industries.	Analytics and process optimisation, Dynamic LCA. Material Flow Analysis.
CSSBoost. (Circular Cities and Regions Initiative – CCRI, upon GAP phase), 2024- 2027 – TUC, ICL	
CSS lifecycle management services for monitoring & optimisation, and cognitive Digital Twin virtualizing regional/ interregional CE markets & ecosystems.	Framework, methodology and tools for circular value chain management and systemic regional solutions.
EVELIXIA (Horizon), Oct 2023-Oct 2027 – CERTH, DHCP	
Integration of DERs across energy networks, transforming buildings into active nodes of power generation & storage.	Lessons learnt from adaptation of CHP from green H2 to DH city system, optimized use of green H2 (Greek pilot) for covering energy needs.

1.2.4 Interdisciplinarity and Regional Science

To promote overall sustainability, systemic and regional approach, H4C implementation of I-US by cross-sectorial cooperation on new circular solutions, waste, energy, water and promote exploitability, Theseus must consider many different aspects and involve different scientific domains. Theseus works in a strong inter-disciplinary field, involving expertise from various disciplines including natural and applied e.g. chemical engineering, physics, ICT, data science, electrical/mechanical engineering and economic and social sciences, e.g. regional science, sociology, economics, political science, ethics, law, business administration, communications, learning, etc. These are interwoven within activities and will be brought together and integrated in pursuit of project objectives forming inter- and trans-disciplinary conditions, used to develop innovations in different fields, and support policy influencing, standardization, business cases, exploitation and upskilling. Theseus will pursue the involvement of all regional actors to consider the complex requirement of the specific markets in terms of standardization and performance. Theseus recognises the difficulties of a trans-disciplinary collaboration (e.g., different language, terminology among different disciplines and methods, different levels of scientific understanding and cognitive structures etc.) and adopts a proactive approach towards them by planning to follow the fundamentals of the Methodology in Interdisciplinary Research framework[40]. Consortium will employ a quality assurance process (T1.1) to ensure that all trans-disciplinary aspects are integrated in the solutions, thus creating a metaknowledge about all the different disciplines, methods and epistemologies, and purposefully and reflectively integrate and synthesise different perspectives, to better understand and contribute to the new processes of H4C. Health related principles[41] will be used to analyse the impact of Theseus activities to healthier cities for the citizens. A special focus will be placed on Regional Science to ensure that solutions contribute to the overall sustainable regional development, including resilience of Attica and peripheral regions and national, by analysing the intra &

[40] Tobi, H., Kampen, J.K. Research design: the methodology for interdisciplinary research framework. (2018). https://doi.org/10.1007/s11135-017-0513-8

[41] https://www.who.int/activities/promoting-health-in-all-policies-and-intersectoral-action-capacities https://iris.who.int/handle/10665/342218

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interregional interactions, the position with other countries and EU, replication potential, synergies upscale and commercialization.

1.2.5 Gender dimension

Theseus will overcome sex and gender bias by disaggregating socioeconomic and biological differences between man, woman or other genders for inclusive research and innovation by integrating activities inspired from gendered innovations proposed by EC[42]. PM (WP1) will be engaged to gender issues including them in period's reports. To overcome gaps (considering occupational and social status of genders) a draft study will be performed with potential actions within project to ensure gender-sensitive objectives, results, activities, partners assessment, management, indicators for monitoring and resource allocation. The motivation of these activities is equality and release of special creativity of all genders and individuals, creating added value for research, innovation and society.

Proportionality and balance in research: equal participation of all genders at all project levels will be ensured and promoted. All partners are encouraged to increase women's and other genders' participation in activities including management, research, analysis, networking etc. to assure equal opportunities, recruitment and arrangements. In addition, there is gender balance in the core project management activities and in the principal investigator per partner. All partners provide equal opportunities on employment and participation in research multinational projects with special care to contributing lessening the disparities being a result of COVID-19 pandemic[43] (e.g. flexible working schedules to parents of all genders according to their special needs).

Scientific and linguistic bias: addressing in texts and material (e.g., personas) will follow gender-neutral language[44], ensuring data, algorithms, digital tools, technologies and visuals are free of gender stereotypes. In all studies gender dimension will be considered. Focus groups dedicated to diversity and gender will be included. Collection of gender statistics employing AI and use of related metadata is also encouraged to identify challenges in policymaking and to better understand gender-specific needs, issues and gaps. Annotating source data for sex and gender will be ensured. Special biological and other factors limiting or exposing to special dangers and gender-specific risk (e.g., specific professions and activities) will considered. Gender participation and real influence will be studied upon the operation and the decision-making in the governance structures of Theseus as well as in the variety of entities and sectors involved.

Equal projects' results: Theseus ensures that new solutions usage (technologies, business models, processes, digital tools, etc.), access to resources and services and the related trainings, foster the involvement of women and other gender scientists in the future landscape.

Equal dissemination: dissemination & communication in public, industry and business, policy recommendations and institutional regulations, will be addressed to all genders equally and adjusted to their needs.

Connection with established gender activities & expertise: Theseus will seek connection with EU organisations and institutes, academia studying gender, national gender equality bodies etc. for knowledge sharing, participation in activities, provision of data in EU gender statistic database (EIGE), policymaking or studies. Existing tools will be considered into the project e.g. checklists (i.e., Yellow Window, IGAR) guidelines and research methodologies.

Gender equality promoting: Theseus focuses on gender equality to be oriented on the participating partners making difference within their workspaces. To assure full integration of gender dimension, a dedicated session will be shared, where general topic knowledge, all necessary information and guidelines (incl. gender-neutral language guidance) will be held for the partners and other participants , early in the project. Partners will be supported to implement their own gender equality plans.

1.2.6 Open science practices

Theseus will establish open science principles, fully aligned with the ambition of the European Open Science Cloud as a trusted environment for hosting and processing research data to support EU science in its global leading role; aligned to Open Innovation 2.0 paradigm 'as open as possible' in the goals of Open Science Policy under Horizon Europe. Appropriate practices will be implemented in terms of coordination, monitoring, management, dissemination, communication replication, cooperation. Specifically:

Research output management and impact to industry: research outputs available to dedicated target groups (i.e., authorities, sectoral, recycle, product and material associations, industry and businesses, SMEs, research

[42] European Commission, Directorate-General for Research and Innovation, (2013) Gendered innovations: how gender analysis contributes to research: report of the expert group 'Innovation through gender'. Publications Office. DOI:10.2777/11868

[43] European Parliament News, Understanding Covid-19's impact on women (infographics), (2021)

[44] Papadimoulis D., GENDER-NEUTRAL LANGUAGE in the European Parliament, (2018)

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community, EU and global associations (e.g., ERSA & OECD[45], A.SPIRE/P4P (in consortium), DGs, SPI, Orgalim, EIT RawMaterials, CEN, CENELEC, standardization bodies, etc.) to be leveraged from results and ensure replicability.

Early and open sharing of research results and data: 5 case studies publications in Open Research Europe, one for each of the pilots. Measures to ensure reproducibility of research outputs: providing standardization deliverables and harmonisation of data of different fields, will ensure that research outputs follow common standards and support mainstream and widespread reporting and documentation methodologies.

Providing open access to research outputs and Participation in open peer-review: partners are committed to use the Open Research Europe scholarly fully open access publishing service for Horizon Europe to enable rapid publication times and publication outputs supporting research integrity, reproducibility, transparency and enable open science practices. The project has dedicated budget for open-access publication to facilitate dissemination and reuse of results. Participants, for various reasons, may need to submit articles to journals (or proceedings) that only offer a lower level of open access, requiring either parallel publication or an embargo period. In this case, Green or Gold Open Access will be chosen depending on each publication. For green one, developers will deposit final peer-reviewed manuscript in a public repository of their choice, ensuring open access to publication within the embargo period of max 6-months.

1.2.7 Data management on FAIR principles and research outputs

Theseus is expected to collect data in the scope of project activities and value chains to assess the technologies and effectiveness of the proposed solutions in real life conditions. All the data, available directly by partners involved or research results like models, algorithms, scripts, code and analysis, will be aggregated and anonymised according to visibility and security level defined. As for experimental measurements and datasets (raw, processed, logs etc.) of processes during demonstration, will be anonymised and secured in Theseus platform. Protocols and data types will be defined according to the needs in T3.5 and constantly monitored through T1.2. Publishing all data generated will be aligned with ethic issues and privacy in a way not affecting exploitation or competitiveness. Aligned with Open science, Theseus will make data not only “as open as possible: but also “as closed as necessary” to encourage collaboration across disciplines for solving today’s complex societal challenges. So, with respect to confidentiality, most of research data generated by the project will be made open and will be offered to the Open Research Data Pilot (Open AIRE). Theseus will generate research results and data with FAIR principles: Findable: with open science research described, proper identifiers e.g., DOIs and standards; Accessible: integrated to databases and expanding; Interoperable and Reusable: standardised, harmonised using common vocabularies, open standards and sharing principles to be easily adjusted and used in further research. Data will be maintained after the project ends for at least 2 years, so that the knowledge management will follow principles of discoverability, accessibility, intelligibility and availability beyond Theseus.

1.2.8 The “Do not harm Principle”

Theseus will ensure full DNSH Principle compliance with activities and expected life cycle impact of innovations (commercialized) contribute in a positive way and/or do not significantly harm any of the six environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation 2020/852. Theseus will apply a life-cycle oriented approach (validated through relevant LCA tool when possible) to ensure that proposed solutions are sustainable, in 3 steps: i) Identification of actual /potential adverse impacts during life

cycle of project activities; ii) Prevention or mitigation of adverse impacts; and iii) Accounting for how adverse impacts are addressed by tracking performance and communicating results (ICT technologies will be validated with Green ICT principles). Activities are expected to have a direct positive impact to objectives defined: Climate Change Mitigation by reducing GHG emissions (use Rec 2013/179/EU or ETSI ES 203 199, ISO 14067:2018 or 14064-2:2019 to calculate lifecycle GHG and net emissions), pollution prevention and control through waste reduction, identifying harmless substances and contaminants, transition to CE through symbiotic flows, whereas is not expected any negative impact to sustainable use and protection of water and marine resources, protection and restoration of biodiversity and Climate Change adaptation. #§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§#

2 Impact #@IMP-ACT-IA@#

2.1 Project’s pathways towards impact

[45] Organization for Economic Co-operation and Development

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Waste management remains one of the most serious environmental issues in Attica holding back transition to circular economy with main figures for municipal solid waste indicated in section 1. Theseus impact within its symbiotic material flows is presented through the lens of regional (Attica) and national (Greece) targets to be reached by 2030 as defined in Regional Solid Waste Management Plan (SWMP)[46][47] and Regional Plan for Climate Change Adaptation.

A. Municipal Solid Waste (MSW) includes the summary of all waste from channels (mainly from trash bins) within city and from industrial (only) packaging wastes. In Greece, the municipal solid waste composition (%) is as follows: paper 28.5, biowaste (organic waste) 39.2, plastic 11.8, metal (Fe) 2.2, metal (Al) 0.9, glass 4.1, textiles 2, wood 3.6, bulky waste 2, waste electrical and electronic equipment 2, small quantities of hazardous waste 0.1, rest 3.6. The special channel of Table 1.

Year	2019	2030	
Total	2.198.746 tn	Est. 2.068.810 tn	
Landfill	1.707.233 tn - 77,7%	268.810 tn -	10%
Recycling rate[54]	272.112 tn - 12,4%	1.277.770 tn -	65%

Attica MSW blue bins (WATT) collects recyclable packaging from citizens, but their misuse results often that their composition is similar to that of common bins, thus Theseus activities can potentially be expanded to all MSWs. Textiles are also included in MSW, collected by dedicated channels (purple Bins by RECYCOM) or disposed in blue or common bins. Thus, both Pilots #1 and #2 concern MSW. For the 2 pilots:

main source is blue bins specialized in packaging including plastic packaging and addressing also biobased

Packaging: blue bins are designed for packaging collection (collected 81.351tn in Attica 2019). Packaging has overall recycling targets 70%w/w by 2030, including i) Plastics: blue bins & MSW with target [48]% by 2030, ii) Paper: blue bins & MSW 85% by 2030 iii) Biobased (organic material): accidently in blue bins & 39,2% of MSW -35% landfill disposal by 2030. A2. Pilot #1: main source is Textiles from Purple Bins (more than 2.500 "purple bins" all over Greece, RECYCOM managed more than 18.500 tons of clothes since 2012 and 4.865 tons in Athens in 2023). Textiles refer to 11,8% of USW with projections collect 27.778 tn from special points common bins and recover 14.958tn by 2030 from common bins by 2030.

B. CDW: 997.742 tons were collected in 2019, and projected value for 2030 is 2.158.177 tn. Recycling/reuse rate is currently estimated to be around 30-35%. For B1. Pilot #3: main source is CDW and metals from and glass bottles, thus impact can potentially affect glass and metals flows in the above streams as well: Glass: 3,1% of MSW & 30% of blue bins target: 75% by 2030, Metals: 4,1% of MSW & 20% of blue bins target: 70% by 2030.

Theseus Contribution

Theseus contribution to the to the above targets is summarized in terms of waste diverted from landfill and GHG emission reduction.

Table 1 estimated saved GHG emissions

Flow	waste diverted from landfill (tons/yr)	tons CO_2e/ yr[55]
Recyclable materials, Blue Bin short term (long-term)	16.270 (38670)	11.063,6 (26295,6)
Textiles	7.784	155,68
CDW	20.055	401,1
Total	44.109 (66509)	11.620 (26852)

MSW: Blue bins collected 81351tn of waste in Attica in 2019. Theseus will increase the accuracy of separation leading to better upcycling and valorization. Theseus sorter will be capable to recover plastics, metal, paper, textiles as well as biowaste from urban waste. Currently 50 % of materials is upcycled by WATT while the Theseus target is to reach 70 % (and consequently meet the regional target), therefore contributing to 16270 t/yr diverted from landfills (11063,6 tCO_2e). The following composition of packaging waste is assumed, based on the report of Hellenic Recovery Recycling Corporation[56]: paper (all types) 45%, glass 13%, plastic 25%, metals 9,5%, wood 8%. In middle and long term, through social activities and education focusing on correct recycling at a source, increasing the predictability of the materials inside the blue bins it is likely, that the input can be increased up to +40%, given that

[46] https://www.edsna.gr/s-m-p-e-3is-anatheorisis-pesda/ (Last updated in June 2022)
[47] Preparation of recycle and recovery defined in https://www.edsna.gr/s-m-p-e-3is-anatheorisis-pesda/ (Last updated in June 2022)
[48] Emission Factors for Greenhouse Gas Inventories (epa.gov) factors for Mixed recyclables, textiles carpets and different CDW materials were used to calculate estimation [56] Έκθεση του Συστήματος Συλλογής Εναλλακτικής Διαχείρισης – Ανακύκλωση (herrco.gr)

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plastics and paper are the 40,3% (thus 833.730 tn by 2030) of MSW and 45% of total packaging waste, therefore 32.000 tn. Recycled by 70% could result in extra recycled 22400 tn/year, thus 38670 tn/yr (26295,6 tCO_2e). disposal saving than current. Also, biobased (organic materials) are a huge part consisting 39,2% of MSW and often accidentally or attached to packaging end up in blue bins. Theseus will contribute to its detection and upcycling in packaging materials.

Textiles Immediate impact in RECYCOM facilities with Theseus interventions is that 50% of the existing sorting line will be automated. The

automated part of the line is estimated to exhibit roughly 30% increase in capacity compared to the previous manual processing (approx. 3162 tn per year compared to the current 2432 t) thus 730 tn per year (15 % increase in total). In middle and long term, Theseus will propose a shift, in how textiles are handled and contributing to a national textile management system, that could potentially handle all the textiles collected yearly in Attica with projections for 2030 being (27.778 tn from special points and14.958 tn common bins) namely around 42.500 tn/yr. After the end of the project and by 2030, RECYCOM could be fully automated increasing the textiles managed by total 30%, to extra 1460 tn/yr. At the same time, with an strategic increase of 100% to the purple bins system capacity (or similar recycling establishments by region), along with training and social awareness raising, extra 6324 tn of textiles (15% of 42.500 tn that could end in landfill as there is no wide textile management system) could be collected, resulting to a total of 7784 tn/yr extra landfill savings (155,68 tCO_2e) from current status.

CDW (excavation not considered) 888270 tons were collected in 2019, and projected values for 2030 are 959606 tn.

Recycling/reuse rate is currently estimated to be around 33-35%, therefore around 302, 000 tn are recycled right now[49]. For CDW sorting Thesues will build on work performed within W2W project, where an adaptive system for wood and glass separation from CDW is developed. Having in mind, that Thesues will focus on adaptation of a well- optimized system for further separation of metals from CDW and that CDW is expected to be accepted from multiple sources, including renovation the target for the recycling rate after completion of Theseus was set to ambitious 50 %. To increase the possibilities of achieving this goal Thesus will work with region of Attica towards optimized system for CDW management, including potential increase of the numbers of permission for SMEs dealing with CDW waste or consider moving the CDW management to neighboring regions. In typical CDW composition for Southern countries we encounter 11,28 % of metal[50] , resulting in (11,28% 100196 t of CDW collected in 2019, with 34067 t recycled. Managing to achieve target recycling rate for 2030: 50 % would result in 54122 t/y recycled, and 20055 t/y diverted from landfill (401,1 tCO_2e).

Energy flow: Renewable energy sources account for 22,68% of total energy generation in 2022 in Greece and for 30,61% of the total energy produced for heating and cooling needs compared to 23,03% and 24,81% in EU-27 respectively for the same year[51]. The impact of Theseus in the energy flow will be multi-fold. Theseus will develop a pilot where an investment of 3.000.000 € in solar thermal technology and fuel cells will introduce more than an additional 3.000 MWh of clean energy into the district heating and cooling network in Greece, showcasing that RES energy for district heating and cooling (DHC) is not only an environmentally friendly, but a financially viable solution as well. Moreover, the potential for a bigger adoption of DHC systems in Greece and the world is huge. In 2018, a little less than 6% of global heat consumption was supplied through DHC networks with fossil fuels dominating as the primary energy source of DHC due to the extensive use of natural gas in Russia and coal in China that combined account for 2% for the global DHC consumption. Energy demand met by DHC networks is projected to increase by +4% globally, whereas RES penetration into the DHC systems is expected to increase by 40%[52] Therefore, Theseus could have a wider impact into the decarbonization of the heating demands on a global scale.

The main stakeholders of the THESEUS are expected to belong to the following target groups: I) Industries, where Ia. associated with P4P partnership and Ib. non-P4P industries, II) Local communities-citizens, general public III) local and regional administration IV) Policy makers V) Partnerships, Associations, NGOs working with circularity, environment, human health, etc. VI) Scientific and academic community. THESEUS Key Exploitable Results (KER)

KER	Focus	KER name	Value chain	Partner	Research output	SO	Target

[49] https://www.edsna.gr/s-m-p-e-3is-anatheorisis-pesda/ (Last updated in June 2022)
[50] https://publications.jrc.ec.europa.eu/repository/handle/JRC130992
[51] Eurostat: https://doi.org/10.2908/NRG_IND_REN
[52] How can district heating help decarbonise the heat sector by 2024? – Analysis - IEA

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Innovation Pillar 1 COLLABORATION							
KER1	Projects & sites	Roadmap for collaboration (i)between sites (GR, SWE, NO) and ii) between Hub & CCRI & Replication Roadmap Tool	All	CIRCE, TUC	Report, software	5, 6	All
KER2		Common hub circularity guidelines	All	A.SPIRE	Guidelines	All	All
KER3	Citizens	AR for social awareness and acceptance	All	ICCS	Report/Framework	6	All
Innovation Pillar 2 SYMBIOTIC FLOWS (MATERIALS)							
KER4	Materials Circular products /technologies	Circular construction material	CDW, Textiles	SINTEF/TUD	Circular product prototype	2	I, VI
KER5		Biobased thin films for packaging	MSW	NCSR	Prototype	2	I, VI
KER6		Electrospinning for thin films	MSW	NTUA	Technology	2	I, VI
KER7		Biological recycling & spinning for upcycling	Textiles	CIRCE, HUD	Technologies	1, 2	ALL
KER8		Upcycled packaging, masterbatches	MSW	HMU	Prototype	2	ALL
KER 9		Strategic Master Plan on Circularity	All	NTUA, EYDAP	Framework	3	I, III, IV, VI
Innovation Pillar 3 SYMBIOTIC FLOWS (WATER/ENERGY)							
KER10	Water	Decentralized sewage mining and NBS for water reuse	Water	EYDAP	Technology/Report	1, 3	I, II, III, VI
KER11		Sewage sludge valorisation	Water	NTUA	Technological scheme	3	I, III, IV,VI
KER 12		Centralized reclaimed water fertigation	Water	NTUA	Technological scheme	3	I, II, IV, VI
KER13	Energy	H2 for production of heat and electricity through a CHP	Energy/Water	CERTH	Technological scheme	3	I,II, III, VI
KER14		New generation of HVFP solar thermal technology for greening District heating and covering cooling loads of buildings	Energy	CERTH, TVP, CLEAN	Technological scheme	3	I, II, III, VI
Innovation Pillar 4 DIGITALIZATION & ROBOTICS							
KER15	Sorting	Automated textiles separation system	textile	ICCS	Prototype, algorithms	1, 2	I, V
KER16		Automated sorting for mixed waste and plastics	MSW	ICCS	Prototype, algorithms	1, 2	I, V
KER17		Industrial sorting systems: Glass/copper & metals sorting system	CDW & Industrial	ICCS	Prototype, algorithms	1, 2	I, V

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KER 18	Digital	XR for human-robot interactions/collaboration	MSW textiles	ICCS	Prototype, software	1,2	I, VI
KER19		Dynamic symbiotic maps, prediction models, DPPs and optimization for smart bins	MSW	ICCS	Framework, prototype, algorithms, aggreg.data	1,2,4	I, II, III, VI
KER20		H4C Marketplace with advanced matchmaking features & secure data exchange	All	IRIS/ICCS	Prototype	1, 4	I, III, VI
KER 21		Models: i) WEF nexus, ii) Cities climate & impacts	All	Imperial, NTUA	Algorithms, aggreg.data	1, 4	I, III,VI
Innovation Pillar 5 GOVERNANCE							
KER22	Recommendations	Analysis of symbiotic flows in the context of EU and local regional plans. Regional and Intraregional development guidelines	All	ICCS, UAGEAN, LiU	Report		1I, III, ,IV, VI 6
KER23		Policy guidelines based on Material Flow Analysis, impact on the health and de-risking assessment	All	LiU	Report		4IV
KER24	Standards	Contributions to CEN and/or ISO standards	All	DS	Standards		1All , 6

2.1.1 Contribution of the project to the expected outcomes specified in the topic

Expected Outcome 1: "… Demonstrate zero urban waste in a near commercial scale environment through systemic resource recovery as alternative material feedstock; a decrease of GHG emissions is also expected by explicitly addressing the reduced flow of goods (due to geographical proximity) ..."
Pathway to impact: Resource recovery as material for feedstock is a critical aspect of THESEUS. THESEUS is addressing materials, water and energy flows. Management, processing and valorization of secondary resource streams will be improved. Implementation of automated sorting systems for different types of waste: MSW, used clothes, construction and demolition will lead to reduction of waste as after processing the waste will serve as feedstock for own or other industries. The resource recovery will be enhanced through systemic approach to reduction of flow of goods. It will be an outcome of deploying diverse methods and addressing the issue from different perspectives. Material flow analysis with project until 2050 will be performed. Results of the study will be combined with the logistics aspect of the good transportation, and with analysis of regional and interregional pathways for symbiotic flows linked to EU and local regional plans. That impact of those activities will be intensified by digital technologies contributing to reduction of flow of goods: GIS tool for dynamic symbiotic maps, matchmaking frameworks symbiotic models, DPPs, Material/Water financial and quantity quality prediction, water-energy-materials-waste nexus model as well as DPPs. All this will allow to optimize existing and propose more efficient future symbiotic collaborations with reduced flow of goods. This approach will be complemented by social awareness increase activities (using engaging AR technology and educational activities). Furthermore, the reduction of the GHG emissions will be complemented by smart grids and sensors used identify waste heat sources, facilitating the recovery of this excess heat which becomes an alternative material feedstock for district heating systems, minimizing waste and reducing the reliance on conventional energy sources. By prioritizing these
sustainable sources, the system reduces reliance on fossil fuels feedstock, contributing to lower GHG emissions and enhancing resource efficiency.

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KERs: KER3-KER21, 22-24, Target Groups: Industries (I), local communities and administration (II, III), Partnerships, Associations, NGOs (V), scientific community (VI)	
Expected Outcomes KPIs and [Target]: KPI1.1 Number of closed loops [5]; KPI 1.2 Number of established cross-sectoral collaborations within geographical proximity [min. 5]; KPI1.3 Increase in efficiency of flows sorting [97%]; KPI1.4 Number of DPPs for circular products [6]; KPI1.5 Number of models developed and validated for prediction: [min 3] KPI 1.6 Cost reduction of transportation [min15 %] KPI1.7 reduction in materials transported from other regions [min 15 %] KPI1.8 Companies implementing upcycling processes and technologies: [min 10]	
Significance: Increase in the use of recovered resources: 30%	Scale: Reduction of GHG emissions by 26852 t CO2e/ yr
Expected Outcome 2: "Reduce the freshwater consumption for urban green irrigation by 50 %, and re-use 90 % of the solid waste generated by the wastewater treatment"	
Pathway to impact: Technical and non-technical innovations will be used within the water use case to generate the impact through valorization of water flows and integration into circular – symbiotic value chains. Centralized and decentralized solutions will be implemented. for the purpose of decentralized water reuse sewer mining technology will be implemented making it possible to reuse water for irrigation. To increase the impact sewage mining implementation will be complemented by assessment of the technology in the context of regulatory requirements, its cost effectiveness and sustainability. Implementation of NBS will contribute to water recovery and availability and to reduced flood risk, which is a serious problem in Attica region. Theseus will also take care of sludge from wastewater treatment plants and use technologies to transform it into hydrogen and biofuel. An important milestone of the pathway toward the impact will be the co-development of the Strategic Regional Master Plan for Circularity in Attica, where the most important actors from public and industrial sectors will collaborate towards improved wastewater management and going beyond the technical aspect with business cases established and pricing proposals. Within plan conclusions, lessons learner, challenges encountered during demonstration will be gathered in the context of future investments and present/upcoming regulations for circular water management for whole region.	
KERs: 9-12 Target Groups: ALL	
Expected Outcomes KPIs and [Target]: KPI1.1 Min. number of actors collaborating towards Attica Reuse Masterplan [min.10]; KPI1.3 Number of sites with sewage mining implemented [min. 3]; KPI1.4 number of NBS implemented [min 2] KPI1.5 Increase in resuse of water for Irrigation of crops and urban area [20%] KPI1.6 Rainfall runoff reduction [by 15%] decreasing flood risk in intervention areas; KPI1.7 Rainfall runoff peak discharge reduction [40%] decreasing flood risk in intervention areas	
Significance: Overall increase in water re-use by 50 %	Scale: re-use 90 % of the solid waste generated by the wastewater treatment, Reduction the freshwater consumption for urban green irrigation by 50 %
Expected outcome 3: "Citizens living in cities will benefit from a healthier environment through industrial/urban symbiosis by lowering emissions through circular an renewable energy sources and waste reduction"	
Pathway to impact: The link between human health and clean environment has been well documented[53] [54]. In huge cities like Athens maintaining clean environment is however a big challenge. The implementation of Attica hub for circularity through industrial/urban symbiosis will introduce a number of actions, which directly and indirectly are linked to healthier environment and therefore healthier citizens. Through implementation of Theseus I-US it is expected that healthier environment will be achieved through i) reduction of waste generation ii) reduction of GHG and other emissions iii) lowering the amount of waste that needs to be landfilled iv) wastewater discharge will be reduced through water reuse activities, v) use of renewable energy sources, vi) improving air quality, vii) mitigating the climate change. In order to evaluate the actual effect, that the implementation of the hub will have on the health of its citizens, THESEUS will perform an assessment based	

[53] https://www.frontiersin.org/articles/10.3389/fpubh.2021.756652/full
[54] https://iris.who.int/bitstream/handle/10665/204585/9789241565196_eng.pdf?sequence=1

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on the framework proposed by WHO on health benefits in the context of transition to circular economy[55] and LCA with focus on health related categories: human toxicity (HT), particulate matter formation (PMF), photochemical oxidant formation (POF), ozone depletion (OD).

KERs: ALL Target Groups: ALL

Expected Outcomes KPIs and [Target]: KPI1.1 Number of positive health benefits for the citizens identified [at least 20]; KPI1.2
Improvement in Air quality index (AQI) [At least by 25%]; KPI1.3 Improvement in health related LCA categories [At least by 30 %];

Significance:	Scale:
Iincreased expected lifespan and lower expenses on healthcare	Almost 4 mln inhabitants of the Attica hub for circularity (~36% of population of Greece) and beyond.

Expected Outcome 4: Use urban/industrial symbiosis and cross sectoral cooperation to pave the way for achieving the EU Green Deal and "Fit for 55" package objectives: providing recommendations for optimized regional framework conditions by highlighting barriers and suitable innovation-oriented policies for possible synergies with the cities selected by the Cities Mission …."

Pathway to impact: The European Green Deal is the strategy on addressing climate and environmental challenges. Through employing urban/industrial symbiosis and enhancing cross sectoral collaboration Theseus is fully in line with EU Green Deal and especially

contributing to its following elements: EU Climate ambition (2030 as a milestone synchronised with project completion and 2050 in the long run), engaging several important industry representatives towards transition to clean and circular economy, contributing to reducing pollution toward toxic-free environment with health impact assessment allowing to identify the relevant benefits, and through the energy flow related activities towards supplying clean and secure energy. 'Fit for 55' package is a set of proposals that were adopted in order to update EU legislation related to climate, energy and transportation, to facilitate achieving EU's climate goals and relate the ambitions described in GD into law. Theseus to several of them including Renewable Energy Directive, Energy Efficiency Directive, Energy Performance of Buildings Directive and Updated EU rules to decarbonize gas markets and promote hydrogen. Athens is on the six Greek cities chosen for the Cities Mission. By working with local authorities, implementing participative approach for all the stakeholders Theseus will significantly contribute to achieving the Missions aims by contributing to achieving climate-neutrality of Athens and Attica and by operating within the framework of hub based on experimentation and innovation. Thesus will propose solutions tailored for Athens at the same time supporting scalability and replication to other cities. Cross-sectoral approach and synergies with existing initiatives (including projects, other H4C, CCRI) will be established. To assure that Theseus generates expected impact, the insights, experiences, knowledge, barriers and challenges and mitigation measures will be synthesized in following thematic reports: Roadmap for collaboration (i) between sites (GR, SWE, NO) and ii) between Hub, Common hub circularity guidelines by A.SPIRE, Strategic Master Plan for Circularity, Analysis of symbiotic flows in the context of EU and local regional plans, Regional and Intraregional development guidelines and Policy guidelines based on Material Flow Analysis, impact on the health and de-risking assessment. All of those documents are expected to contribute to paving the way towards innovation-oriented policies for Athens, but also other 5 Greek cities selected for the Cities Mission as well as Malmo.To facilitate further policies creation, Theseus will make an attempt to contribute to CEN and/or ISO standards.

KERs: 1, 2, 9, 22, 23, 24 Target Groups: ALL

[55] https://iris.who.int/handle/10665/342218

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Expected Outcomes KPIs and [Target]: KPI1.1 Policy guidelines shared with regional policy makers of [min 3 regions] KPI1.2 Number of innovations highlighted in the policy guidelines: [min 5]	
Significance: Contribution to achieving climate neutrality	Scale: Number of cities from the Cities Mission on which Theseus might have direct impact: 7

2.1.2 Contribution of the project to wider impacts specified in the destination

Expected Impact: "Global leadership in clean, climate-neutral and resilient industrial value chains, circular economy and climate-neutral and human-centric digital systems and infrastructures."
Theseus project is fully aligned with the topic by working towards clean, climate-neutral and resilient industrial value chains, that are supported by digital systems and infrastructures facilitating tasks and providing support in decision making in the context of I-US and CE. Theseus is fully aligned with the key objectives of the destination and goals of the EU policy. Theseus deploys the HC4 concept involving and motivating European industries in achieving Green Deal and "Fit for 55" objectives. Through the development of Key Enabling Technologies and their validation, Theseus contributes towards achieving the goal of the EU Climate Neutrality, with digital, low-carbon and circular industry being one of core elements, as described in the Green Deal. Theseus proposes innovative products and technologies supported by robotics (incl. XR for human-robot interactions/collaboration) and digitalization supporting informed decision making, optimization and enhancing collaboration, while establishing secure and sustainable IT infrastructures in line with Digital Decade. Those activities are accompanied by exploitation strategies, business models, financial solutions. The implementation of the hub will have direct effect on the environment and health, therefore supporting Zero pollution action plan. Global leadership is additionally boosted through intensified collaboration cross-sectoral, between other regions, H4C and CCRI as well as guidelines for regional and interregional development, policy guidelines and contributions to future standards. Theseus introduces H4C in one of the EU cities of Cities missions, supporting its goals with replication cases, which will facilitate replication also beyond the project.
Topics Serving the Objectives of the Destination and relevant to Theseus Project
Manufacturing Industry: Theseus implements I-US, re-integrates diverse flows in the manufacturing of circular products working towards negative carbon footprint, keeping resources in the loop and aiming to boost circular, carbon-neutral industries operating on new, circular terms. This new model introduced by Theseus is supported by educational activities for the workforce to assure smooth transition.
A new way to build, accelerating disruptive change in construction: CDW is one of the flows of focus of Theseus, therefore the project engages several important industrial representatives of the CDW sector. Theseus aims to improve the resource efficiency in the sector by expanding separation and sorting of CDW, supporting digitalization and upskilling.
Energy efficient and climate neutral process industries: Theseus aims to contribute to achieving P4P partnership objectives. Theseus will work on closing circular loops involving flows connecting different manufacturing sectors and P4P industries (cement, chemicals, non-ferrous metals, water) towards climate neutrality.
Circularity and Zero Pollution in process industries: Theseus will directly contribute towards the optimization of resources and keeping them in the loop, reducing landfilling, improved waste management and lowering emissions.
Hubs for Circularity (H4C): Theseus is directly addressing this topic, implementing I-US, enhancing collaboration at maximum levels within and beyond value chain, researching synergies, implementing innovative technologies (digital or non) supporting actions maximizing the impact in heavy industrialized urban area of Attica region to demonstrate a first-of-kind shift towards circularity and climate neutrality.

2.1.3 Quantification of potential long-term impact Theseus

Baseline in EU	2021[56]: i) 527 kg generated in EU i)12.6 mln t textile waste generated/ year, per capita, ii) 49 % recycled and out of which 5.2 million tonnes is clothes composted, iii) trend towards and footwear, ii) Collected for re-use & decrease in landfilling in the	l a s t r

[56] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Municipal_waste_statistics

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	ecycling: 22% of post-consumer textile years iv) EU target on landfilled waste, iii) the rest is incinerated or waste: 10 % by		CDW
Assumption[67] Europe,	2035 landfilled, iv) Theseus technologies to allow decrease in landfilling to 10 % for Theseus technologies to increase recycling EU countries staying behind in rate to 37 % the target.	to be introduced: requirement to collect textiles separately (2025) and Extended Producer Responsibility[57] the Theseus approach adopted across	Theseus technologies to increase recycling rate by 15%
CDW (metals) i) average for CDW (per capita) [58:] wood are the most precious materials in	metal accounts	for 4,3 %, ii) metals together with	
Environmental impact	4 sorting technologies contributing to i) 0.73 billion tons of CO2 that could be save globally ii) the target of minimizing landfilling to 10 % of MSW generated by 2030, iii) improved waste management and resource efficiency, symbiotic flows energy contributes to reaching targets i) minimum of 42.5% of renewable energy by 2030 in EU5 , ii) Install 600GW of solar photovoltaic by 2030 in EU7 In Greece, 7.7GW of solar panels are aimed to be installed for electricity generation and appx. 6,978 MWh for heating and cooling aims to come from solar energy by 2030 iii) The RES share for heating/cooling to be 46% (or ~27 GWh) by 2030 and 100% by 2050[59,] iv) Overall increase in water re-use by 50 %, re-use 90 % of the solid waste generated by the wastewater treatment at EU scale v) all in digital pillar increased decision making capacity in terms of circularity & I-US flows iii) Collaboration pillar contributing to environmental education, guidelines, and awareness raising about H4C		
Financial impact	i) economic grow with sustainable and competitive markets, contributing to almost 73 bln euro savings across EU28[60] ii) new (incl. green-& higher quality) job opportunities –(e.g. for textiles sector, around 100 jobs in GR only until 2030) & further contributing to potential 120k jobs in EU[61] & 6.5 mln new jobs in the construction sector [62] , decreasing green skills gap and increasing circular skills and green-task intensity[63] iii) new circular business models spreading, expanding and adjusting across Europe according to the local needs.		
Social impact	Better health as a result of improved environmental indexes resulting from H4C implementation (~3,7 million citizens of Athens metropolitan area), in long -terms influencing whole country and replication regions (14 million European citizens in total) and emerging across EU with further citizens affected.		
Unfolding Theseus Impacts to EU	The project will unfold the outlined impacts across Europe via: 1)The replication activities, including the 2 locations, the collaboration (IP1) and KER1: Roadmap for collaboration (i)between sites (GR, SWE, NO) and ii) between Hub & CCRI & Replication Roadmap Tool and KER2: Common hub circularity guidelines. 2)KER22: Recommendations, analysis of symbiotic flows in the context of EU and local regional plans,		

[57] https://ec.europa.eu/commission/presscorner/detail/en/ip_23_3635

[58] JRC130992_01.pdf Background data collection and life cycle assessment for construction and demolition waste (CDW) management

67 an estimation based on technological progress we expect impact, current statistics as well as reports such as: https://circulareconomy.europa.eu/platform/sites/default/files/emaf_ce-in-cities-factsheetsproducts_all_mar19.pdf

[59] National energy and Climate plan (in Greek)

[60] https://op.europa.eu/en/publication-detail/-/publication/174996c9-3947-11e8-b5fe-01aa75ed71a1/language-en

[61] https://www.sitra.fi/app/uploads/2021/03/sitra-how-does-the-circular-economy-change-jobs-in-europe-v2.pdf

[62] https://www.sitra.fi/app/uploads/2021/03/sitra-how-does-the-circular-economy-change-jobs-in-europe-v2.pdf

[63] https://www.eesc.europa.eu/sites/default/files/files/qe-07-23-330-en-n.pdf

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Regional and Intraregional development guidelines, KER23: Policy guidelines based on Material Flow Analysis, impact on the health and de-risking assessment and KER24: Contributions to CEN and/or ISO standards.
3)Synergies with the cities selected by the Cities Mission and other networking activities (expected Outcome 4).
4)The Dissemination & Exploitation activities to be performed.

2.1.4 Potential barriers and Mitigation Measures

The implementation of Theseus objectives is an ambitious undertaking. To assure success, analysis of barriers for implementation of H4C for Attica and peripheral regions encountered at the proposal writing stage have been summarized together with respective mitigation measures to address them. Potential barriers have been grouped into general and region specific having to do with the local context and current status quo of Attica region and Greece.

Barriers	Theseus mitigation measures
General	
Systemic: Management of complex value chains, uncertainties of innovation Assessing impacts. Issues due to lack of related Political directions. The diversity of existing or planned waste management schemes in Europe	i)Theseus covers full circular flows from feedstock as secondary resource to final end-user, ii) Consortium has significant scientific/technical expertise to propose mitigation measures to any barriers related to technology/process iii) Theseus will monitor relevant risks to mitigate them on time (List of critical risks) iv) Development of Strategic Master Plan on Circularity to enable investment proposals that will consider topics like pricing matters, systems design, Regulatory Framework for included sectors. v)multiple analysis of the current situation in technological, environmental and social level and analysis of EU and local regional plans -general strategic or related to wastes-, to align and optimize Theseus activities, maximize impact and produce policy recommendations and standardization activities, while having the Region of Attica as a member of the consortium.
Technological/Processes related: Issues related to specific technologies /processes and their relevant reconfigurations and adjustments in existing assets and procedures. Novel uncertainties and lack of data.	i)In order to assure efficient barriers mitigation at technology/process level Theseus Consortium consists of partners of significant scientific and technical expertise, that will propose measures to mitigate barriers identified for respective relevant technology/process, ii) Theseus will monitor relevant risks as part of management activities to mitigate on time.
Social/public acceptance: Low acceptance of novel technologies and methodologies, due to lack of awareness, low knowledge and digital skills, scepticism. lack of capacity or/and personnel specialized in recycling technologies	i)Raising social awareness using AR app, ii) promotion of Theseus activities through diverse communication and dissemination actions aiming at spreading information about project, results, benefits, iii) training activities in the regions cities and universities aiming at enhancing circular and digital upskilling for the population, the workforce and high-level decisionmakers, iv) enlightenment, guidance, training, and mentoring services for decision-makers, employees, and citizens, v) knowledge sharing by the active involvement of a diverse groups of stakeholders through THESEUS Bubbles, vi) Replication activities aiming at promoting Thesus H4C in other locations, vii) standardization system facilitating the acceptance.

Legal: lack of legislation (or overlapping) on FoaK pioneering technologies not standardized yet. Unknown normative and differences between national and local legislations.	i)Participation in Consortium of local actors of decisive level , ii) Development of standardization schemes where needed, iii) Active monitoring of ongoing legislation schemes at EU and national level, participation in technical committees (e.g. DS in CEN/CLC/JTC 24) and relevant events, ii) Implementation of flexible approach where applicable, allowing for future adjustments, iii) Cooperation with other H4C, learning from their experiences
Economical: feedstocks & final circular product costs due collection, sorting, processing	i)promote collaboration among value chain actors and knowledge exchange ii) Contribution to development of common standards iii) Raising awareness on sustainability and circularity benefits among the end users contributing to more stable market of circular products iv) Development of Strategic Master Plan on Circularity, which will consider pricing matters among others
Region specific	
Textiles flow: Currently no recognized management system for textile waste by the Hellenic Recycling Agency, in Recycom facilities textiles are currently manually separated.	i)Automated textiles separation system will greatly increase the efficiency of textiles separation accelerating sorting procedures and enabling recovery of different fibre types, ii) policy briefs recommendations to facilitate future actions.
Urban waste flow: Although blue bins in Greece are intended to gather only packaging, currently most household recyclable materials end up in them making it a compilation of different materials.	i) Advanced separation of mixed urban waste through fine-tuning Smart Bins for the needs of Attica region to recover plastics, metal, paper, and textiles from urban waste streams, ii) automated sorting for mixed waste and plastics, iii) awareness raising activities
Urban waste - Packaging: Currently only paper, plastic and metal are collected as packaging waste but not glass and wood.	Theseus activities will facilitate to expand collection of packaging materials.
Biowaste: Separate collection of biobased trash has been introduced only in some municipalities. A portion ends up in blue bins by accident.	Sorting technologies and mass flow analysis for optimal operation and placement of smart bins for Attika with special focus on space constraints typical for Attica region will pave the road towards better separation of biowaste.
Energy: The integration of biogas to the natural gas network has not yet been implemented in Greece, legislation missing	Theseus will study RES technologies application incl. biogas into DH systems, preparing further implementation, considering aspects like origin of biogas and end users/consumers perception.
Water: Attika region is especially vulnerable to heat and water stress especially in the summer	i) perform water needs analysis for Attica for better planning, introduce ii) decentralized water reuse for gardening iii) rainwater harvesting for urban farming iv) decentralized NBS for recovery and reuse v) Development of Strategic Master Plan on Circularity incl. water and wastewater services.

2.2 Measures to maximise impact #@COM-DIS-VIS-CDV@#

To achieve its stepwise methodology, the Theseus consortium will utilize a concrete D&C plan together with an exploitation/commercialization plan, taking advantage of the in-project user community and external stakeholders. Theseus Consortium will implement an agile strategy for the dissemination, communication, and exploitation (DEC) actions. The application of these actions stretches from the onset and covers the duration as well as the period after the end of the project. DEC activities aim to accomplish Theseus's goals by targeting specific audiences

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to achieve greater impact. The actions approach and communication material and channel will differentiate according to each of the target groups. The agility of DEC strategy lies upon a flexible responsive approach, with each iteration cycle relying on regular reviews and revisions. This allows the consortium to respond directly to any unanticipated risks that may influence the goals of the project and facilitates the opportunity to adapt to any new situations. The DEC plan will unfold chronologically in three phases: 1. Activation phase (M1-M6): This phase is dedicated to the formulation of an initial DEC plan and producing preliminary awareness for relevant sectors and markets. During this phase, the consortium will create a common project identity and raise interest for the expected results, through the preparation and distribution of customized dissemination and communication material and the project's website promotion. Furthermore, the consortium will prepare an indicative schedule list of communication activities. This list may include events such as scientific conferences, webinars and cultural and business fairs that the consortium partners are involved as well as a set of indicative public events, workshops and training sessions that could be organized during the next phase of the DEC plan. In favor of securing the coordinated and successful engagement of key stakeholders a preliminary exploitation plan will be established. 2. Execution phase (M7-M32): This phase focuses on disseminating, communicating, and exploiting the produced outcomes to key stakeholders, with a particular focus on endusers. The consortium will establish more targeted awareness campaigns, disseminate the key exploitable results as delivered and investigate the exploitation opportunities of each partner through dedicated training sessions, and tools such as surveys, and interviews. 3. Final phase (M33-M48 and beyond): In this phase the consortium will focus on promoting the final key exploitable results and developed business models, with the purpose of establishing the necessary prerequisites for broader scalability/replication, further engagement of new end-users and reach out to wider target audiences. Another significant objective of this phase is to establish a feasible go-to-market strategy and ensure that the promotion of exploitable results is maintained after the end of the project.

2.2.1 Dissemination Measures

While focused on Awareness, Interest, Desire and Action (AIDA) activities, the project's dissemination measures will follow a multi-actor, multi-step and multi-channel approach. Table 2.3 identifies the target audiences, specific dissemination objectives and target values, tailored to the needs and profile of each target audience.

Table 2.3: Dissemination Plan – Measures, Target Audience and Target Values

Scientific Publications	Principal Decision Influencers, Decision-Making Actors, Industry Associations, Environmental Agencies, Academia, Researchers, Relevant Consortiums (such as Greek ministry of Environment/Development, Hellenic Recovery Recycling Corporation, Attica Waste Management Union, Hellenic Federation of Enterprises, All Greek Regions, Municipalities and Universities, Association of Municipalities in the Attica Region, Solid Waste Management (EDSNA), Hellenic Institute for Occupational Health and Safety (ELINYAE))
Scientific Conferences and showcasing events (examples): World Circular Economy Forum, Eur. Conference of Waste Management (ECWM), Digital Recycling Conference for Circular Economy, Sustainability, and Waste Management (eREC), Symposium on Circular Economy and Sustainability,/, Eur. Conference on Process Safety and Big Data, Int. Conference on Sustainable Solid Waste Management, Materials Science and Technology in Europe, International Conference on Waste Recycling and Management, etc; Peer-Review OpenAccess Journal (preselected): Recycling, Waste Management & Research, Polymer, Sustainability, Waste Management, Journal of Material Cycles and Waste Management, , Resources Conservation and Recycling, etc.	
Target Values: Publishing in Peer-reviewed Journals: > 15; Presenting in Scientific Conferences: > 20.	
Technical Publications	European Commission Relevant Offices, European Environment Agency, Public and Private Actors, ICT SMEs and large Enterprises, Formulators, Recyclers.
THESEUS will publish and contribute to technical blog posts, articles, position/white papers, books, or any other reference from technology suppliers (bottom-up), in addition to references linked to the application scenario domains under consideration. The forthcoming issues of Institute of Environment and Development	

Blog and Spire database of the outputs from projects of SPIRE portfolio are excellent instances of this type of work.	
Target Values: Publishing Technical Publications: > 25; Participation in Industrial Exhibitions with Booths: > 9	
Source Code	Ecosystem of Open-Source Developers, Incubators, Associations and Digital Innovation Hubs, Data Spaces for
Repository	circular material use, ICT SMEs, Researchers, relevant Consortia. (such as Greek DIHs, Greek Universities, Greek Open Source Association, Athens Digital Lab)
Stimulating a transparent and open-source culture, THESEUS will make widely available its software repositories through well-known distributed source code platforms such as GitHub and Bitbucket.	
Target Values: Publicly Available Deliverables: > 15; Source code: publish to >2 different repositories.	
Q & A Platform	Public Authorities, Associations, Open-source Developers, NGOs, SMEs, relevant Hubs.
Discussions regarding the project repository will be centered around reference sites. The designated Q&A platform, developed as an integral part of the project website, will be Stack Overflow - the most extensive and reputable online community for developers to exchange knowledge and acquire new insights.	
Target Values: No. of Scientific Conversations to Participate/Contribute: > 20.	
Workshops and Trainings	Industrial Organisations, SPIRE Associations, Public Authorities, Potential Investors, Funding Organisations, ICT SMEs, Academia and research Organisations (such as Greek ministies, Hellenic Recovery Recycling Corporation, Attica Waste Management Union, Hellenic Federation of Enterprises, All Greek Regions, Municipalities and Universities, Association of Municipalities in the Attica Region, Solid Waste Management (EDSNA), Greek DIHs, Greek Universities, Greek Open Source Association, Athens Digital Lab, Hellenic Institute for Occupational Health and Safety (ELINYAE) , Chamber of Commerce and Industry of Athens, E.B.E.A.))
Particular emphasis will be placed on collaboratively designing training activities and lifelong learning programs to ensure that THESEUS actively contributes to skills development and standardisation, as outlined in WP10.	
Target Values: Technical Workshops: > 10; Online Training Tutorials: > 8; Webinars: > 12.	
Joint Activities with relevant Initiatives	All Stakeholders with special focus on secondary resources in process industry, Policy-making Actors, Key Environmental Influencing Organisations, relevant Public Bodies.
To further amplify and maximise the impact and outreach of project's results and to feed them into related international work streams, joint dissemination activities will be contracted. Theseus will seek out opportunities to collaborate with other relevant projects and initiatives within Europe and beyond, and will organise and participate in events promoting knowledge sharing as outlined in WP11.	
Target Values: Symposiums/Fairs to Participate: > 9; Banners in each event: > 3; Procures to be distributed: > 300.	
Standards and Policy Contribution	Standards Developing Organisations, Certification Committees, Open-Source Communities, Policy Makers, Relevant Public Bodies, Local Agencies.

THESEUS aims to support the standardisation of secondary material upcycling technologies. Collaborative efforts with relevant Technical Committees will be established to ensure seamless integration into current and future standardisation initiatives. The project's contributions will be incorporated into new or revised standards via one of the following channels: i) sharing information with Technical Committees, ii) facilitating mutual knowledge exchange, iii) submitting technical proposals, and iv) promoting the creation of new standard documents, based on the maturity of the specific solution, as outlined in T11.5.	
Target Values: Standards Contribution (new or revised): ≥ 3; Policy Briefs: > 2.	
On-site Demonstrations and Presentations	Industry Stakeholders, Academic Institutions, General Public
Demonstrate practical application of project results, engage stakeholders; On-site demonstrations and presentations showcasing project outcomes and methodologies	
Target Values: On-site Demonstrations=5;	

2.2.2 Communication Measures

The following design elements will be used to strengthen the project uniformity and identity and to deliver clear messages to the target audience: Theseus naming, logo, presentations template, templates for reports, project newsletter, project posters/leaflets, etc. A recognisable visual identity is provided at the proposal stage, to be adapted if required. Theseus communication strategy is built upon the Promotion Mix, the 4th element of the Marketing Mix, focuses on creating awareness and persuading the audience to initiate the engagement, as outlined in Table 2.4.

Table 2.4: Communication plan – Communication Channels/Target Audience/Timeline and Target Values

Project Website	ALL End-users.	From M06 to 3y after
Theseus dynamic project website will be the backbone of all the D&C activities, allowing easy retrieval of main action data with few clicks, while acting as the core source of information about the project's objectives, activities and results. The website will be launched by M06, and it will be lined with all Theseus's social media accounts. It will be constantly maintained and updated regularly, running also for 3years after the project end.		
Target values: Online by: M06; Unique Visitors from M12: > 1,500; Unique Visitors from M48: > 6,500		
Social Media	ALL End-users.	From M06 to M48
Theseus will create and actively maintain its online presence across many social media platforms, utilizing Twitter and LinkedIn when connecting with research and innovation networks and disseminating new publications, and Facebook, and Podcasts for broad community participation.		
Target Values: Size of online community (by M48): >5,000; No. of impressions (monthly average) :>500		
e-Newsletters and e-mail Campaigns	All End-users.	Bi-yearly Issues starting from M06
When project findings are presented, relevant stakeholders will get e-newsletters. Emails are a very effective engagement tool, particularly for promoting events and outcomes across domains; hence, project communications will be delivered to a selected set of contact points. Professional marketing solutions (such as Sendinblue and Mailchimp) will be utilised to automate the dissemination to contact points.		
Target Values: e-Newsletters contributed/released: 8; Total Mailing List Contact Points: >2,000		

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Policy Fora and Debate Hubs	Societal Actors, Debaters, Recycle Operations, Sustainable Product Organisations, Workers' Associations, Alliances (such as Greek ministries of Economics, Internal Affairs, Environment, Development, Hellenic Recovery Recycling Corporation, Attica Waste Management Union, Hellenic Federation of Enterprises, All Greek Regions, Municipalities and Universities, Association of Municipalities in the Attica Region, Solid Waste Management (EDSNA), Greek DIHs, Greek Universities, Greek Open Source Association, Athens Digital Hub, Delphi Economic Forum, Greek Consumers Union, GSEE and other workers associations and unions, Hellenic Institute for Occupational Health and Safety (ELINYAE), Chamber of Commerce and Industry of Athens, E.B.E.A.)	From M18 to M48
With the Science-Policy dialogue as its primary objective, Theseus will contribute to discussions and make recommendations regarding how the developed solutions could contribute to the sustainable, balanced, and inclusive development of secondary material upcycling and waste management across various industry sectors and products. Theseus intends to contribute to and support the activities of relevant hubs and forums, such as the World Economic Forum, Science Policy Forum, EIT RawMaterials Expert Forum, and GS1 linked DPP forums.		
Target Values: No. of Fora and Hubs to interact: >7; Contributions to Policy Fora/Debate Hubs: >15		
Blogs	Digital Innovation Hubs, Relevant Associations, Academia (such as Greek DIHs, Greek Universities, Greek Open Source Association, Athens Digital Lab)	From M18 to M48
Theseus will link and execute periodic wide marketing operations in open communities, DIHs, blogs, and non-profit organisations and channels, such as Gitter Communities, Nature.com, and non-profit organisations, such as RawMaterials Innovation Centers of the EIT, etc.		
Target Values: Unique Publications/Contributions to Blogs: >15; Different Blogs to Post: >8		
Printed Material	ALL end users attending physical events	From M06 to M48
Theseus will develop a distinct visual identity and reference promotional instrument when participating in events, including brochures, catalogues, posters and any other laid out paper-based resource. Most of the content will be in digital format (QR-codes will be leveraged to allow visitors to see project brochures/flyers in digital format). Printed material will be avoided highlighting on the overall “green” aspect of the project.		
Target Values: No. of QR - scans per material (i.e., brochures, flyers, posters, etc.): >1500; Roll-up Banners: >6		
Multimedia	ALL end-users focusing on Regional and Local Authorities	From M09 to M48
The project will develop multimedia content for a self-explanatory and aesthetically pleasing presentation of the project, utilizing various distribution channels for marketing (e.g., YouTube, Vimeo). In addition, in close connection with the Theseus capacity building efforts, a series of video interviews will be conducted with scientific partners to collect feedback, taking use of plenary sessions and other events.		
Target Values: No. of short videos and interviews produced in total: 7; No. of views: >5,000		

Exploitation Plan and sustainability

Turing brilliant R&I into successful commercial products is a complex task with more failures than successes. Most failures occur because there is a lack of Product/Market fit or to put it simply, the product that is built is not what the market wants to buy. To orientate the outcomes from Theseus towards viable commercial solutions and assets, the consortium will adopt a focused commercialization strategy with measurable outcomes by embedding best practice into the project's lifecycle. The exploitation and commercialization tasks in WP11 will focus on delivering solutions that have a clear market opportunity that can be proven and validated through the pilots with key stakeholders and in close collaboration with formulators, recyclers, public authorities, and standardization actors. The completed business models and marketing plan will aim to outline viable and commercially relevant products and assets from the project by taking a broad cross-consortium perspective, with credibility built on a foundation of embracing industry best practice for bringing new products to market. To achieve sustainability for the KERs and prepare the ground for market entry, the consortium has drafted the exploitation plan presented in Figure 9, which is split in 3 distinct phases: Phase 1: setting the necessary conditions for the exploration of the business plan; Phase 2: validating the developed solution, aligns with market and prepares the go to market strategy; Phase 3: featuring the sustainability and commercialization roadmap. It is important to highlight that Theseus shall have a complete exploitation plan with a clear analysis of the whole value chain and the opportunities that arise across the 5 analyzed value chains. Potential market regulations that are needed will also be considered through the participation of the local public authorities. The replication potential in Europe will be achieved through the participation of International Companies like Thrace Group, Halcor etc. and further supported by a common H4C replication scheme provided by SPIRE and all the SPIRE members. In addition seeking of additional investments shall be performed at this stage of commercialization If the technology and partners are ready, it is possible to create a legal structure (new start-up), set up organizational procedures (IPR contracts) and team (job creation), to market the solution and signed the first commercial contracts before the end of the project.

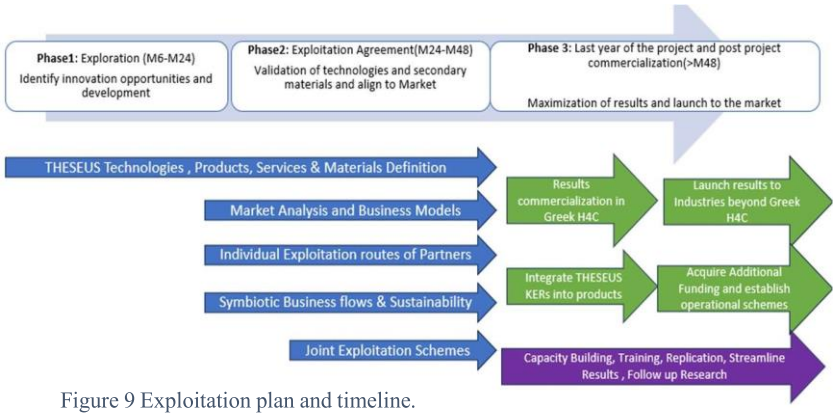

Figure 9 Exploitation plan and timeline.

Key Exploitable Results (KERs) and individual exploitation plans

The consortium partners have decided to take a multi-level approach towards managing the knowledge produced, where the respective knowledge producer holds the rights to the knowledge and shares the software and material implementation with the rest of the consortium. The initial exploitation intentions towards the Theseus KERs are presented in Table 2.5, which will be further refined during the project.

Table 2.5: KERs (Type: Product (P), Service (S), Other (O); Services: Support (Su), Training (T), Consulting (C), Extension (E)) ; Licensing: Open source(O), Commercial(CO)

KERs	Owner	Type	Licensing	Services	Time to market
KERs resulting from Pilar 1:					
Roadmap for collaboration	CIRCE	S	O/CO	C, T, E	0.5y after project
Common hub circularity guidelines	A.SPIRE	S	O/CO	C, T, E	0.5y after project
AR for social awareness and acceptance	ICCS	S/P	O/CO	Su, C, T, E	0.5y after project
KERs resulting from Pilar 2:					
Circular construction material	SINTEF/TUD	P	O/CO	Su, C, T, E	0.5y after project
Biobased thin films for packaging	NCSR	P	O/CO	Su, C, T, E	0.5y after project
Electrospinning for thin films	NTUA	P	O/CO	Su, C, T, E	0.5y after project

Biological recycling & spinning for upcycling	CIRCE, HUD	P	O/CO	Su, C, T, E	0.5y after project
Upcycled packaging, masterbatches	HMU	P	O/CO	Su, C, T, E	0.5y after project
Strategic Master Plan on Circularity	NTUA, EYDAP	S	O/CO	Su, C, T, E	0.5y after project
KERs resulting from Pilar 3:					
Decentralized sewage mining and NBS for water reuse	EYDAP	P	O/CO	Su, C, T, E	0.5y after project
Sewage sludge valorisation	NTUA	P	O/CO	Su, C, T, E	0.5y after project
Centralized reclaimed water fertigation	NTUA	P	O/CO	Su, C, T, E	0.5y after project
H2 for production of heat and electricity through a CHP	CERTH	P	O/CO	Su, C, T, E	0.5y after project
New generation of HVFP solar thermal technology	CERTH/TVP	P	O/CO	Su, C, T, E	0.5y after project
KERs resulting from Pilar 4:					
Automated textiles separation system	ICCS	P	O/CO	Su, C, T, E	0.5y after project
Automated sorting for mixed waste and plastics	ICCS	P	O/CO	Su, C, T, E	0.5y after project
Industrial sorting systems: Glass/copper & metals sorting system	ICCS	P	O/CO	Su, C, T, E	0.5y after project
XR for human-robot interactions/collaboration	ICCS	P	O/CO	Su, C, T, E	0.5y after project
Dynamic symbiotic maps, prediction models, DPPs and optimization for smart bins	ICCS	P	O/CO	Su, C, T, E	0.5y after project
H4C Marketplace with advanced matchmaking features & secure data exchange	IRIS/ICCS	P/S	O/CO	Su, C, T, E	0.5y after project
Models: i) WEF nexus, ii) Cities climate & impacts	Imperial, NTUA	S	O/CO	Su, C, T, E	0.5y after project
KERs resulting from Pilar5 :					
Regional & Intraregional guidelines	UAGEAN, LiU	O	O/CO	C, E	0.5y after project
Policy guidelines based on Material Flow Analysis, impact on the health and de-risking assessment	RISA	O	O/CO	C, E	0.5y after project
Contributions to CEN and/or ISO standards	DS/SN	O	O/CO	C, E	0.5y after project

Preliminary market analysis with needs, value propositions, and target stakeholders for exploitation

EU alone produces more than 2 billion tn of waste per year[64] or 4.8 tn/capita, primarily from construction, mining and industry. Out of these quantities of waste generated annually in the EU only 50% is recycled and only 10% of those recycled materials are returned back to the economy[65], while in Greece the recycling ratio is lower and close

[64] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Waste_statistics#Total_waste_generation

[65] Hedberg, A., & Šipka, S. (2020). The circular economy: Going digital. In European Policy Centre (Issue March). https://circulareconomy.europa.eu/platform/sites/default/files/drce_final.pdf.

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to 30% according to the approved National Waste Management Plan (NWMP[66]). Given that the world's raw materials consumption is expected to nearly double by 2060[67] the environmental pressures from materials extraction, processing, use and waste management lead to very significant environmental pressures contributing to 70% of global GHG emissions and extraction and use driving more than 90% of biodiversity loss and water stress[68]. Between 2015 and 2021, the average circularity rate for all EU member states (EU-27) increased by only 0.4 percentage points while, the 2021 EU circularity rate of 11.7 % is higher than the most recent global circularity rate which stands at 7.6 %, down from 9.1 % in 2018[78]. In this framework, the following analysis addresses the market potential for each waste flow, based on the suggested symbiotic flows.

In terms of dynamics, In EU-27 and Switzerland, the production of waste from textiles is estimated between 5.8[69] to 7.5[80] million tn, corresponding to an annual 11.3 to 15 kg/person, which is expected to increase to 20 kg/person/year[81]. In EU separate collection-toconsumption ratios range from 11% (Italy) to more than 70% (Germany) and remains very low with few exceptions[70]. For the Theseus project, used textiles will be collected from the private system of the approximately 2.500 purple bins in Greece that operates in close cooperation with cities, as well as the blue bin, where mixed recycled waste is collected and monitored by the local authorities. Textile wastes refer to 11,8% of USW with projections to collect up to 27.778 tn from purple bins and recover 14.958 tn by 2030 from common bins by 2030. Considering that, an average of 20 to 35 jobs are created for every 1.000 tn of textiles collected for re-use[71] (such as selling them secondhand) the effect of the Theseus project is expected to be significant, especially in the framework of local communities. In terms of environmental protection, the positive impact is also expected to be significant, since GHG emissions would decrease by approximately 44% if a If the number of times a garment is worn were doubled, compared to the production of a new garment[72].

The amount of urban solid, mixed waste approximates at 3.2 billion tons. It is estimated that from this amount 47% is landfilled, 31% recycled and 22% incinerated[73]. Thus, almost 70% of waste is currently not reused or recycled. According to the 2030 EU targets, recycling rates must come up to 65% for MSW, 70% for CDW and 75% for packaging waste, while of industrial symbiosis and eco-design must facilitate this process. Theseus will facilitate the urban waste metabolism from generation to treatment, following a material flow analysis by proposing innovative ways of reuse for waste stream. Urban mixed waste including packaging, biobased waste and plastics will be collected from municipal bins, and industrial waste from private project partners' waste streams, following the specific targets of the National Waste Management Plan (NWMP 2020 as revised in 2022).

CDW and other industrial waste streams, including metals, glass and marble will be collected from various sources such as civil engineering works, demolition projects, municipal waste and industrial waste (e.g. incineration fly ash for the recovery of metals). The construction and demolition industry is responsible for the production of immense quantities of waste. In the EU alone, it accounts for approximately 30% of the total waste generated[86]. About 374 million tons of CDW were generated in 2016[74], making it the largest waste stream in the EU by weight. According to the findings of the current Study, the respective amount for Greece was estimated up to 4,5 million tons for 2019. The recovery and recycling rate of CDW in Greece ranges from 12%-15%, out of which, only a very limited amount of CDW

is currently treated for reuse or recycling[75]. The existing CDW treatment facilities in Greece treat mainly the mineral fraction of CDW, while materials such as metals, plastics and glass (when present) are sent to other recycling facilities that handle each specific material fraction. For the Theseus project CDW will be sorted dependent on particle sizes for an optimal reactivity environment and either used directly for the production of cement, or mixed with other streams such as fibers, glass and be further used for the production of insulation panels

[66] https://ypen.gov.gr/wp-content/uploads/2021/02/%CE%B5%CE%B3%CE%BA%CF%81%CE%B9%CF%83%CE%B7-%CE%95%CE%A3%CE%94%CE%91.pdf

[67] https://www.oecd.org/env/global-material-resources-outlook-to-2060-9789264307452-en.htm

[68] International Resource Panel (IRP). (2020). Global resources outlook 2019: Natural resources for the future we want. Paris: IRP: https://www.resourcepanel.org/reports/global-resources-outlook [78] Circle Economy. (2023). The circularity gap report 2023. Amsterdam: Circle Economy: https://www.circularity-gap.world/

[69] Beasley, J. and Georgeson, R., 2014, Advancing resource efficiency in Europe. Indicators and waste policy scenarios to deliver a resource efficient and sustainable Europe, European Environmental Bureau, [80] Scaling textile recycling in Europe–turning waste into value, McKinsey &Company, July 2022; (Eurostat Prodcom database; Intecus, Germany report, 2020; ISPRA, 2021; ISPRA, 2022;) [81] European Environment Agency (EEA), 2019 Textiles and the environment in a circular economy:

[70] Watson, D., et al., 2017, Business Models Extending Active Lifetime of Garments Supporting Policy instruments:

[71] Sustainable and Circular Textiles by 2030, Factsheet, European Commission: https://environment.ec.europa.eu/strategy/textiles-strategy_en

[72] Durable, repairable and mainstream, ECOS Aril 2021: https://ecostandard.org/wp-content/uploads/2021/04/ECOS-REPORT-HOW-ECODESIGN-CAN-MAKE-OUR-TEXTILES-CIRCULAR.pdf

[73] Tisserant et al., 2017. Solid Waste and the Circular Economy: A Global Analysis of Waste Treatment and Waste Footprints: https://doi.org/10.1111/jiec.12562 [86] Eurostat (2019): Generation of waste by waste category, hazardousness and NACE Rev. 2 activity

[74] European Environmental Agency (EEA)/European Topic Centre on Waste and Materials in a Green Economy (2020). Construction and Demolition Waste: challenges and opportunities in a circular economy: https://www.eea.europa.eu/themes/waste/waste-management/construction-and-demolitionwaste-challenges

[75] Bio by Deloitte, 2014, Construction and demolition waste management in Greece.

ass well as other construction materials, such as bricks. Other innovative solutions, expected to be developed in the framework of Theseus, include innovative recovery of metals from CDW and involves automated processes that employ magnetic and eddy current configurations. Latest advances demonstrate use of visual recognition systems coupled with robotics, with instances exhibiting capacity to handle waste up to 30kg. Based on this preliminary market analysis, the use of innovative solutions like those to be developed by Theseus targeting the more resource efficient use of are vital for the waste management sector in Europe and the role of the H4C will be to lead systemic change in the chosen waste streams and facilitate multistakeholder engagement, circular-oriented decision-making, circular value capture models, co-creation, and diffusion of a sustainability culture and, thus enabling the development of efficient strategies by extending the network of and governance. Since, circular economy can foster renewable material source to enhance economic growth and create employment opportunities, which in turn while minimize landfill and other management costs, the status of the market offers favourable framework conditions to position the innovative circular technologies of Theseus across a diverse array of stakeholders' groups, as in Table 2.6. Table 2.6: Preliminary outline of customers' needs and value propositions

Customers	Key Needs	Main Value Propositions
Waste Suppliers	Sound management of by-products engaging lowcost solutions; Valorisation of by-products streams; Increased profitability; Reduced carbon footprint; foster employment ; Reduce disposal costs.	Sustainable pathways for utilising by-product/waste; Alternative and increased revenue streams along with reduced taxes due to decreased waste management; Compliance with the EU regulation frameworks regarding waste, packaging, emissions and circular economy.
Producers Processors	Innovative technologies to incorporate new streams of waste as raw material; Scale-up production to meet increased demand; Return on investment on sustainable technologies; Creation of new green jobs; Compliance with regulations; Novel opportunities for scaling up circular solutions	Develop innovative technologies and processes in order to manage streams of incoming waste-raw material; Engage Best Available Techniques (BATs); Change of mindsets regarding circularity and industrial symbiosis models; Increased efficiency in production processes with cost-effective technologies; Improved sustainability and marketable products with lower carbon footprint; New business models serving as concrete demand, improving economic performance; Access to green funding for investment in circular innovation;
End-users	New economic opportunities; Diversification of sources of products and minimization of risks springing from monopoly markets; Circular products with a lower environmental and carbon footprint; Reduced waste;	Increased quality of the produced products; Improved corporate ESG performance; Affordability of the products; Improved sustainability with lower carbon footprint; Access to green funding for green investment;

For the value propositions to effectively address the needs of the potential cust
and products need to be
introduced to market and keep its value chains sustainably operational. To this end, an initial set of requirements and potential obstacles that the exploitation strategy needs address, as well as key relevant actions and levers that will be employed were identified, presented in Table 2.7. Table 2.7: Initial set of requirements, potential obstacles, actions and supporting factors for exploitation

Requirements & Obstacles	Relevant Actions and Supporting Factors for Exploitation
Lack of interest in the innovative services and digital tools produced	The benefits of innovative services and digital tools will be demonstrated and validated, alongside their pilot implementation. Theseus solutions will be designed with end-users in complete value chains, safeguarding, that they will cover real needs. User acceptance will be validated at several stages to build confidence, that solutions and products can respond to market demands.

Further advancement of techs in terms of TRL	Partners (researchers, technology developers and value chain players) are highly motivated by the potential of the Theseus solutions and are committed to exploit the project's results, investing the resources required to develop them further, to reach the commercialisation stage. Go-to-market strategies (with identified options for acquiring external financing / funding if needed) to this end will be delivered, before the end of the project.
IPR management	An IPR management strategy will be developed during the project, describing the baseline for managing the IP produced within Theseus and setting the basis for the exploitation of project results after the end of the grant. Specifically, for IP protection, the proper protection measures will be identified and all the necessary actions to apply them will be carried out (starting within the project's duration), to facilitate exploitation at least within the target market of Theseus.
Needs for standardization	As any new technology, compliance and refinement (if needed) to existing standards and upcoming standardization activities are key requirements for its release to the market. As the further development of related standards can affect market growth, Theseus will contribute to standardization activities, to support easy-to-market solutions.
Sustainability of financing for exploitation after the project end	Aside from investing own resources, partners are willing to continue the development and exploitation of the Theseus technologies and products, sustainably operating the value chains set up and demonstrated during the project. Further financing sources will be investigated as well, either from national and EU funds, and/or by approaching private investors, building on the go-to-market strategies developed during the project. More specifically additional sources of funding coming from Innovation Fund , InvestEU and ESIF will be considered. A dedicated strategy for the additional investment will run at the last year of the project.
Availability of initial demand for driving product sales and market shares	The consortium's pilot partners will provide the initial demand needed to successfully market the solutions beyond the end of the grant. As early adopters, they are also committed to further promote the technologies/products to their ecosystem for identifying additional customers.

2.2.3 Strategy and procedures for IP management

IPR issues have been discussed by the consortium during the proposal preparation phase for the commitment of the offered technical solutions and by consulting their respective partners' legal offices. In conformance with the rules for participation of current European Programs, a minimum set of basic principles concerning ownership of knowledge and access rights have been fixed and will be detailed with the signature of a consortium agreement (CA), before the start of the project, which will regulate rights and obligations by all partners and will be further updated this during the project. The IPR provisions distinguish between 2 basic types: i) Background Knowledge: Access rights necessary for the execution of the project will be granted on a royalty free basis. ii) Foreground Knowledge: Intellectual property protection of foreground, i.e., the new technologies and knowledge developed in the project and its applications will be filled, owned and funded by the partner(s) that developed the particular results. Each partner shall inform the Project Coordinator and the project partners 4 weeks in advance of the filing of a patent application. Subject to the provisions of the EU Grant Agreement and the CA any owner of any result shall be free to use such results and to move it to the market. In case of joint ownership each owner shall be free to use such results in its own organisation but adhere to the CA and exploitation strategy with regards to commercialisation. Relevant to knowledge management Theseus will employ the following strategy: • Intra-Consortium Knowledge Management: 1) Technical and scientific progress: create documents and place them in a document repository to ensure that every partner has access to them. 2) Exploitation perspectives: the WP exploitation leader will structure, and map required knowledge and extra-knowledge needed to enhance exploitation (i.e., patents, technology transfer agreements, potential users, other projects) and propose an exploitation strategy. • Extra-Consortium Knowledge Management: The exploitation leader will be in charge of

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collecting and proposing to the consortium all matters referring to the results for dissemination and will monitor the productivity of the project in terms of knowledge publications (scientific papers, posters, etc.). #§COM-DIS-VIS-CDV§#

2.3 Summary

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Specific Needs	Expected Results (per pillar)	D & E & C Measures
Implementation of H4Cs concept Systemic solutions Cutting edge solutions for circularity Systemic recovery of resources GHG emissions reduction Zero urban waste Effectively implemented UIS Effective cross-sectoral collaboration Improved waste management Better valorisation of secondary materials Solutions for closing loops Healthy environment Safe IT infrastructures Involvement of local communities	COLLABORATION: Roadmap for collaboration (i) between sites (GR, SWE, NO) and ii) between Hub & CCRI; Replication Roadmap Tool ; Common hub circularity guidelines; AR for social awareness and acceptance SYMBIOTIC FLOWS – MATERIALS : Circular construction material; Biobased thin films for packaging; Electrospinning for thin films; Chemical recycling & Spinning for upcycling; Upcycled packaging, masterbatches SYMBIOTIC FLOWS – WATER: Decentralized sewage mining and NBS for water reuse; Centralized sludge residues valorisation; Masterplan for water reuse at regional industrial scale; Centralized treated water networkSYMBIOTIC FLOWS – ENERGY: H2 for production of heat and electricity through a CHP; New generation of solar thermal technology for greening District heating and covering cooling loads of buildings DIGITALIZATION&ROBOTICS: Automated textiles separation system; Automated sorting for mixed waste and plastics ; Industrial sorting systems: Glass/copper & metals sorting system ; XR for human-robot interactions/collaboration; Dynamic symbiotic maps, prediction models, DPPs and optimization for smart bins ; H4C Marketplace with advanced matchmaking features and secure data exchange ; Models: i) WEF nexus, ii) Cities climate & impacts GOVERNANCE: Analysis of symbiotic flows in the context of EU and local regional plans. Regional and Intraregional development guidelines; Policy guidelines based on MFA, impact on the health and de-risking assessment; Contributions to CEN and/or ISO standards	Dissemination (D): • Peer-review Journals >15; Scientific Conferences >20 • Technical Publications >25; Industrial Booths >9 • Relevant initiatives: >15; Scientific Conversations: 20 • Workshops >10; Training Tutorials: >8; Webinars >12; • Fairs to Participate: >9• Standards Contribution: ≥3; Policy Briefs: >2Communication (C): • Website Online by M06 and running for 3years after the project end; >6,500 unique visitors by M48 • Size of online community >5,000; Impressions >500 • Newsletters: 8; Policy Fora/Debate Hubs: >15; Publications to Blogs: >15 • Hard Copies Distributed >1,500; Roll-up Banners >6 • Videos produced: 7; Exploitation (E): Open version of the Frameworks, Procedures, Models, Tools and Data Repository, and Practical Services / Guidelines under a fee (post-project) to ensure sustainability.

Target Groups	OUTCOMES	Impacts
Industries associated with P4P partnership Other industries Local communities-citizens, general public (especially in Greece, Sweden, Norway and EU) Local and regional administration Policy makers (national and EU) Partnerships, Associations, NGOs working with circularity, environment, human health, etc. (e.g. European and National standardization bodies, Ellen McArthur Foundation, European Public Health Alliance) Scientific and academic community.	Recovered resources use ↑ 30%, GHG emissions ↓26852 tons CO2e/ yr, Sorting efficiency ↑ to 97%, water re-use ↑ 50 %, freshwater consumption for urban green irrigation ↓50 %, solid waste generated by WWT: reuse 90% reuse of water for irrigation ↑20%, transport cost ↓ min. 15 %,materials transported ↓min. 15 %5 closed loops, min. 5 close cross-sectoral collaborations, 6 DPPs for circular products, min. 3 prediction models dev & valid., min. 10 companies implementing upcycling processes& technologies, min 10 actors collaborating towards Attica Reuse Masterplan, min. 3 sites with sewage mining & min 2 NBS, 15 % rainfall runoff reduction Rainfall runoff peak discharge reduction 40%, ↓ flood risk in intervention areas, ++ health min 20 benefits, Improved AQI min. 25%; health LCA imporv. Min. 30 %], ↑ expected lifespan, ↓expenses on healthcare, min 5 innovations highlighted in the policy briefs, nr of cities from the Cities Mission impacted min. 8	Contribution to aims of EU Mission: Climate-Neutral and Smart Cities, P4P objectives, EU Green Deal and "Fit for 55" objectives Environmental impact: contribution to climate neutrality, GHG emissions reduction, landfilling rates decrease, increase in recycling rates Scientific impact: Advancement in state-of-the-art in 5 Theseus pillars, 23 KERs accompanied by documentation, that can serve as basis for further expansion and reference Economic impact: Cost savings, new (incl. green) job opportunities; new circular business models available; Societal impact: better health as a result of improved environmental indexes, improvement in awareness and skills, co-created educational materials available

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3 Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

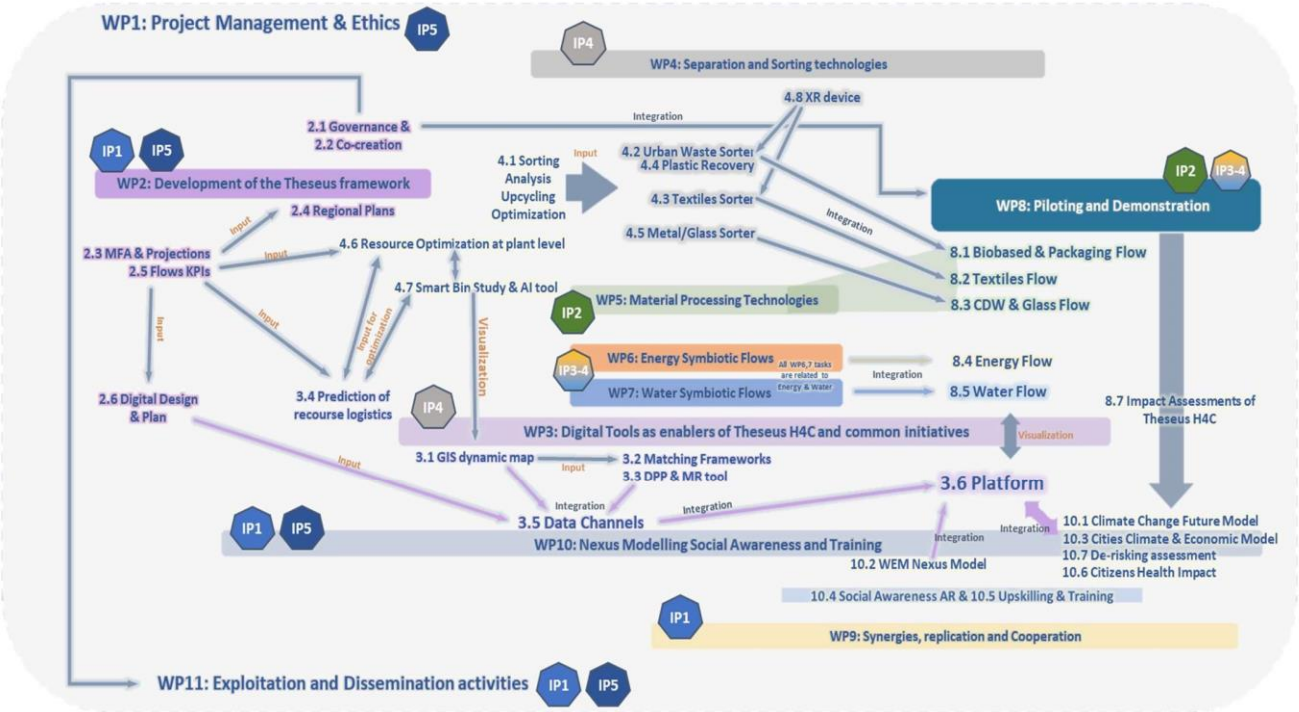

Figure 10 Theseus PERT chart and WP interactions

The work plan is displayed in the PERT chart. WP1, WP9 and WP11 are being developed transversely within Theseus. WP2 is mainly the baseline studies required to establish and align the H4C activities with the regional, while WP10 is the models developed from Theseus practices and experiences, measuring impacts, along with related but separated social awareness, training and de-risking activities. Specifically for WP10, AR (T10.4) and Training (T10.5) are transversal and side activities, while Tasks 10.1, 10.2, 10.3 are developing the models as the result of the Theseus research and innovation.Tasks 10.6 and 10.7 are more focused separated assessments in Health Impact and Risk Analysis. WP3 and WP4 are developed in parallel with WP5, WP6 and WP7 which are the main developments in materials, energy and water to be integrated for demonstration in WP8. WP4 is supporting WP5 and the relevant pilots, while WP3 is supporting WP3,4,5 and all the pilots.

The interlinkages and the interdependencies between work packages and activities are also displayed, e.g. the study of Smart Bins in Athens City (T4.7) will be fed with valuable data from the prediction of resource logistics of T3.4 and the resources optimization of T4.6, while its results will be visualized in the dynamic map of T3.1. Also, the impact assessment of each pilot (T8.7) will be aligned with T10.1, T10.3 produced models.

3.1.1 Work Packages Gantt and Tasks Description

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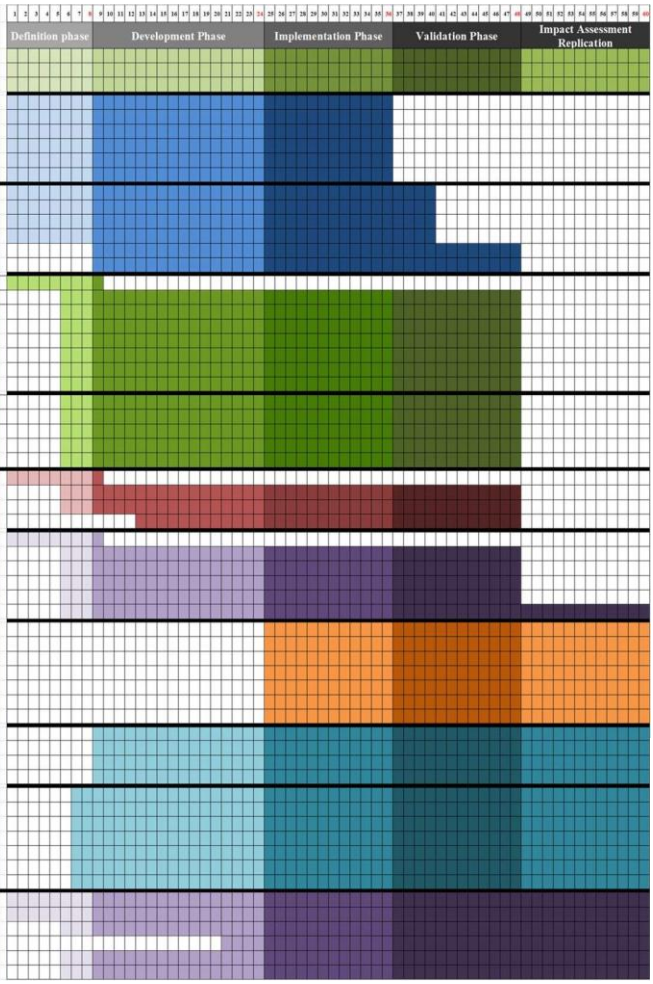
THESEUS Gantt Chart

WP	Task	Name of the task	Leader
WP1: Project Management & Ethics	1.1	Project Coordination and Management (incl. quality assurance and risk management)	ICCS
	1.2	Technical Management and Scientific Connection (incl.data management)	NTUA
	1.3	GDPR, Gender Balance, Legal and Ethical Issues Management	ICCS
WP2: Development of the Theseus framework	2.1	Governance for Regional and Intra Regional I-US	LiU
	2.2	Participatory approaches and co-creation	LiU
	2.3	Material flow analysis and 2050 projections	ICCS
	2.4	Regional & interregional pathways for symbiotic flows linked to EU & local regional plans	UAEGEAN
	2.5	Characterization of value chains & KPIs definition	RISE
	2.6	Digital transformation planning & implementation	CORE
WP3: Digital Tools as enablers of Thesens H4C and common initiatives	3,1	GIS tool for dynamic symbiotic maps	TLNV
	3,2	Matching frameworks and development of symbiotic models	ICCS
	3,3	DPPs and end-users interactions	ICCS
	3,4	Prediction of resource logistics	CORE
	3,5	Data Harmonization and Secure data exchange	ICCS
	3,6	Marketplace & modules integration across digital H4C	IRIS
WP4: Separation and Sorting technologies	4,1	Separation and sorting at multiple levels analysis and Materials Upcycling, optimization and design	NTUA
	4,2	Mixed urban waste separation	ICCS
	4,3	Textiles sorting	ICCS
	4,4	Plastics recovery from mixed waste	ICCS
	4,5	Metals Separation/ Glass separation	ICCS
	4,6	Resource optimization for symbiotic flows management at plant level	DUTH
	4,7	Smart bins for Attica	ICCS
	4,8	XR for human-robot interaction/collaboration	ICCS
WP5: Material Processing Technologies	5,1	Textiles upcycling	CIRCE
	5,2	Packaging Upcycling	HMU
	5,3	Innovative construction/cement materias	TUD
	5,4	Biobased thin films for packaging	NCSR
	5,5	Electrospinning for thin films	NTUA
WP6: Energy Symbiotic Flows	6,1	Study for Introduction of innovative RES technologies to District heating (DH) systems	CERTH
	6,2	Innovative solar thermal technology for DH covering heating and cooling purposes	CERTH
	6,3	Utilization of green H2 for production of heat and electricity through a CHP fuel cell system	CERTH
	6,4	Use of digital smart technologies for DH towards improved energy management and control	CERTH
WP7: Water Symbiotic Flows	7,1	Analysis for water needs per use for the whole Attica region	NTUA
	7,2	Decentralized reclaimed water valorisation for different uses	NTUA
	7,3	NBS for water recovery and reuse	NTUA
	7,4	Centralized water treatment for reuse in Eastern Attica	NTUA
	7,5	Novel Valorisation Pathways for Sewage Sludge in Attica	NTUA
	7,6	Strategic Master Plan on Circularity	NTUA
WP8: Piloting and Demonstration	8,1	Biobased materials and Packaging flow implementation	HMU
	8,2	Textiles flow implementation	TUC
	8,3	CDW & Glass flow implementation	ICCS
	8,4	Energy flow implementation	CERTH
	8,5	Water flow implementation	NTUA
	8,6	Validation of digital governance & collaboration schemes	ICCS
	8,7	Impact Assessment of Theseus H4C	CIRCE
WP9: Synergies, replication and Cooperation	9,1	Synergies with existing H4C and ECoP towards common tools and knowledge exchnage	A-SPIRE
	9,2	Synergies with Cities mission and CCRI initiative towards tools and methodologies	CIRCE
	9,3	Replication to Malmö and MO	RISE
	9,4	Roadmap for replicability to additional sites	ICCS
WP10: Nexus Modelling Social Awareness and Training	10,1	Climate Change in relation to the future of H4C	ICCS
	10,2	Water-Energy-Materials-Waste nexus decision support module	Imperial
	10,3	Cities climate modelling and evaluation of environmental and economic impacts	NTUA
	10,4	Social Awareness increase through AR	ICCS
	10,5	Upskilling & Training activities in region cities and universities	Imperial
	10,6	Impact on the health of citizens	RISA
	10,7	De-risking assessment	HSRT
WP11: Exploitation and Dissemination activities	11,1	Dissemination and communication activities	HSRT
	11,2	Policy recommendations and barriers	LiU
	11,3	Exploitation, Business models & IPR Management	HSRT
	11,4	Commercialization of solutions & trading opportunities	EMVIS
	11,5	Standardization of novel technologies and activities	DS
	11,6	Stakeholder engagement and Theseus bubbles	Imperial



3.1.6 Subcontracting costs' items

The beneficiaries must base their contracts/subcontracts according to the principles for best value for money and absence of any conflict of interest (according to Articles 6.2 and 9.3 of GA). Beneficiaries that are 'contracting authorities' or 'contracting entities' (within the meaning of the EU public procurement Directives 2004/18/EC and 2004/17/EC or any EU legislation that replaces these Directives) must moreover comply with the applicable national law on public procurement.

Table 3.1g

Partner	Cost (€)	Justification
CIRCE	20,000	for laboratory test for material characterization to optimize glycolysis process.

3.1.7 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

All depreciation costs for equipment, infrastructure or other assets in the project are in compliance with Article 6 and will be recorded in the appropriate beneficiary's accounts, purchased in accordance with Article 6.2.C of the grant agreement and written off in accordance with international accounting standards and the beneficiary's usual accounting practices.

Table 3.1h:

1/ICCS	Cost (€)	Justification	17/A.SPIRE	Cost (€)	Justification
Equipment	431,000	4 * HIS Camera(4*40=160K); Conveyor Belts for sorting 4*15K	Other Goods	30,000	Organization of common events between
		(60K); 2 industrial robots (800K); 2 Delta Robots (100K); Electric equipment for metals movement(31K)	and services		Theseus and other H4C(30K)
Rem.(<15%)	165,000	Total : 596,000	Rem.(<15%)	11,000	Total : 41,000
3/NTUA	Cost (€)	Justification	27/EYDAP	Cost (€)	Justification
Equipment	322,000	2 Sewer mining system for water treatment(2*100K) , Electrospinning Equipment(45K) , Gas Mixture system for fast pyrolysis (15K) & online monitoring device for gaseous emissions (25K), 7K Pumps, Pipes and valves for rainwater harvesting systems, 15K tanks for rainwater harvesting system and reclaimed water storage, 3K meteorological station, 10K sensors for water quality, 2K soil moisture meters	Equipment	65,000	Pumping stations for the sewer mining system (45K), Sensors for system monitoring (20K)
Other Goods and services	71,000	Chemical Regents & consumables for analysis(40K) , Chemical regents for advanced biological analysis(21K) (10K) development of drip irrigation system with precision irrigation	Other Goods and services	30,000	Irrigation system for reclaimed water (30K)
Rem.(<15%)	91,000	Total : 484,000	Rem.(<15%)	24,500	Total : 119,500
5/ADDMA	Cost (€)	Justification	22/IRIS		
Other Goods and services	83,000	Rain Gardens in the city(18K), Bioswale system(25K) , Green Corridor Configuration(40K)	Other Goods and services		
Rem.(<15%)	6,000	Total : 89,000	Rem.(<15%)		
7/DHCP	Cost (€)	Justification	9/CERTH	Cost (€)	Justification
Other Goods and services	30,000	Materials for the connection of piping from solar thermal system to (screws, nuts, washers, supports,electrodes, gases, grinding and cutting discs) (30K)	Equipment	200,000	An adsorption chiller for solar cooling from solar thermal panels (200K)
Rem.(<15%)	7,5000	Total : 37,500	Rem. (<15%)	96,500	Total : 296,5000
8/CLEAN	Cost (€)	Justification	33/Panchart	Cost (€)	Justification
Other Goods and services	120,500	Insulated steel pipe , electrowelded for underground hot water 1200m, (75K), Special pieces (t-connector, elbow, etc) (18K) , Cosumables for support bases, sand, connections, electro welding (27,5K)	Equipment	45,000	(i) Tensile testing machine (15K) και (ii) drop test Machine(10K), (iii) Elmendorf Tear Tester (10K), Film thickness laser measurement system (10K€)

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Rem. (<15%)	7,0000	Total : 127,500	Rem. (<15%)	22,500	Total : 67,500
28/Halcor	Cost (€)	Justification	34/Fibran	Cost (€)	Justification
Equipment	132,500	copper sample melting furnace (5.000) & power unit supply (20K), air compressor (7.5K)Shredder (100K)	Equipment	105,000	Granulometry (6K), Fiber diameter analyzer (55K) Particle size analyzer used for analysis at industrial and CDW recycling (50K)
Other Goods and services	50,000	Technical on site preparation for the sorting system(30K), Intefration with the existing lines(20K)	Other Goods and services	13,500	2 biosolubility evaluations (5K), 4 Reaction to fire evaluations (8.5K)
Rem.(<15%)	36,000	Total : 218,500	Rem. (<15%)	7,500	Total : 126,,500
10/NCSR	Cost (€)	Justification	36/AB	Cost (€)	Justification
Other Goods and services	90,000	Robotic consumables for the development of a continuous testing line of biobased films(90K)	Other Goods and services	88,000	Electromechanical preparation for the installation of the equipment (40K) , test trials of the system (40K) , Publication fees(8K)
Rem. (<15%)	7,5000	Total : 97,500	Rem. (<15%)	17,000	Total : 105,000

3.2 Capacity of participants and consortium as a whole

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Figure 12 Theseus H4C main activities

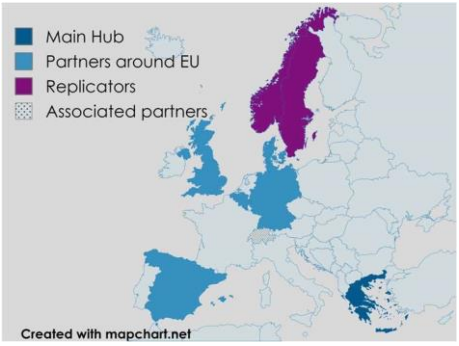

Figure 11 Theseus Consortium across EU

Theseus will be executed by a well-balanced and multidisciplinary consortium composed by 47 partners and 5 associated, from 9 European countries, including 13 industries / Large companies, 12 SMEs, 18 Academia/RTOs (including also prestigious both Greek and European universities: TUD, LiU, ILC, HUD and RTOs: CIRCE, RISE, SINTEF), 8 public institutions and 1 non-profit organisations. ICCS will act as coordinator, its wide experience in several H2020 and HE successful projects including I-US related ones, ensures the correct execution and implementation of the project. NTUA will be the technical coordinator ensuring scientific quality in the pilots and the value chains. The consortium is a high-quality partnership fully capable to attain the targeted objectives including key actors, ensuring results will maximize I-US symbiosis in the region and replication of the solutions across Europe. Region of Attica is the main public body occupying the central hierarchical position, ADDMA is the development company of Athens Municipality and SusCITY is a city network of more than 50 Greek Municipalities. EYDAP is the water utilities manager, WATT and RECYCOM are managing an important part of recyclable packaging and textiles waste respectively in Attica, while DHCP is the relevant municipal heating network manager in Prolemaida. A total of 25 Universities, Research organizations and innovative SMEs from both Greece and Europe, will develop the innovative solutions and exploitation plans, while the nexus of the symbiotic value chains, with local industry and stakeholders is completed with and 9 strong industrial partners (ThraceMarble, COMO, TSIMIS, ThracePoly, Panchart, AB, Halcor, KEBE, Fibran) having activities around Greece and Europe, thus enabling potential replication and upscaling, used as waste sources, end-users and validators of circular product or both. A.SPIRE will directly link Theseus Activities with H4C ECoP and P4P maximizing its impact and the upscaling potential to EU regions, while RISE and SHAS will coordinate the activities of the 2 Nordic replication ecosystems.

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ICCS: Coordinator, Sorting & Digital Technologies Provider; NTUA: Technical & Pilots' manager; Innovation provider in water & materials.	Attica: Regional authority, H4C hierarchical 1st; ADDMA: Athen's municipal development company; SusCITY: Greek cities network, involved in in smart bins, policy making and engagement.	A.SPIRE: managing P4P partnership, representing EU innovative process industries. In charge of synergies with H4Cs ECoP and upscaling across EU; DS, SN: standardization bodies (DK,NO) in charge of relevant activities.

Universities/RTOs: CERTH: energy innovations and piloting; DUTH: algorithms for sorting; UAEGEAN: regional science research; TUC, HUD: textiles innovation providers; NCSR, HMU: packaging innovations; TUD, SINTEF: construction materials innovations; LiU: governance innovations; ICL: modelling and training activities; CIRCE: textile and assessment activities.	Innovative SMEs: CIBOS: validation and exploitation of sorting systems; CORE: Digital planning and prediction of resource logistics development; IRIS: Platform development and Matchmaking; HSRT: C&D&E activities; TWI: XRT/XRF innovations for sorting; EMVIS: Commercialization expertise and activities; RISA: Modelling experts, health; TLNV: GIS tool development; CLEAN: Certh's spin-off in RES innovations development.	Sources & Utilities Managers: WATT: Blue Bins provider of MSW; RECYCOM, EYDAP: Attica's water utility manager; DHCP: DH manager of energy pilot area; ThraceMarble: source of marble dust;

Replicators: RISE, MALMO: Malmo replication; SHAS, KPH, HFB: MO replication -- Associated partners: TVP: energy innovations; MIP: Industrial Park Manager in MO replicator NFK RANA: municipal authorities in MO replicator, RU: Utility manager in MO replicator.	End users of CPs: COMO: textiles fibersvalidation; TSIMIS, ThracePoly: thin films use and validation; Panchart: packaging materials validation; AB: provider and end user (reuse) recycling of , sorted glass bottles; Halcor: metal croppers; KEBE, Fibran: construction materials validation;

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3.2.1 Associated Parners

THESEUS involves five associated partners, namely Tvp Solar Sa based in CH that will receive complementary funding from their state government and Mo Industripark As, Nordland Fylkeskommune, Rana Utvikling As, Rana Kommune based in NO who will not receive complementary funding from their state government.

The total estimated costs of the associated partner Tvp Solar Sa to implement the project tasks in WP1, WP6, WP8, WP9, WP10, WP11 amount to EUR 1,744,750.00 as shown below.

The Associated Partner listed will not receive any EU funding and the budget set out in this sub-section will be provided by the Associated Partner.

Table 2 Budget Table Of Tvp Solar Sa

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A.Personnel Costs	B.Subcontracting Costs	C.Purchase Costs/C1 Travel and subsistence	C.Purchase Costs/C2 Equipment	C.Purchase Costs/C3 Other goods, works and services	D.Other cost categories/ D2.Internally invoiced goods and services	E.Indirect costs	Total Costs	Max Eu contribution	Requested Eu Contribution	Maximu Grant amount
310,800€	1,280,000€	25,000€	0.00€	36,000€	0.00€	92,950€	1,744,750€	0.00€	0.00€	0.00€

The below clarifications are needed:

a. TVP subcontracting:

The exact scope of the subcontract and the cost breakdown:

TVP	Amount [€]	Justification
Subcontracting	800.000 480.000	Subcontracting is related to WP6 and T8.4 Manufacturing of the HT-Power panels: 1600m2 of HT-Power panels for 500€/m2 Balance of System (including the monitoring and control system of the solar thermal system) for one system with a total of 1600m2 of panels; cost 300€/m2 Note: The solar thermal panels will be manufactured and the Balance of System components will be procured by TVP' Italian subsidiary, Tresol S.r.l. According to the Swiss funding authority such costs fall under the Subcontracting category.*
TOTAL	1.280.000	

*Tresol - TVP's full subsidiary (TVP owns 99% of Tresol) - is the only manufacturer worldwide of our High Vacuum Flat Panels (HVFPs). Therefore, there is no alternative supplier for our proprietary technology. For the balance of system (BoS): They confirm that the procurement of the components will be made according to their corporate procedure.

b. Norwegian associated partners:

As for the associated partners of Norwegian (NO) Replicator, as already indicated in section 3.2, they are:

- MIP: Industrial Park Manager
- NFK, RANA: municipal authorities
- RU: Utility manager

They have been included in Task 9.3 in which they will be providing input. Specifically, they have an important role to the Norwegian replicator and its defined activities. The associate Norwegian partners are public authorities (communities/utility manager) as well as the industrial park manager, therefore their role (institutional and operational) in the local area is important and they will operate as multipliers to the results and the impact of the Norwegian replicator. They have agreed to participate and multiply the activities of the replicator as associate partners with no funding and no administrative overheads.

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Section 4 ETHICS

The consortium confirms compliance with the ethical principles and the applicable EU, international, and national laws for the ethics issues identified in the Ethics Summary Report and any additional ethics issues that may emerge in the course of the grant, will be ensured. The Consortium also confirms that for any applicable ethics issue, the guidance provided in the European Commission Ethics Self-Assessment Guidelines will be rigorously followed

4.1 Ethical dimension of the objectives, methodology and likely impact

Each member of Theseus leading or participating in research task that involves human participants will ensure the respect for people for human dignity, fair distribution of benefits and burden of research,

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protection of the values, rights and interests of participants. Ethical standards and guidelines will be rigorously followed and applied. Ethics approvals must be obtained, if required and fully informed consent of the participants. It is the responsibility of activity leaders to obtain the relevant approvals and to ensure that their research complies with the laws and regulations of related EU Member States. Human participants will be required to provide information e.g. via interviews, online forms and questionnaires for user engagement, definition of scenarios, gathering inputs for new solutions and technologies. Participation is voluntary requiring agreement to participate with a clear affirmative act. Theseus will apply clear informed consent procedures with dedicated forms. Although no adverse side effects are expected, a plan for collecting, reporting, and managing adverse and other unintended effects will be undertaken. All partners are fully aware of this Charter and Directive and have abided by these regulations in past and on-going research projects and any individual partners belonging to professional bodies will abide by their ethics rules and regulations. Theseus consortium will keep a record of all potential ethical issues that arise and evaluated through discussion with the stakeholders. Ethics approval for the project will be sought from all necessary and applicable review panels and relevant committees in all partner countries undertaking any work. Consortium will examine the following aspects in every related activity: •Composition, roles, and responsibilities of the partners in charge;
•Description of research questions, specific objectives and justification of study; study design, including type, settings and sites where data will be collected; Inclusion and exclusion criteria; Time schedule;
•Estimated number of participants needed to achieve study objectives and how these were determined; Plans for assessment and collection of outcome, baseline, and other trial data and study instruments; •Who will obtain informed consent or assent from potential trial participants or authorized surrogates, and how; How personal information will be collected, shared, erased/anonymized if necessary and maintained; Theseus objectives also include an important ethical dimension related to its overall methodology and regarding the use of AI and advanced data analysis for the development of the sorting, classification and optimization systems. The AI systems developed will follow a transparency and accountability principle. The data sets used will be documented in English and all participants and end-users will be trained on their proper use and informed about their limitations. Theseus will use artifical intelligence for the development for some of the digital tools of the project as well as the seperation and sorting solutions. The above usage if AI is of limited risk based on the EC's AI risk classification and regulatory framework [1]. It is not expected to raise any ethical concerns about human rights and values, or to have any negative impact such as environmental, social, political or financial. The interaction with the AI systems, their abilities, limitations and benefits will be presented at meetings, and through the design of technical documents in English to explain their concepts to the end-users. The performance of the solutions will also be performed with deterministic systems following know standards and ISOs.

4.2 Compliance with ethical principles and relevant legislations

All the local legislations for virgin raw or secondary materials, waste, wastewater treatment, energy, social and governance, digitization including its AI systems, will be analyzed and respected in collaboration with the local and regional authorities, Environmental or other relevant Ministries, in harmony with the respective European directives and regulations. Also, Greek legislation will be followed and, in case of any differences with EU legislation, the strictest will be taken into consideration. Theseus will ensure that it is fully compliant with the Do No Significant Harm (DNSH) Principle since its activities as well as the expected life cycle impact of the innovations at a commercialization stage, substantially contribute in a positive way and/or do not significantly harm any of the six environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation (EU) 2020/852. Theseus will apply a life-cycle oriented approach (validated through the utilization of relevant LCA tool when possible) to ensure that proposed solutions are sustainable and consisting of 3 steps: i)Identification of actual and potential adverse impacts during the life cycle of the project activities ii)Prevention or mitigation of adverse impacts iii)Accounting for how adverse impacts are addressed by tracking performance and communicating results. Theseus activities are expected to have a direct positive impact to the environmental objectives defined: Climate Change Mitigation by reducing GHG emissions (use Recommendation 2013/179/EU or ETSI ES 203 199,ISO 14067:2018 or ISO 14064-2:2019 to calculate lifecycle GHG and net emissions), to Pollution prevention & control through waste reduction and identifying harmless substances and contaminants in pilots, to Transition to Circular

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Economy through the establishment of symbiotic and circular flows between urban environment and same or different industries for materials, wastes, energy and water, whereas is not expected any negative impact to the sustainable use and protection of water and marine resources, protection and restoration of biodiversity & ecosystems and the Climate Change adaptation. Specifically, Theseus will make sure that the related DNSH criteria (on Climate Adaption; climate adaptation; water; circular economy; pollution prevention and biodiversity) from the EU Taxonomy Compass will be applied for the below project related activities:

-Data processing, hosting and related activities
-Data-driven ICT solutions for GHG emissions reduction
-Computer programming, consultancy and related activities
-Close to market research, development and innovation

And if applicable in the below as well: ·

-Processing and Manufacturing of Plastics, Textiles, ·
-Processing of Water and Energy. ·
-Engineering activities and related technical consultancy dedicated to adaptation to climate change · - Any other activity that may arise during the project

ANNEX 2